Multi-Armed Bandits with Arriving Arms: Sequential Screening, Dynamic Regret, and Sublinear Guarantees

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Abstract

We study a stochastic multi-armed bandit problem in which the set of available arms expands over time. This setting arises in sequential experimentation when new actions or treatments become available during an ongoing study, making regret against a single best arm in hindsight inappropriate. We instead evaluate performance relative to the best arm currently available, leading to a dynamic-regret criterion for arriving-arm environments. To address the resulting challenges of arrival information discrepancy (AID) and a drifting benchmark (DB), we propose UCB for Arriving Arms (UCB-AA), an elimination-based procedure with an aiding preliminary screening step for newly arrived arms before full competition with incumbent arms. We show that UCB-AA attains regret bounds that depend explicitly on the arrival process, achieves sublinear dynamic regret under regularity conditions on gap evolution, and admits an online extension for unknown horizons. Simulation results show that UCB-AA reduces wasted pulls and maintains a smaller active arm set while preserving competitive regret performance.

Keywords: Sequential experimentation; adaptive screening; expanding action sets; stochastic bandits; arrival information discrepancy; drifting benchmark

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1 Introduction

The classical multi-armed bandit framework assumes that the set of available actions is fixed in advance. In many modern applications, however, the action set expands over time. Precision oncology provides a representative example: as genomic profiling has advanced, the set of approved targeted therapies for non-small cell lung cancer has grown substantially, so a trial designed around a fixed collection of treatments may become outdated before it is completed. Similar issues arise in online advertising and other adaptive decision problems in which new options appear sequentially.

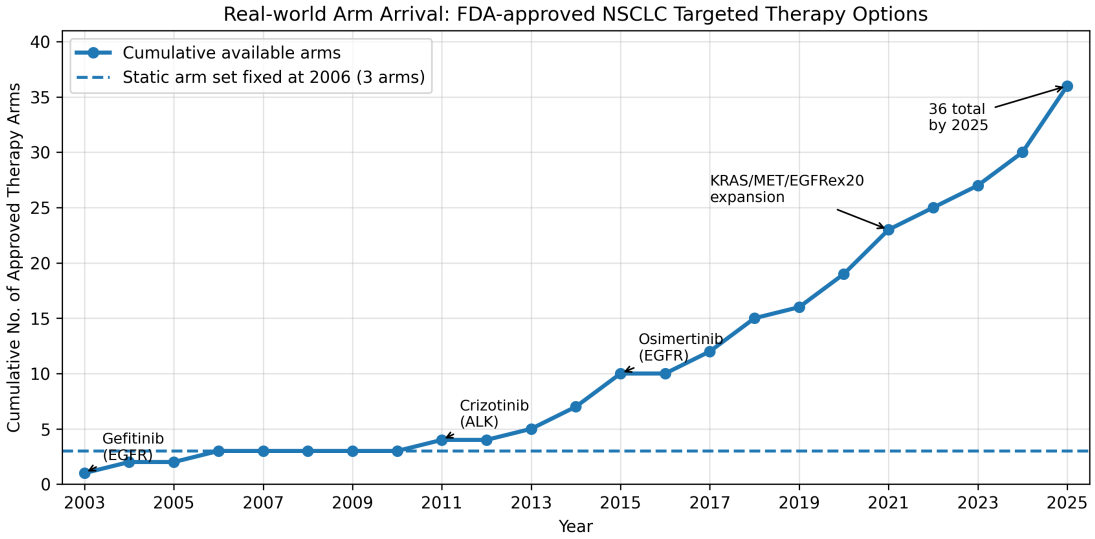


Figure 1: **The rapid expansion of the decision space in precision oncology.** FDA approval data for NSCLC (2003–2025). The solid line shows the cumulative number of targeted and immunotherapies (arms), from 1 in 2003 (Gefitinib) to 36 by 2025. A static algorithm fixed to the 2006 arm set would fail to incorporate most subsequently available treatments, illustrating the need for methods that accommodate arriving arms.

Motivated by such problems, we study the **multi-armed bandit problem with arriving arms** (MAB-AA). Let A_t denote the set of arms available at time t, and suppose that $A_t \subseteq A_{t+1}$, so that new arms arrive over time and remain available thereafter. In this

setting, performance is more naturally evaluated against the best arm currently available than against the single best arm in hindsight, since the latter may not have been available earlier.

Relative to the classical stochastic bandit setting, MAB-AA introduces two main difficulties. The first is *arrival information discrepancy (AID)*: newly arrived arms must be explored, which repeatedly diverts samples away from well-understood incumbent arms. The second is a *drifting benchmark (DB)*: as better arms enter the system, both the identity of the optimal arm and the regret benchmark may change over time.

The arriving-arm setting is related to, but distinct from, several existing bandit formulations. Unlike infinite-armed bandits (Berry, Chen, et al., 1997), where arms are actively sampled from a reservoir, new arms here arrive exogenously and remain available thereafter. Unlike mortal bandits (Chakrabarti et al., 2008), the defining feature is not arm death, and unlike sleeping bandits (Kleinberg, Niculescu-Mizil, and Sharma, 2010), the available set evolves monotonically over time. Our setting is also related to bandits with moving comparators: in nonstationary and switching bandits, the best arm may change because reward distributions vary over time (Besbes, Gur, and Zeevi, 2014; Auer, Gajane, and Ortner, 2019), whereas in MAB-AA each arm has a stationary reward law after arrival and benchmark changes are driven only by the exogenous arrival of new arms.

Most closely related, Ghalme et al. (2021) studies ballooning bandits with regret measured against the best currently available arm, but focuses on feasibility of sublinear regret under restricted arrival patterns. Relatedly, Qi et al. (2025) considers an expanding-arm extension in graph-feedback bandits with similar arms. In contrast, our focus is the unstructured arriving-arm case, which is more suitable in applications where new options are introduced over time, previously introduced options remain available, and reliable similarity or contextual structure across options is unavailable. Examples include online product or content recommendation with continuously added items, adaptive A/B testing with newly

launched treatments or designs, and sequential decision problems in which candidate actions are generated by an external pipeline rather than chosen by the learner.

From a statistical perspective, multi-armed bandit problems have long been studied as models for sequential allocation and adaptive experimentation, dating back to Robbins (1952) and the asymptotically efficient allocation theory of Lai and Robbins (1985); see also Lai (1987) for connections to adaptive treatment allocation. This viewpoint has motivated a broad statistical literature, including Bayesian and index-based approaches to sequential allocation (Gittins, 1979; Berry and Fristedt, 1985), response-adaptive randomization in clinical trials (Zelen, 1969; Wei and Durham, 1978; Villar, Wason, and Bowden, 2015; Williamson and Villar, 2020; Aziz, Kaufmann, and Riviere, 2021), covariate-dependent and nonparametric bandit methods (Woodroofe, 1979; Yang and Zhu, 2002; Rigollet and Zeevi, 2010; Perchet and Rigollet, 2013), and recent high-dimensional, semiparametric, and transfer-learning formulations (Qian and Yang, 2016; Qian, Ing, and Liu, 2024; Cai, Cai, and Li, 2024). These works share with ours the broad goal of learning from accumulated evidence to reduce allocation to inferior alternatives, but they focus on a fixed set of treatments or actions, and thus are not applicable to the arising arm setting that we study.

Methodologically, the two challenges call for different but complementary design principles. To address AID, our approach combines elimination steps (Even-Dar, Mannor, and Mansour, 2006; Auer and Ortner, 2010; Perchet and Rigollet, 2013; Qian and Yang, 2016) with a pre-elimination mechanism tailored to the expanding action set. The key idea is to prevent the main competition stage from being repeatedly disrupted by every newly arriving arm. Instead, late arrivals are first screened against incumbent candidates, so that clearly inferior arms can be removed before receiving the full exploration budget assigned to serious contenders. To handle DB, the algorithm repeatedly updates the relevant comparison set as new arms arrive, and the analysis evaluates regret against the best arm currently available rather than against a fixed hindsight oracle. Based on these ideas, we

propose **UCB for Arriving Arms (UCB-AA)**. For UCB-AA, we derive regret upper bounds that depend explicitly on the arrival process, establish sublinear dynamic regret under suitable conditions on gap evolution, study representative lower-bound regimes, and develop an online extension for unknown horizons.

Our contributions are threefold. First, we formulate the arriving-arm stochastic bandit problem using a dynamic-regret benchmark relative to the best currently available arm. Second, we propose a screening-based elimination procedure that is tailored to the information imbalance between incumbent and newly arrived arms. Third, we provide regret guarantees, lower-bound insights in representative regimes, and an online extension, together with simulations showing that UCB-AA reduces wasted pulls and maintains a compact active arm set.

The remainder of the paper is organized as follows. Section 2 formulates the problem and introduces the regret criterion. Section 3 presents UCB-AA, its theoretical guarantees, and an online extension for unknown horizons. Section 4 develops lower bounds in representative regimes. Section 5 reports numerical results. Section 6 concludes.

2 Multi-Armed Bandit Problem with Arriving Arms

2.1 Problem Formulation

We consider a sequential decision process over time steps $t = 1, 2, \ldots$. Let $\mathcal{A}$ denote the (possibly infinite) set of all arms. At each time step t, only a subset $\mathcal{A}_t \subset \mathcal{A}$ is available, and we assume that the available set is non-decreasing: $\mathcal{A}_t \subseteq \mathcal{A}_{t+1}$. For any arm $i \in \mathcal{A}$, define its arrival time as $\tau_i := \inf\{t \geq 1 : i \in \mathcal{A}_t\}$. Each arm i is associated with an unknown reward distribution on $[0, 1]$ with mean r_i. When arm i is pulled at time t, a reward $X_{t,i}$ is generated; for each fixed i, we assume $\{X_{t,i}\}_{t\geq 1}$ are i.i.d. with mean r_i, and rewards are independent across arms. The arrival process is treated as exogenous to the learner.

A policy is a sequence $\pi = (\pi_t)_{t\geq 1}$, where each decision rule π_t maps the past history to an available arm $I_t := \pi_t(H_{t-1}) \in \mathcal{A}_t$, where $H_{t-1} := \Big(\mathcal{A}_s, I_s, X_{s,I_s}\Big)_{s=1}^{t-1}$ denote the observed history up to time $t-1$. Then the learner observes only X_{t,I_t}. Because the available set changes over time, the best arm may also change. We define the best mean reward at time t as $r^*(t) := \max_{i\in\mathcal{A}_t} r_i$.

For any (possibly random) stopping time τ, define the cumulative regret by the (expected) regret against the best available arm at each time:

$$R_\tau(\pi) := \sum_{t=1}^{\tau}\Big(r^*(t) - r_{I_t}\Big).$$

For a deterministic horizon T, this reduces to $R_T(\pi)$. For the round-based procedure analyzed below, the relevant stopping time is $T_0(\pi)$, the total number of pulls used to complete M rounds. This benchmark differs from the usual regret against the single best arm in hindsight. In the arriving-arm setting, the terminal best arm in $\mathcal{A}_T$ may not have been available in earlier periods, so comparing the learner with that arm throughout the entire horizon may overstate the achievable performance. The criterion above instead compares the learner at each time t with the best arm that was actually available at that time. When $\mathcal{A}_t$ is constant over time, this formulation reduces to the classical stochastic multi-armed bandit problem.

2.2 Main Statistical Challenges

Relative to the classical fixed-arm stochastic bandit model, the arriving-arm setting introduces two related sources of difficulty.

Arrival Information Discrepancy (AID): A first difficulty is arrival information discrepancy. With a finite sampling budget up to time T, the learner must allocate samples both to incumbent arms and to newly arriving arms whose rewards are still highly uncertain. Under continual arrivals, this repeated re-exploration can prevent the learner from

concentrating pulls on the best currently available arm. Consequently, without additional regularity conditions on the arrival process, sublinear regret may be unattainable.

Drifting Benchmark (DB): In contrast to the classical setting where the best mean reward is fixed, the benchmark in MAB-AA is time-dependent: $r^*(t) = \max_{i \in \mathcal{A}_t} r_i$. Since $\mathcal{A}_t$ expands over time, both the identity of the best arm and the value $r^*(t)$ may change when new arms arrive. Consequently, for a fixed arm i, the suboptimal gap $r^*(t) - r_i$ is generally time-varying (and non-decreasing in t). This drifting benchmark complicates both algorithm design and analysis, because the learner must continuously compare well-sampled incumbent arms with newly arriving arms whose estimates are still highly uncertain.

3 UCB for Arriving Arms (UCB-AA) Algorithm

3.1 Algorithm Description

UCB-AA is an elimination-based procedure organized by rounds. Its main idea is simple: newly arriving arms are not allowed to compete immediately with well-sampled incumbent arms. Instead, they are first buffered and screened at lower cost. Only those that survive this screening step enter full competition in the next stage. This design aims to reduce wasted exploration on clearly suboptimal late arrivals while preserving the ability to detect genuinely competitive new arms.

Fix a finite number of rounds M. For each round $m = 0, 1, \ldots, M$, let A_m denote the set of arms arriving during round $m - 1$, with A_0 the initial arm set, and define $A(m) := \bigcup_{j=0}^{m} A_j$. Let C_m denote the set of arms surviving round m. Then round m starts from the surviving incumbent arms C_{m-1} together with the newly admitted batch A_m. We write B_m for the active set in that round.

We assume that the cumulative number of arrived arms is bounded by a known

nondecreasing function $K(m)$:

$$|A(m)| \le K(m), \qquad |A_{m+1}| \le K(m+1) - K(m).$$

For simplicity, assume that the best arm is unique in each round. Let $r_{*(m)} := \max_{i \in B_m} r_i$, and let $*(m) \in B_m$ denote the corresponding optimal arm. For each arm i, define its arrival round $t_i := \min\{m : i \in A_m\}$, and its round-$m$ suboptimality gap

$$\Delta_i(m) := \begin{cases} r_{*(m)} - r_i, & m \ge t_i, \\ 0, & m < t_i. \end{cases}$$

In round m, UCB-AA proceeds in two stages. As shown in Algorithm 1, the earlier phases $p = 0, \ldots, m-2$ are used for pre-elimination of newly arrived arms, while the last two phases $p = m-1, m$ are used for elimination on the merged active set.

3.1.1 Stage I: Pre-elimination of newly arrived arms

At the beginning of round m, the newly admitted batch A_m is screened against the surviving incumbent set C_{m-1}. During the preliminary phases $p = 0, \ldots, m-2$, only arms in A_m are subject to removal: each surviving new arm is sampled to the phase-specific target n_p, and is discarded if its upper confidence bound falls below the largest lower confidence bound of the reference set. This stage exploits the information asymmetry between incumbents and arrivals. Incumbent arms have typically accumulated more samples, whereas newly arrived arms are still highly uncertain. Screening arrivals before full competition therefore prevents clearly inferior late arrivals from consuming the exploration budget assigned to serious contenders.

3.1.2 Stage II: Formal elimination on the active set

After pre-elimination, the surviving new arms are merged with the incumbent arms to form the active set for round m. UCB-AA then applies elimination to the full active set.

Algorithm 1 UCB for Arriving Arms (UCB-AA)

Input: number of rounds M, parameter $N > 0$.

Initialization: $\tilde{\Delta}_0 \coloneqq 1$, $B_{-1} \coloneqq \varnothing$, $C_{-1} \coloneqq \varnothing$, $T_0 \leftarrow 0$.

procedure ELIM($S,\ R,\ p$)

  $n_p \coloneqq \left\lceil \frac{2\log(N\tilde{\Delta}_p^2)}{\tilde{\Delta}_p^2} \right\rceil$

  Pull each $j \in S$ to n_p total pulls; update T_0 after each pull

  Let $b_p \coloneqq \sqrt{\frac{\log(N\tilde{\Delta}_p^2)}{2n_p}}$; set $\mathrm{LCB}_j = \hat{r}_j(n_p) - b_p$, $\mathrm{UCB}_j = \hat{r}_j(n_p) + b_p$

  Remove from S any j with $\mathrm{UCB}_j \le \max_{i \in R} \mathrm{LCB}_i$

  return S

end procedure

for $m = 0, 1, \ldots, M$ **do**

  Receive A_m and set $B_m^0 \coloneqq C_{m-1} \cup A_m$

  if $|A_m| \neq 0$ and $m \ge 2$ **then**

    $A_m^0 \coloneqq A_m$

    for $p = 0, 1, \ldots, m-2$ **do**

      $A_m^{p+1} \leftarrow \text{ELIM}(A_m^p,\ B_m^p,\ p)$; $B_m^{p+1} \coloneqq C_{m-1} \cup A_m^{p+1}$

    end for

  end if

  if $m = 0$ **then**

    $B_m^1 \leftarrow \text{ELIM}(B_m^0,\ B_m^0,\ 0)$

  else

    for $p = m-1, m$ **do**

      $B_m^{p+1} \leftarrow \text{ELIM}(B_m^p,\ B_m^p,\ p)$

    end for

  end if

  $C_m \coloneqq B_m^{m+1}$, $\tilde{\Delta}_{m+1} = \frac{1}{2}\tilde{\Delta}_m$

end for

At this stage, both incumbent and newly arrived arms may be removed. The surviving set is denoted by C_m and is carried into the next round. Thus, unlike a naive elimination rule, UCB-AA does not allow newly arrived arms to enter the main elimination stage immediately upon arrival. They must first pass the preliminary screening step in phases $p = 0, \ldots, m-2$. Once merged into the active set, however, all surviving arms are handled in the same way.

3.2 Regret Decomposition

Our analysis is indexed by the number of rounds M, rather than by fixing the total number of pulls in advance. We assume that the sampling budget is sufficient for UCB-AA to complete all M rounds, so the total number of pulls used up to the end of round M is a policy-dependent random variable. For a policy π, let $T_0(\pi)$ denote this total, and write

$$T := \mathbb{E}[T_0(\pi)].$$

Let $N_{i,m}^{\pi}$ be the number of times arm i is pulled during round m. Then $T_0(\pi) = \sum_{m=0}^{M} \sum_{i \in B_m} N_{i,m}^{\pi}$, where B_m is the active set at the beginning of round m.

Because UCB-AA updates its active set only at round boundaries, it is convenient to first analyze regret against the benchmark fixed at the beginning of each round. Define the *roundwise regret*

$$R_M(\pi) := \sum_{m=0}^{M} \sum_{i \in B_m} N_{i,m}^{\pi} \Delta_i(m).$$

This quantity charges each pull in round m against the best arm in the active set at the start of that round. Our target criterion, however, is the original dynamic regret

$$R_{T_0}(\pi) = \sum_{t=1}^{T_0(\pi)} \Big(r^*(t) - r_{I_t} \Big),$$

where $r^*(t) = \max_{i \in A_t} r_i$ is the best mean reward among all arms available at time t. These two regret notions differ because a benchmark-improving arm may arrive within a round but is not incorporated into the active set until the next round.

Accordingly, we decompose the dynamic regret as $R_{T_0}(\pi) = R_M(\pi) + D_M(\pi)$, where

$$D_M(\pi) := R_{T_0}(\pi) - R_M(\pi)$$

is the *delay regret*. The term $R_M(\pi)$ captures the main statistical cost of identifying competitive arms under the round structure, while $D_M(\pi)$ captures the additional loss caused by delayed incorporation of within-round benchmark improvements. The remainder of this section controls these two terms separately: we first bound the roundwise regret $R_M(\pi)$, and then bound the delay regret $D_M(\pi)$.

3.3 Upper Bound for the Roundwise Regret

We now bound the principal term in the decomposition of Section 3.2. The main question is how much regret is incurred before a suboptimal arm is removed under the elimination schedule of UCB-AA. To formalize this, define

$$m_i := \min\Big\{m \in \{t_i, t_i+1, \dots, M\} : \tilde{\Delta}_m < \tfrac{1}{2}\Delta_i(m)\Big\},$$

the first round in which arm i becomes statistically eliminable under the target resolution sequence. Also define

$$\tilde{m}_i := \min\Big\{m \in \{0, 1, \dots, M\} : \tilde{\Delta}_m < \tfrac{1}{2}\Delta_i(m_i)\Big\},$$

so that $\tilde{m}_i \le m_i$. We classify suboptimal arms according to the phase in which they are eliminated with high probability:

- G_1 ($m_i = t_i$): arms eliminated when they first appear;
- G_2 ($m_i > t_i,\ \tilde{m}_i = m_i$): arms are eliminated in the formal elimination phase $p = m_i$ of round m_i;
- G_3 ($m_i > t_i,\ \tilde{m}_i < m_i$): arms are eliminated in the formal elimination phase $p = m_i - 1$ of round m_i.

For a threshold $\lambda \geq 2^{-M+1}$, define $A'(\lambda) := \{i \in A(M) : \Delta_i(M) > \lambda\}$. Then $A'(\lambda)$ admits the partition $A'(\lambda) = G_1(\lambda) \cup G_2(\lambda) \cup G_3(\lambda)$. Moreover, define

$$\bar{m}_i := \begin{cases} \tilde{m}_i, & i \in G_1(\lambda) \cup G_2(\lambda), \\ m_i - 1, & i \in G_3(\lambda). \end{cases}$$

Thus $\bar{m}_i$ is the elimination scale at which arm i is removed with high probability.

Theorem 1 (Upper bound for the roundwise regret)**.** *For any $\lambda \geq 2^{-M+1}$, the regret of UCB-AA up to round M satisfies*

$$\mathbb{E}R_M(\pi) \leq \sum_{i:\Delta_i(M)\geq\lambda} \left(\tilde{S}_i + \frac{4\Big(K(m_i)+5\Big)\, R_i}{N\tilde{\Delta}^2_{\tilde{m}_i}} \right) + T \max_{i:\Delta_i(M)\leq\lambda} \Delta_i(M),$$

where $T = \mathbb{E}[T_0(\pi)]$ denotes the expected total number of pulls,

$$R_i := n_{t_i}\Delta_i(t_i) + \sum_{m=t_i+1}^{M} \Big(n_m - n_{m-1}\Big)\Delta_i(m),$$

and

$$\tilde{S}_i := \begin{cases} n_{\tilde{m}_i}\, \Delta_i(t_i), & i \in G_1(\lambda), \\ n_{t_i}\Delta_i(t_i) + \sum_{m=t_i+1}^{\overline{m}_i} \Big(n_m - n_{m-1}\Big)\Delta_i(m), & i \in G_2(\lambda) \cup G_3(\lambda). \end{cases}$$

Theorem 1 decomposes the roundwise regret into three statistically interpretable components. The leading term, $\tilde{S}_i$, is the identification cost of certifying that a suboptimal arm should be removed at the resolution dictated by the elimination schedule. The second term quantifies the additional regret incurred on low-probability events where statistical fluctuations delay elimination or allow an incorrect survivor to persist. The final truncation term collects arms whose terminal gaps are below the target resolution λ, and may therefore be treated as effectively indistinguishable at the scale of the analysis.

To obtain a simpler rate statement, we impose the following stability condition.

Assumption 1 (Gap stability)**.** *For every suboptimal arm $i \in A'(\lambda)$, there exists a universal constant $\tilde{C} > 0$, independent of λ, such that*

$$\Delta_i(M) \leq \tilde{C}\, \Delta_i(m_i).$$

Assumption 1 does not restrict the arrival times of arms; rather, it limits how much the benchmark can improve after an arm has already become statistically distinguishable. Equivalently, the post-detection benchmark increase cannot be larger than a constant multiple of the gap already visible at m_i. This is a mild condition in several common regimes. It holds with $\tilde{C} = 1$ if the best arm available at the terminal round has already arrived by m_i, or more generally if no later arrival improves the benchmark. It also holds whenever later improvements are moderate relative to the current detectable gap, for example if $r_{*(M)} - r_{*(m_i)} \le c\,\Delta_i(m_i)$ for a constant c, in which case $\tilde{C} = 1 + c$. This covers environments with bounded multiplicative benchmark growth after detection, as well as settings in which only a finite number of post-detection benchmark-improving arrivals occur and their cumulative improvement is comparable to the detection scale. Another simple case is a separated-gap regime: if every arm in $A'(\lambda)$ has $\Delta_i(m_i) \ge \delta_0 > 0$ for a fixed δ_0, then bounded rewards imply $\Delta_i(M) \le 1 \le \delta_0^{-1}\Delta_i(m_i)$. Thus the assumption mainly rules out extreme post-detection drift, where an arm is already clearly inferior but many substantially better arms arrive later and inflate its terminal gap by an unbounded factor.

Corollary 1. *Suppose that Assumption 1 holds for all* $i \in A'(\lambda) = \{i \in A(M) : \Delta_i(M) > \lambda\}$. *Let* $K \coloneqq K(M)$ *and choose*

$$N = 10K\, n_M, \qquad \lambda \asymp \sqrt{\frac{K \log K + K}{T}}.$$

Then the roundwise regret of UCB-AA satisfies

$$\mathbb{E}R_M(\pi) \lesssim \sqrt{K \log K \cdot T}.$$

Corollary 1 shows that UCB-AA achieves sublinear roundwise regret when the effective number of arriving competitors remains moderate relative to the available sample size. Here $K = K(M)$ is the cumulative number of arms that may have appeared by round M, rather than the total size of the global arm universe. In particular, the bound is informative

in regimes where $K(M) = o\left(\frac{T}{\log T}\right)$, so that the AID effect created by continual arrivals remains controllable.

Remark 1. *The parameter N must be fixed before running UCB-AA. In the theoretical analysis, we use the conservative calibration $N = 10K(M)n_M$, $n_M = \left\lceil \frac{2\log(N\widetilde{\Delta}_M^2)}{\widetilde{\Delta}_M^2} \right\rceil$, where $K(M)$ is a prior upper bound on the cumulative number of arrived arms. Although this implicit relation can be solved approximately, its role is mainly analytical: it ensures uniform control of low-probability failure events over all arms, but may lead to overly large pull targets n_m, and hence excessive forced exploration in both the pre-elimination and formal elimination stages. For implementation, we use a less conservative calibration based only on the monotonicity of the target sequence $\{n_m\}_{m=0}^{M}$. Since $f(x) = \frac{\log(Nx)}{x}$ is decreasing on $[\widetilde{\Delta}_M^2, 1]$ whenever $N\widetilde{\Delta}_M^2 \geq e$, a natural lower-end choice is $N_{\min} = \frac{e}{\widetilde{\Delta}_M^2}$. This optimistic calibration is not used in the regret proof, but provides a practical baseline with substantially smaller pull targets, and is adopted in the simulations.*

3.4 Control of the Delay Regret

Delay regret arises because UCB-AA updates its active set only at round boundaries. If a benchmark-improving arm arrives in the middle of round k, then the true dynamic benchmark improves immediately on the original timeline, whereas the roundwise benchmark used in $R_M(\pi)$ is updated only at the beginning of round $k+1$. Thus $D_M(\pi)$ captures the additional loss caused by this deferred incorporation of within-round benchmark improvements.

Fix a round k. Suppose that within this round there are q_k benchmark-improving arrival events, with increments

$$\beta_{k,1}, \ldots, \beta_{k,q_k} \geq 0, \qquad \sum_{j=1}^{q_k} \beta_{k,j} =: \delta_k,$$

where $\delta_k = r_{*(k+1)} - r_{*(k)}$ is the total benchmark increase associated with round k. Let L_k be the total number of pulls in round k, and let $1 \leq \tau_{k,1} < \cdots < \tau_{k,q_k} \leq L_k$ denote the

within-round locations of these improving arrivals. We define the exposure factor

$$\rho_k := \frac{1}{L_k \delta_k} \sum_{j=1}^{q_k} (L_k - \tau_{k,j} + 1)\beta_{k,j}, \qquad \delta_k > 0,$$

so that $0 \le \rho_k \le 1$. The factor ρ_k measures how much of round k is exposed to the benchmark increase: improvements that occur early contribute more to delay regret than those that occur late. For $\ell \ge k+1$, define the cumulative benchmark improvement after round k by

$$\Delta_{*(k)}(\ell) := r_{*(\ell)} - r_{*(k)} = \sum_{i=k}^{\ell-1} \delta_i.$$

We also define $m_{*(k)} = \min\Big\{\ell \ge k+1 : \tilde{\Delta}_\ell < \frac{1}{2}\Delta_{*(k)}(\ell)\Big\}$, the first round at which the cumulative improvement after round k becomes statistically detectable under the elimination schedule.

Remark 2. *For analytical convenience, we take the analyzed horizon to end at a checkpoint immediately after the last batch of arrivals has been admitted. Equivalently, one may append a terminal bookkeeping round with no further sampling. This convention does not require the learner to determine whether the final arrivals improve the benchmark; it only ensures that no arrivals remain unincorporated at the endpoint. Under this convention, the terminal delay contribution is null, so the endpoint condition involving $\rho_M \delta_M$ becomes vacuous.*

To bound $D_M(\pi)$, we impose a condition that limits how aggressively benchmark improvements can accumulate before they become statistically detectable. This condition is used to close the upper bound and should be viewed as a sufficient condition, not a necessary one.

Assumption 2 (Cumulative jump control)**.** *Assume that for every round k satisfying $\rho_k \delta_k > \lambda$, one has $m_{*(k)} > k+1$, and at least one of the following conditions holds:*

$$\Delta_{*(k)}\big(m_{*(k)}\big) \le \tilde{\Delta}_k \sqrt{\frac{M - m_{*(k)} + 1}{3\rho_k K(k)(M-k+1)}}, \tag{1}$$

$$\frac{\delta_k}{\Delta_{*(k)}(m_{*(k)} - 1)} \le \frac{2^{\,m_{*(k)}-k-2}}{\rho_k K(k)}. \tag{2}$$

Assumption 2 is most naturally satisfied when benchmark improvements remain moderate until they become detectable at the current resolution scale. This includes, for example, settings in which improvements occur gradually, settings in which large improvements are rare, or settings in which most benchmark-improving arrivals occur late within a round and therefore have small exposure factors ρ_k.

Proposition 1 (Delay regret under cumulative jump control)**.** *Let* $\lambda \geq 2^{-M+1}$. *Under Assumption 2,*

$$\mathbb{E}\Big[D_M(\pi)\Big] \leq \sum_{i:\Delta_i(M)>\lambda} \tilde{S}_i + \lambda T.$$

Proposition 1 shows that, under cumulative control of benchmark jumps, the delay regret is bounded by the same certification-type quantity that appears in the main regret analysis, up to the truncation term λT. Hence the delayed admission of within-round arrivals does not create a new leading-order source of statistical difficulty. More broadly, the result clarifies how the dynamic benchmark enters the regret analysis: the regret of MAB-AA can be viewed as consisting of a principal identification term and an additional delay term induced by benchmark improvement. Under Assumption 2, the latter remains controlled, so benchmark movement acts as a structured perturbation of the elimination problem rather than as a fundamentally separate source of regret.

3.5 General regret bound

We now return to the original dynamic regret $R_{T_0}(\pi) = \sum_{t=1}^{T_0(\pi)} \Big(r^*(t) - r_{I_t}\Big)$. By the decomposition in Section 3.2,

$$R_{T_0}(\pi) = R_M(\pi) + D_M(\pi).$$

Theorem 1 controls the principal term $R_M(\pi)$, and Proposition 1 controls the delay term $D_M(\pi)$. Combining the two results yields the following bound for the original regret criterion.

Theorem 2 (General regret bound)**.** *Let $\lambda \geq 2^{-M+1}$. Suppose that Assumptions 1 and 2 hold. Then the dynamic regret of UCB-AA satisfies*

$$\mathbb{E}R_{T_0}(\pi) \leq \sum_{i:\Delta_i(M)\geq\lambda} \left(2\tilde{S}_i + \frac{4\Big(K(m_i)+5\Big)R_i}{N\tilde{\Delta}^2_{\tilde{m}_i}} \right) + 2\lambda T.$$

In particular, under the conservative calibration of N in Corollary 1,

$$\mathbb{E}R_{T_0}(\pi) \lesssim \sqrt{K \log K \cdot T}.$$

3.6 Online UCB-AA

We briefly discuss the case in which the terminal round M is unknown. To remove the fixed-horizon requirement, we use a geometric-restart scheme and run UCB-AA over successive stages with geometrically increasing stage lengths. This yields an anytime version of the procedure, referred to as Online UCB-AA. The idea is standard: each stage is long enough to cover the possibility that the effective horizon is of that order, and restarting across stages incurs only a logarithmic overhead relative to the fixed-M analysis. We retain the algorithm here for completeness and defer the more detailed stage-wise analysis to the Supplementary Material.

At each restart, Online UCB-AA reinitializes the internal state of the base procedure for the new stage, while all pulls across stages are still counted in the cumulative regret. The next corollary summarizes the resulting regret guarantee in the unknown-horizon setting.

Corollary 2. *Suppose that Assumption 1 holds. The expected roundwise regret of Online UCB-AA satisfies*

$$\mathbb{E}R_M(\pi) \lesssim \sqrt{K(M)\,T\,\log T}.$$

If a stagewise version of Assumption 2 holds uniformly across the restart stages, then

$$\mathbb{E}R_{T_0}(\pi) \lesssim \sqrt{K(M)\,T\,\log T}.$$

Algorithm 2 Online UCB-AA

Input: Base algorithm UCB-AA

Initialize stage index $l \leftarrow 1$

while interaction continues **do**

  $M_l \leftarrow 2^l$, $\tilde{M}_l \leftarrow \sum_{i=1}^{l} M_i$, $\tilde{\Delta}_{M_l} \leftarrow 2^{-M_l}$

  $n_{M_l}(N_l) := \left\lceil \frac{2\log(N_l \tilde{\Delta}_{M_l}^2)}{\tilde{\Delta}_{M_l}^2} \right\rceil$

  Compute N_l from $N_l = 10K(\tilde{M}_l) n_{M_l}(N_l)$ (approx.)

  Run UCB-AA(M_l, N_l) for M_l rounds

  $l \leftarrow l + 1$

end while

Corollary 2 shows that the online extension preserves the same square-root-type behavior as the fixed-M procedure, at the cost of only a logarithmic overhead due to geometric restarts. Thus, lack of prior knowledge of the terminal round does not alter the qualitative statistical difficulty of the arriving-arm problem. A more detailed stage-wise bound is given in the Supplementary Material.

4 Lower Bounds in Representative Regimes

As mentioned earlier, without structural restrictions on the arrival process, sublinear regret may be unattainable (Ghalme et al., 2021). A fully general lower bound for the unrestricted arriving-arm model is therefore not our goal here. Instead, we focus on three representative regimes that isolate distinct hardness mechanisms in MAB-AA: static identification, repeated distractor certification, and near-tied benchmark updates under a drifting comparator.

The statistical hardness of MAB-AA is therefore not monolithic. In the static limit, the problem reduces to the classical identification barrier of stochastic bandits. When a

globally optimal incumbent is present from the outset and all late arrivals are inferior, the dominant cost is repeated certification of distractors. When the benchmark improves through a sequence of near-tied arrivals, the difficulty becomes a succession of local tests under a drifting comparator. This section isolates these three mechanisms and relates them to the upper-bound structure in Section 3.

To avoid repeating regime-specific growth conditions, we use the following scale-sensitive extension of the usual notion of uniform goodness.

Definition 1 (Uniform goodness at scale $h(T)$)**.** *Let $\mathcal{V}_T$ be a family of bandit instances indexed by T, and let $h(T) \to \infty$. A sequence of policies $\{\pi^{(T)}\}_{T\geq 1}$ is* uniformly good on $\{\mathcal{V}_T\}$ at scale $h(T)$ *if, for every $a > 0$,*

$$\sup_{\nu\in\mathcal{V}_T} \mathbb{E}_\nu[R_T(\pi^{(T)};\nu)] = o\Big(h(T)^a\Big) \qquad \text{as } T \to \infty.$$

When $h(T) = T$, it reduces to the classical requirement considered in Lai and Robbins (1985) and Burnetas and Katehakis (1996). For related information-theoretic change-of-measure techniques in best-arm identification, see also Kaufmann, Cappé, and Garivier (2016).

4.1 Static Reduction

We first consider the degenerate no-arrival case. Here $h(T) = T$, so Definition 1 reduces to the usual notion of uniform goodness. The purpose of this subsection is mainly calibrational: when no new arms enter, MAB-AA reduces to the classical stochastic bandit problem, so the arriving-arm framework should recover the familiar logarithmic identification barrier. Accordingly, we restate a standard lower-bound implication from the classical bandit literature (Lai and Robbins, 1985; Burnetas and Katehakis, 1996) in a form adapted to our notation. This result is included for comparison with our arriving-arm analysis, rather than as a new contribution. To avoid unnecessary generality, we use a local formulation under a

quadratic KL condition, which is sufficient for our comparison with the upper bounds in Section 3.

Consider a static bandit family with rewards supported on $[0,1]$. Let $\nu^{(0)}$ be a baseline instance in which arm 0 is the unique optimal arm with mean $r^\star \in (0,1)$. For $i = 1, \dots, K$, let arm i have reward law P_i and mean $r_i = r^\star - \Delta_i$ where $\Delta_i > 0$.

Theorem 3 (Static lower bound)**.** *Assume that for each $i = 1, \dots, K$ there exists a distribution $Q_i \in \mathcal{P}([0,1])$ such that $\eta_i := \mathbb{E}_{Q_i}[X] - r^\star > 0$ and $D(P_i \| Q_i) \le C_i \Delta_i^2$ for some finite constant $C_i > 0$. Let $\nu^{(i)}$ denote the instance obtained from $\nu^{(0)}$ by replacing only the reward law of arm i from P_i to Q_i. If a policy π is uniformly good on the local family $\mathcal{V} := \{\nu^{(0)}, \nu^{(1)}, \dots, \nu^{(K)}\}$, then*

$$\mathbb{E}_{\nu^{(0)}}[R_T(\pi; \nu^{(0)})] \ge c \sum_{i=1}^{K} \frac{\log T}{C_i \Delta_i}$$

for all sufficiently large T, where $c > 0$ is an absolute constant. In particular, if $\sup_{1 \le i \le K} C_i \le C_\star < \infty$, then

$$\mathbb{E}_{\nu^{(0)}}[R_T(\pi; \nu^{(0)})] \ge c\, C_\star^{-1} \sum_{i=1}^{K} \frac{\log T}{\Delta_i}$$

for all sufficiently large T.

Example 1 (Static specialization)**.** *In the absence of arrivals, UCB-AA specializes to an elimination-style stochastic bandit procedure and achieves $\tilde{O}\left(\sum_i \frac{\log(T\Delta_i^2)}{\Delta_i}\right)$ instance-dependent regret. Together with Theorem 3, this shows that the arriving-arm reduction does not sacrifice statistical efficiency in the static limit, up to logarithmic refinements.*

4.2 Sequential Distractors

We next consider a regime in which a globally optimal incumbent is present from the outset, while every newly arriving arm is suboptimal under the baseline instance. The difficulty is therefore repeated certification: each arrival creates a local alternative under

which that arm would become optimal, so the learner must accumulate logarithmic evidence before discarding it.

For each horizon T, let $\nu^{(0)}$ be a baseline instance in which arm 0 is the unique optimal arm, available from time 1, with mean $r^\star \in (0,1)$. Let $\mathcal{I}_T$ be a collection of arriving distractor arms. For each $i \in \mathcal{I}_T$, suppose arm i arrives at time $\tau_i \le T$, has reward law P_i, and mean $r_i = r^\star - \Delta_i$ where $\Delta_i > 0$. Write

$$H_i(T) := T - \tau_i + 1, \qquad H_{\min}(T) := \min_{i \in \mathcal{I}_T} H_i(T),$$

and assume $H_{\min}(T) \to \infty$. In this regime we evaluate uniform goodness at the effective scale $h(T) = H_{\min}(T)$.

Theorem 4 (Distractor lower bound under local alternatives)**.** *Suppose that for each $i \in \mathcal{I}_T$ there exists a law $Q_i \in \mathcal{P}([0,1])$ such that $\eta_i := \mathbb{E}_{Q_i}[X] - r^\star > 0$ and $d_i := D(P_i \| Q_i) < \infty$. Let $\nu^{(i)}$ denote the instance obtained from $\nu^{(0)}$ by replacing the reward law of arm i from P_i to Q_i, and set $\mathcal{V}_T := \{\nu^{(0)}\} \cup \{\nu^{(i)} : i \in \mathcal{I}_T\}$. Assume that, for all sufficiently large T,*

$$\inf_{i \in \mathcal{I}_T} \Delta_i \ge \underline{\Delta} > 0, \qquad \inf_{i \in \mathcal{I}_T} \eta_i \ge \underline{\eta} > 0, \qquad \sup_{i \in \mathcal{I}_T} d_i \le \overline{d} < \infty.$$

Then any sequence $\{\pi^{(T)}\}_{T \ge 1}$ uniformly good on $\mathcal{V}_T$ satisfies

$$\mathbb{E}_{\nu^{(0)}}[R_T(\pi^{(T)}; \nu^{(0)})] \ge c \sum_{i \in \mathcal{I}_T} \frac{\Delta_i}{d_i} \log H_i(T) \ge c\, \underline{\Delta}\, \overline{d}^{-1} \, |\mathcal{I}_T| \, \log H_{\min}(T)$$

for all sufficiently large T, where $c > 0$ is an absolute constant.

Corollary 3 (Two canonical distractor regimes)**.** *Under the assumptions of Theorem 4, suppose in addition that one of the following holds:*

(i) Equal-distance arrival: *each arm in $\mathcal{I}_T = \{1, \dots, K_T\}$ has the same arrival interval $H := \lfloor T/K_T \rfloor$ which means $\tau_i = (i-1)H + 1$,and H tends to infinity.*

(ii) Early arrival: *there exists a fixed $\alpha \in (0,1)$ such that $\tau_i \le \alpha T$ for every $i \in \mathcal{I}_T$.*

Then

$$\mathbb{E}_{\nu^{(0)}}[R_T(\pi^{(T)}; \nu^{(0)})] \ge c \sum_{i \in \mathcal{I}_T} \frac{\Delta_i}{d_i} \log H_{\min}(T)$$

for all sufficiently large T. In particular, if

$$\sup_{i\in\mathcal{I}_T} d_i \le \overline{d} < \infty, \qquad \inf_{i\in\mathcal{I}_T} \Delta_i \ge \underline{\Delta} > 0, \qquad |\mathcal{I}_T| \asymp K_T,$$

then

$$\mathbb{E}_{\nu^{(0)}}[R_T(\pi^{(T)};\nu^{(0)})] = \Omega\Big(K_T \log H_{\min}(T)\Big).$$

The lower bound reflects a repeated certification cost: although the arriving arms are inferior in the baseline instance, a uniformly good policy cannot ignore them, because each arrival induces a local alternative under which that arm is optimal.

Example 2 (Fixed-gap distractors)**.** *Consider an initial set of arms $A_0 = \{*, 0\}$, where arm $*$ is globally optimal with mean $\mu_* = \frac{3}{4}$ and arm 0 has mean $\mu_0 = \frac{1}{4}$. For each round $m = 1, \dots, M$, a single new arm $i = m$ arrives with $\mu_i = \frac{1}{2} - 2^{-i-2}$. Let $K(m) = m + 2$ denote an upper bound on the number of arms available up to round m. All new arrivals are distractors: their means converge to $1/2$, while the optimal arm remains fixed at $3/4$, so the gaps stay bounded away from zero. Applying Theorem 1 with $N = 10K(M)n_M$ yields* $\mathbb{E}R_T(\pi_{\text{UCB-AA}}) = \mathbb{E}R_M(\pi_{\text{UCB-AA}}) = \widetilde{O}\Big(K(M)\log T\Big)$.

Combined with Theorem 4, this gives the comparison

$$\widetilde{O}\Big(K_T \log T\Big) \qquad \text{versus} \qquad \Omega\Big(K_T \log(T/K_T)\Big),$$

so in the canonical sparse-arrival regime $K_T = o(T)$ the rates agree up to logarithmic factors. A complementary failure-mode construction showing that the absence of pre-elimination can lead to linear regret is deferred to the Supplementary Material.

4.3 Progressive Near-tied Arrivals

We finally consider a regime in which the benchmark improves through a sequence of near-tied arrivals. Unlike the distractor setting, the main difficulty is no longer certifying clearly inferior arms, but repeatedly deciding whether a new challenger lies slightly below or

slightly above the current reference level. Because this mechanism is intrinsically local and horizon dependent, the natural lower bound here is a finite-horizon minimax statement.

Fix a horizon T and integers $K_T \geq 1$, $H_1, \ldots, H_{K_T} \in \mathbb{N}$ such that $\sum_{m=1}^{K_T} H_m = T$. Define checkpoints and blocks by

$$s_0 := 0, \qquad s_m := \sum_{\ell=1}^{m} H_\ell, \qquad \mathcal{B}_m := \{s_{m-1}+1, \ldots, s_m\}, \qquad m = 1, \ldots, K_T.$$

Let $\gamma \in (0, 1/2]$, and choose $\theta_1, \ldots, \theta_{K_T} \in (0,1)$, $\varepsilon_1, \ldots, \varepsilon_{K_T} > 0$ such that

$$\theta_m - \varepsilon_m,\ \theta_m,\ \theta_m + \varepsilon_m \in [\gamma, 1-\gamma], \qquad m = 1, \ldots, K_T, \tag{3}$$

$$\theta_{m+1} \geq \theta_m + \varepsilon_m, \qquad m = 1, \ldots, K_T - 1. \tag{4}$$

For each sign vector $\sigma = (\sigma_1, \ldots, \sigma_{K_T}) \in \{-1, +1\}^{K_T}$, define an arriving-arm instance ν_σ in which, at time $s_{m-1}+1$, two Bernoulli arms c_m, d_m arrive and remain available thereafter, with means $\mu(c_m) = \theta_m$ and $\mu(d_m) = \theta_m + \sigma_m \varepsilon_m$. Let $\mathcal{V}_T^{\mathrm{prog}} := \{\nu_\sigma : \sigma \in \{-1,+1\}^{K_T}\}$.

Theorem 5 (Deterministic-checkpoint minimax lower bound)**.** *Under (3)–(4), every non-anticipating policy π satisfies*

$$\sup_{\nu \in \mathcal{V}_T^{\mathrm{prog}}} \mathbb{E}_\nu[R_T(\pi;\nu)] \geq \frac{1}{2} \sum_{m=1}^{K_T} H_m \varepsilon_m \left(1 - \sqrt{\frac{2H_m\varepsilon_m^2}{\gamma(1-\gamma)}}\right)_+. \tag{5}$$

In particular, if there exists a constant $\kappa \in (0, 1/2)$ such that

$$H_m \varepsilon_m^2 \leq \kappa\, \gamma(1-\gamma), \qquad m = 1, \ldots, K_T,$$

$$\textit{then} \quad \sup_{\nu \in \mathcal{V}_T^{\mathrm{prog}}} \mathbb{E}_\nu[R_T(\pi;\nu)] \geq c_\kappa \sum_{m=1}^{K_T} H_m \varepsilon_m, \qquad c_\kappa := \frac{1}{2}(1 - \sqrt{2\kappa}) > 0. \tag{6}$$

The same lower bounds remain valid after taking $\inf_\pi$ *on the left-hand side.*

Each block creates a fresh one-sided local test: the learner must determine whether the new challenger lies slightly below or slightly above the current benchmark. When $H_m\varepsilon_m^2$ is of constant order, the resulting regret contribution is of order $H_m\varepsilon_m$, and these local-testing costs add across blocks.

Corollary 4 ($\sqrt{K_T T}$ lower bound under near-homogeneous local tests)**.** *Under the assumptions of Theorem 5, assume in addition that there exist constants $c_H, C_H, c_\varepsilon, C_\varepsilon > 0$ and $\kappa \in (0, 1/2)$ such that, for every $m = 1, \dots, K_T$,*

$$c_H \frac{T}{K_T} \le H_m \le C_H \frac{T}{K_T}, \qquad c_\varepsilon \sqrt{\frac{K_T}{T}} \le \varepsilon_m \le C_\varepsilon \sqrt{\frac{K_T}{T}},$$

and $C_H C_\varepsilon^2 \le \kappa\, \gamma(1-\gamma)$. Then

$$\inf_{\pi} \sup_{\nu \in \mathcal{V}_T^{\mathrm{prog}}} \mathbb{E}_\nu[R_T(\pi; \nu)] \ge c \sqrt{K_T T}$$

for all sufficiently large T, where $c > 0$ depends only on c_H, c_ε, and κ.

Corollary 4 identifies the canonical square-root minimax scale in a near-homogeneous local-testing regime. Our general upper bound in Section 3 matches this dependence up to logarithmic factors. The example below is not a literal specialization of the deterministic-checkpoint lower bound, because UCB-AA operates with endogenous random checkpoints, but it exhibits the same qualitative mechanism.

Example 3 (Monotone near-tied ladder)**.** *Fix $M \ge 5$ and define $\xi_m := \frac{1}{2} - 2^{-m-2}$ for $m = 0, 1, \dots, M$. At round 0, there is a single arm a_0, so that $A_0 = \{a_0\}$ and $\mu(a_0) = \xi_0$. For each round $m = 1, \dots, M$, two new Bernoulli arms arrive, namely $A_m = \{c_m, d_m\}$, with means $\mu(c_m) = \xi_{m-1}$ and $\mu(d_m) = \xi_m$. Hence the unique optimal arm in round m is d_m, with $r^*(m) = \xi_m$, and the cumulative number of available arms is $K(m) = |A(m)| = 2m+1$. Choosing $N = 10K(M)n_M$ yields $\mathbb{E}R_T(\pi_{\text{UCB-AA}}) = \mathbb{E}R_M(\pi_{\text{UCB-AA}}) = O\Big(\sqrt{T \log K(M)}\Big)$.*

Example 3 should be interpreted as an endogenous-checkpoint illustration of the same local-testing mechanism. The deterministic-checkpoint lower bound gives the canonical minimax benchmark, while the UCB-AA analysis yields the same square-root-type behavior up to an additional logarithmic factor along its own random checkpoint schedule.

Taken together, these lower-bound examples show that MAB-AA contains several distinct sources of statistical difficulty. They are not intended as term-by-term converses to the upper bounds in Section 3; rather, they clarify why the analysis must separately control

the costs of identifying good incumbent arms, certifying arriving competitors, and tracking benchmark changes.

5 Simulation

We evaluate UCB-AA on two synthetic arriving-arm environments. The first contains abrupt improvements in the best available arm, while the second contains frequent near-optimal arrivals. Both settings use horizon $T = 60000$. We report three metrics:

- Cumulative dynamic regret R_t: measures loss relative to the best currently available arm, $R_t = \sum_{s=1}^{t}(r^*(s) - r_{I_s})$, where $r^*(s) = \max_{i \in \mathcal{A}_s} r_i$ and I_s is the arm pulled at time s.
- Wasted pulls W_T: measures exploration spent on clearly suboptimal choices, $W_T = \sum_{s=1}^{T} \mathbb{I}\{\Delta_{I_s}(s) \geq 0.1\}$.
- Active-set size: measures the pruning effect of elimination, defined as the number of arms retained after elimination steps.

We consider two distinct arrival scenarios designed to stress-test different challenges inherent to the MAB-AA setting; for specific distribution parameter settings, refer to the supplementary materials:

Scenario 1 (Progressive Step-up Arrivals)

The environment starts with $K_0 = 10$ arms and background arrivals following a Poisson process with rate $\lambda = 1/200$, with means drawn from a bi-modal mixture. To create benchmark shifts, high-performing step arms are introduced at $t/T \in \{0.3, 0.6, 0.9\}$, with $\mu_k = \min(1, r_0 + k\Delta_{\text{step}})$, $r_0 = 0.6$, and $\Delta_{\text{step}} = 0.05$. This setting tests whether an algorithm can quickly detect and switch to newly arrived superior arms.

Scenario 2 (Near-tied Top Set)

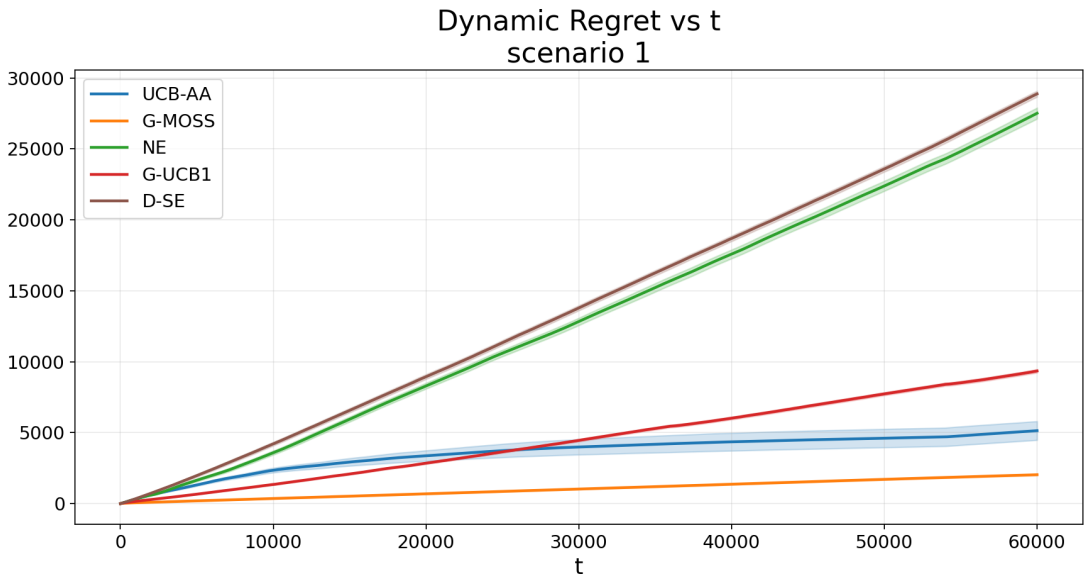
The environment starts with $K_0 = 20$ arms, including $m_0 = 5$ near-optimal arms with means from Uniform$(0.7, 0.8)$. New arms arrive according to a Bernoulli process with probability $\lambda = 1/200$, and 30% of them are near-optimal. This setting represents mature decision problems with many near-tied top arms, where small performance differences must be detected without excessive exploration.

To ensure a fair comparison, we adapt classical algorithms to handle the continuous arrival of new arms. Their implementations are described below; additional details are provided in the supplementary materials.

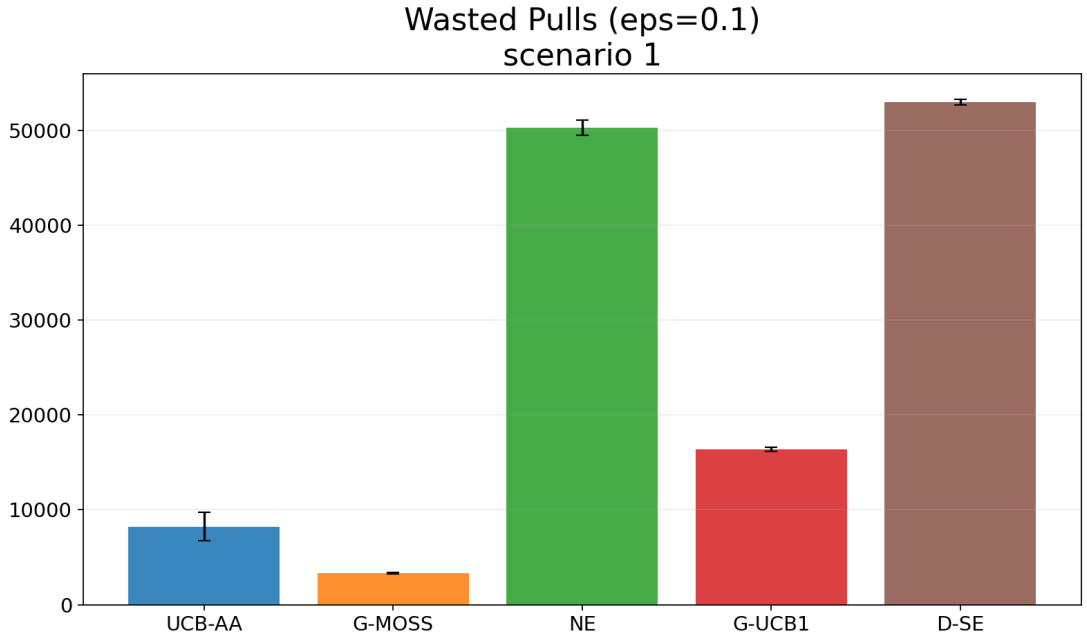
1. UCB-AA (Proposed): the proposed method, which buffers newly arrived arms and screens them through pre-elimination before full competition. We set $N = e/\tilde{\Delta}_M^2$ in the simulations.
2. Growing UCB1 (G-UCB1): UCB1 (Auer, Cesa-Bianchi, and Fischer, 2002) with immediate admission of new arms, using index $\hat{\mu}_{i,n} + \sqrt{2 \ln t / n}$.
3. Growing MOSS (G-MOSS): MOSS (Audibert and Bubeck, 2009) with immediate admission of new arms, using index $\hat{\mu}_i + \sqrt{\max\{0, \log(T/(K_t n_i))\}/n_i}$, where $K_t = |\mathcal{A}_t|$.
4. Dynamic Successive Elimination (D-SE): an SE-style baseline (Perchet and Rigollet, 2013) that admits new arms at each round, samples them to match incumbent pull counts, and then applies the SE elimination rule.
5. Naive Elimination (NE): UCB-AA with the same formal elimination rule but without the pre-elimination stage.

For each scenario, we ran 100 independent simulations. The figures below report the average cumulative dynamic regret and wasted pulls, with 95% confidence intervals.

The simulation results show that UCB-AA substantially reduces the number of wasted pulls (W_T) relative to Growing UCB1 and the elimination-based baselines. The main reason

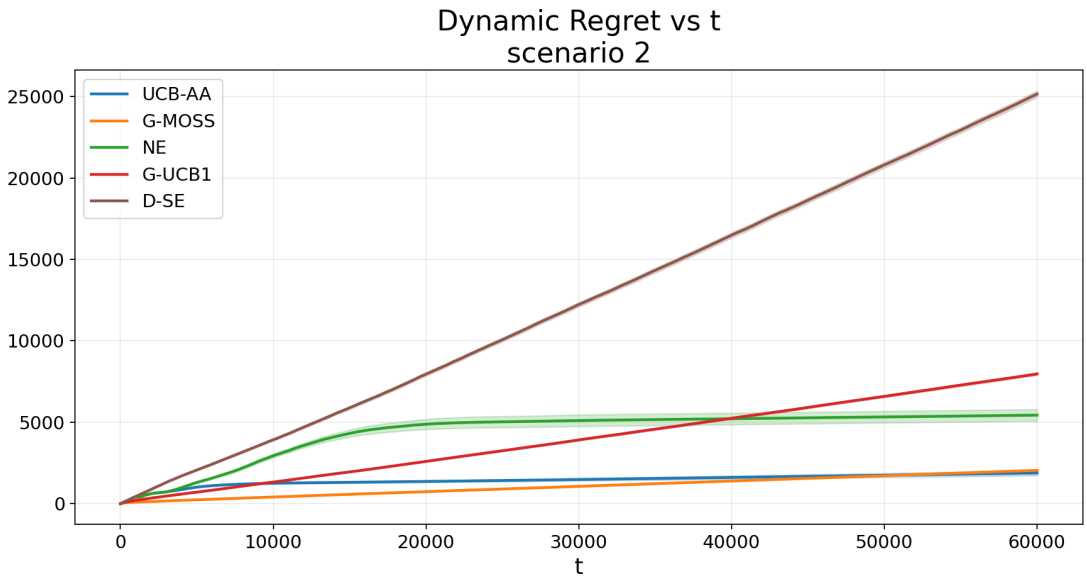


(a) Cumulative dynamic regret (Scenario 1)

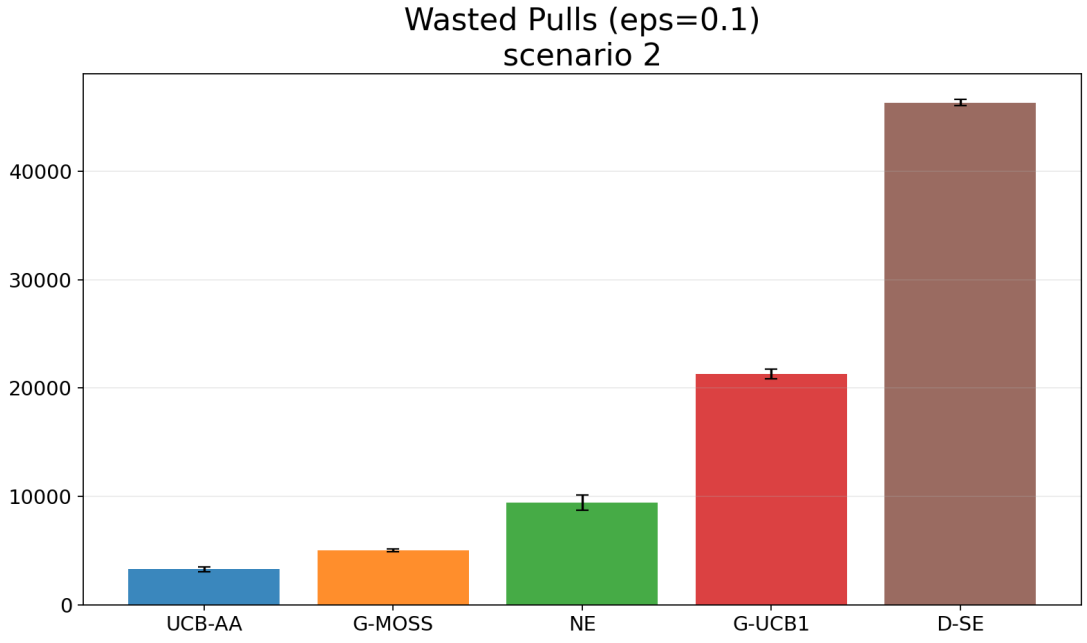


(b) Wasted pulls (Scenario 1)

Figure 2: Experimental results for Scenario 1 (Progressive step-up arrivals). Panel (a) shows cumulative dynamic regret, and panel (b) compares wasted pulls (W_T) and the final number of active arms. Results are averaged over 100 independent runs.

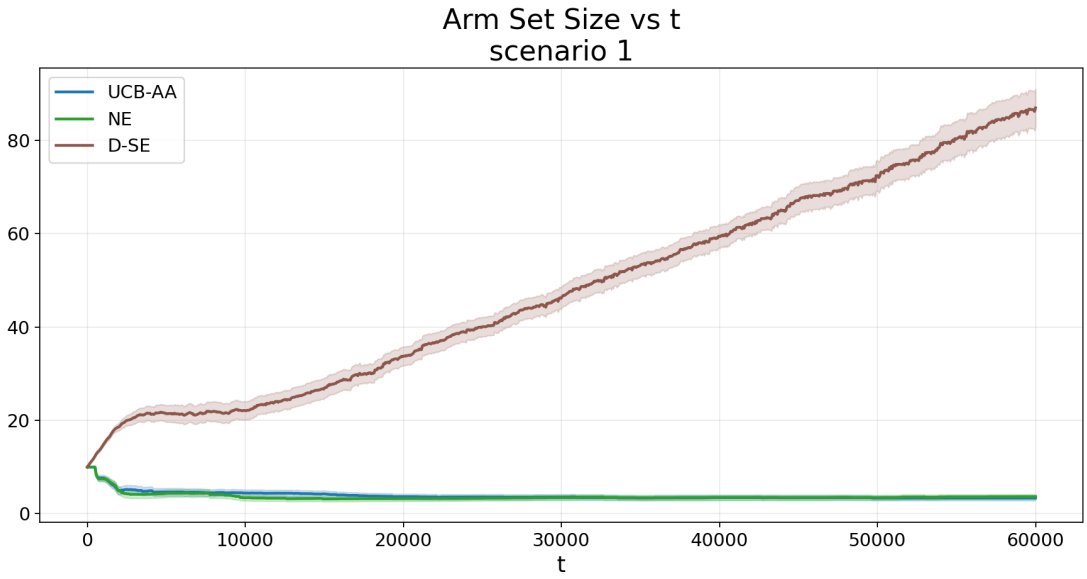


(a) Cumulative dynamic regret (Scenario 2)

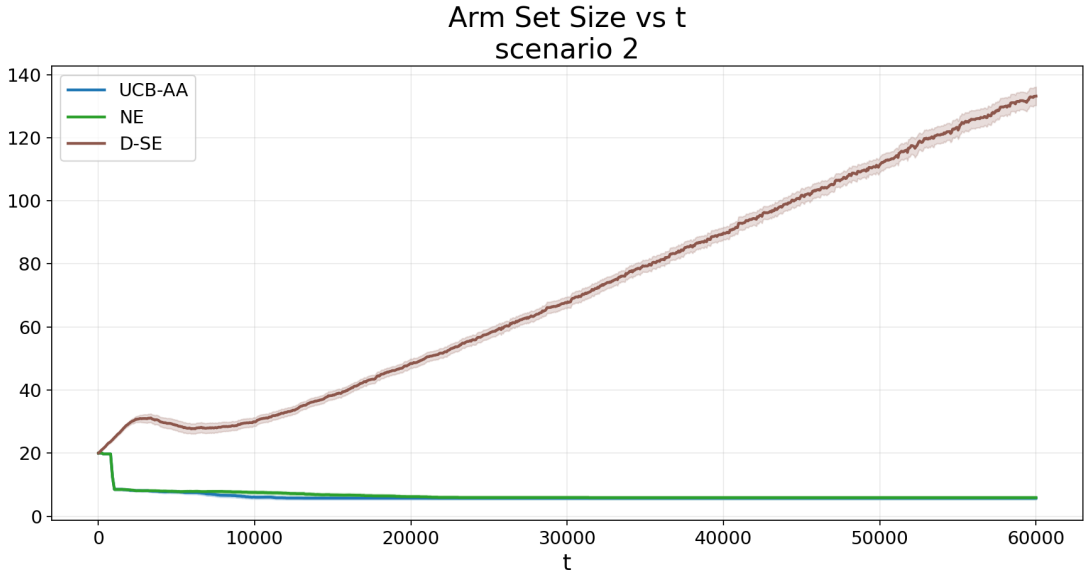


(b) Wasted pulls (Scenario 2)

Figure 3: Experimental results for Scenario 2 (Near-tied top set). UCB-AA's explicit elimination mechanism maintains a compact decision space while achieving regret performance comparable to Growing MOSS.



(a) Active arm size (Scenario 1)



(b) Active arm size (Scenario 2)

Figure 4: Active arm size for two scenarios. UCB-AA's explicit elimination mechanism maintains a compact decision space while achieving regret performance comparable to Growing MOSS.

is that newly arriving arms are not allowed to compete immediately on equal footing with well-sampled incumbents. Instead, UCB-AA uses a buffered pre-elimination step to screen out clearly inferior late arrivals before full competition, thereby protecting the sampling budget of the currently most competitive arms. By contrast, Growing UCB1 continues to allocate exploration to each new arrival through its uncertainty bonus, while Naive Elimination and Dynamic SE treat new and old arms too symmetrically at entry, leading to larger regret.

A second advantage of UCB-AA is that it maintains a compact active set whose members have survived repeated statistical comparisons over time. Methods such as Growing UCB1 and Growing MOSS do not explicitly prune the arm set once arms arrive, whereas UCB-AA actively shrinks its working pool $|\mathcal{C}_t|$ through pre-elimination and formal elimination. As a result, UCB-AA not only controls regret well, but also returns a smaller set of arms that have survived repeated statistical comparisons and are therefore more likely to be genuinely competitive.

At the same time, UCB-AA does not uniformly dominate Growing MOSS in cumulative dynamic regret (R_T); instead, their regret levels are often comparable, and Growing MOSS can be slightly better in settings where many arms are near-tied. This reflects the cost of explicit hard elimination. In UCB-AA, an arm is removed only when its upper confidence bound falls below the largest lower confidence bound among the surviving arms, $UCB_j < \max_i LCB_i$. Because this rule requires strong evidence, UCB-AA may spend additional samples to certify that a borderline arm is truly inferior before discarding it. By contrast, Growing MOSS performs only a soft form of elimination: weak arms receive fewer pulls when their indices fall behind, but they remain in the available set and can be sampled again if their uncertainty remains large.

This trade-off also indicates when hard elimination is useful. If the only objective is cumulative regret over a moderate horizon and keeping all arrived arms active is essentially

free, then soft index-based methods such as Growing MOSS can be highly competitive. However, in many arriving-arm applications, the active set itself is costly to maintain. Examples include recommendation systems with continuously added items, adaptive A/B testing with many newly launched designs, biomedical screening or dose-finding studies with costly or risky candidates, and sequential model or policy selection where each surviving candidate requires repeated evaluation. In such settings, reducing the active set is not merely a computational convenience; it is part of the statistical objective. UCB-AA therefore trades a modest amount of additional exploration for certified pruning of clearly inferior late arrivals, thereby controlling both wasted pulls and the operational burden created by an expanding action space.

6 Discussion

We studied stochastic multi-armed bandits with arriving arms, where the available action set expands over time and the performance is measured against the best currently available arm. The distinct features of arrival information discrepancy (AID) and drifting benchmark (DB) depart from the classical fixed-oracle benchmark and creates two coupled statistical challenges: repeated certification of newly arriving competitors and a moving target to catch up.

Our main contribution shows that these challenges can be handled in a structured way. The round-based reduction separates the principal certification burden from the additional delay induced by benchmark movement, while UCB-AA uses a pre-elimination stage to screen late arrivals before they enter full competition. Under suitable control of gap evolution, this yields sublinear regret together with a compact active set. The lower-bound regimes further clarify that the hardness of MAB-AA has multiple sources: the static regime recovers the classical logarithmic identification barrier, distractor arrivals induce cumulative

certification costs, and near-tied improvements create a local-testing difficulty under a drifting benchmark.

Several extensions are natural. One direction is to incorporate contextual information or arm-level covariates when available, which may improve reward via individualization as well as the screening efficiency for newly arriving arms. Another direction is to study settings with post-arrival reward drift, while retaining the central focus on how an expanding action set affects exploration and regret. Finally, the arriving-arm framework can be extended to settings involving delayed feedback, operational resource constraints, and arrival processes calibrated from real application data.

Data availability statement

No original empirical or human-subject data were collected in this study. The numerical experiments are based on synthetic simulations whose design, parameter settings, and algorithmic details are reported in the article and Supplementary Material. The processed data used to produce Figure 1 were derived from publicly available FDA drug-approval resources.

References

Audibert, J.-Y. and S. Bubeck (2009). "Minimax Policies for Adversarial and Stochastic Bandits". *Proceedings of the 22nd Annual Conference on Learning Theory (COLT)*. Proceedings of Machine Learning Research, 217–226.

Auer, P., N. Cesa-Bianchi, and P. Fischer (2002). "Finite-time Analysis of the Multiarmed Bandit Problem". *Machine Learning* 47, 235–256.

Auer, P., P. Gajane, and R. Ortner (2019). "Adaptively Tracking the Best Bandit Arm with an Unknown Number of Distribution Changes". *Proceedings of the 32nd Conference on Learning Theory*. Vol. 99. Proceedings of Machine Learning Research, 138–158.

Auer, P. and R. Ortner (2010). "UCB revisited: Improved regret bounds for the stochastic multi-armed bandit problem". *Periodica Mathematica Hungarica* 61, 55–65.

Aziz, M., E. Kaufmann, and M.-K. Riviere (2021). "On multi-armed bandit designs for dose-finding clinical trials". *Journal of Machine Learning Research* 22, 686–723.

Berry, D. A., R. W. Chen, A. Zame, D. C. Heath, and L. A. Shepp (1997). "Bandit problems with infinitely many arms". *The Annals of Statistics* 25, 2103–2116.

Berry, D. A. and B. Fristedt (1985). *Bandit Problems: Sequential Allocation of Experiments*. London: Chapman and Hall.

Besbes, O., Y. Gur, and A. Zeevi (2014). "Stochastic multi-armed-bandit problem with non-stationary rewards". *Proceedings of the 28th International Conference on Neural Information Processing Systems*. Vol. 1. NIPS'14, 199–207.

Burnetas, A. N. and M. N. Katehakis (1996). "Optimal Adaptive Policies for Sequential Allocation Problems". *Advances in Applied Mathematics* 17, 122–142.

Cai, C., T. T. Cai, and H. Li (2024). "Transfer learning for contextual multi-armed bandits". *The Annals of Statistics* 52, 207–232.

Chakrabarti, D., R. Kumar, F. Radlinski, and E. Upfal (2008). "Mortal multi-armed bandits". *Proceedings of the 22nd International Conference on Neural Information Processing Systems*. NIPS'08, 273–280.

Cover, T. M. and J. A. Thomas (2006). *Elements of Information Theory*. 2nd ed. Wiley-Interscience.

Even-Dar, E., S. Mannor, and Y. Mansour (2006). "Action Elimination and Stopping Conditions for the Multi-Armed Bandit and Reinforcement Learning Problems". *Journal of Machine Learning Research* 7, 1079–1105.

Ghalme, G., S. Dhamal, S. Jain, S. Gujar, and Y. Narahari (2021). "Ballooning multi-armed bandits". *Artificial Intelligence* 296, 103485.

Gittins, J. C. (1979). "Bandit Processes and Dynamic Allocation Indices". *Journal of the Royal Statistical Society: Series B (Methodological)* 41, 148–164.

Kaufmann, E., O. Cappé, and A. Garivier (2016). "On the complexity of best-arm identification in multi-armed bandit models". *Journal of Machine Learning Research* 17, 1–42.

Kleinberg, R., A. Niculescu-Mizil, and Y. Sharma (2010). "Regret bounds for sleeping experts and bandits". *Machine Learning* 80, 245–272.

Lai, T. and H. Robbins (1985). "Asymptotically efficient adaptive allocation rules". *Advances in Applied Mathematics* 6, 4–22.

Lai, T. L. (1987). "Adaptive Treatment Allocation and the Multi-Armed Bandit Problem". *The Annals of Statistics* 15, 1091–1114.

Lattimore, T. and C. Szepesvári (2020). *Bandit Algorithms*. Cambridge University Press.

Perchet, V. and P. Rigollet (2013). "The multi-armed bandit problem with covariates". *The Annals of Statistics* 41, 693–721.

Pinsker, M. S. (1964). *Information and Information Stability of Random Variables and Processes*. San Francisco: Holden-Day.

Qi, H., F. Guo, L. Zhu, and Q. Zhang (2025). "Graph Feedback Bandits on Similar Arms: With and Without Graph Structures". *arXiv preprint*. arXiv: 2501.14314 [cs.LG].

Qian, W., C.-K. Ing, and J. Liu (2024). "Adaptive Algorithm for Multi-Armed Bandit Problem with High-Dimensional Covariates". *Journal of the American Statistical Association* 119, 970–982.

Qian, W. and Y. Yang (2016). "Randomized allocation with arm elimination in a bandit problem with covariates". *Electronic Journal of Statistics* 10, 242–270.

Rigollet, P. and A. Zeevi (2010). "Nonparametric bandits with covariates". *Proceedings of the 23rd Annual Conference on Learning Theory (COLT)*. Proceedings of Machine Learning Research, 54–66.

Robbins, H. (1952). "Some aspects of the sequential design of experiments". *Bulletin of the American Mathematical Society* 58, 527–535.

Tsybakov, A. B. (2009). *Introduction to Nonparametric Estimation*. Springer Series in Statistics. Springer.

Villar, S. S., J. Wason, and J. Bowden (2015). "Response-Adaptive Randomization for Multi-arm Clinical Trials Using the Forward Looking Gittins Index Rule". *Biometrics* 71, 969–978.

Wei, L. J. and S. Durham (1978). "The Randomized Play-the-Winner Rule in Medical Trials". *Journal of the American Statistical Association* 73, 840–843.

Williamson, S. F. and S. S. Villar (2020). "A Response-Adaptive Randomization Procedure for Multi-Armed Clinical Trials with Normally Distributed Outcomes". *Biometrics* 76, 197–209.

Woodroofe, M. (1979). "A One-Armed Bandit Problem with a Concomitant Variable". *Journal of the American Statistical Association* 74, 799–806.

Yang, Y. and D. Zhu (2002). "Randomized Allocation with nonparametric estimation for a multi-armed bandit problem with covariates". *The Annals of Statistics* 30, 100–121.

Zelen, M. (1969). "Play the Winner Rule and the Controlled Clinical Trial". *Journal of the American Statistical Association* 64, 131–146.

Supplementary Material for "Multi-Armed Bandits with Arriving Arms: Sequential Screening, Dynamic Regret, and Sublinear Guarantees"

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This supplementary material is organized as follows.

- **Section A** contains the proofs of the main theoretical results:
  - **Section A.1**: Present some auxiliary lemmas and necessary proofs;
  - **Section A.2**: Proof of Theorem 1;
  - **Section A.3**: Proof of Corollary 1;
  - **Section A.4**: Proof of Proposition 1;
  - **Section A.5**: Analysis of Online UCB-AA and Proof of Corollary 2;
  - **Section A.6**: Proof of Theorem 3;
  - **Section A.7**: Proof of Theorem 4;
  - **Section A.8**: Proof of Theorem 5.
- **Section B** provides the analysis of four concrete cases:
  - **Section B.1**: Analysis of Example 1;
  - **Section B.2**: Analysis of Example 2;
  - **Section B.3**: Analysis of Example 3;
  - **Section B.4**: Analysis of UCB-AA in the Case that Gaps Vary Dramatically.
- **Section C** presents the details of the numerical experiments, including the experimental design, environment parameters, implementation details, and pseudocodes for algorithms.

A Proofs

A.1 Auxiliary Lemmas

We first collect several standard information-theoretic facts used in the lower-bound arguments. Lemma A.1 follows from the data-processing inequality for relative entropy applied to the binary statistic $\mathbf{1}_A$ (see, e.g., Cover and Thomas, 2006; Lattimore and Szepesvári, 2020). Lemmas A.2 and A.3 are elementary bounds for the Bernoulli KL divergence; in particular, Lemma A.3 follows from the standard comparison $D(P\|Q) \le \chi^2(P,Q)$ specialized to Bernoulli laws (see, e.g., Tsybakov, 2009; Lattimore and Szepesvári, 2020). Lemma A.4 is Pinsker's inequality (Pinsker, 1964; Tsybakov, 2009). Lemma A.5 is the standard change-of-measure, or divergence-decomposition, identity for adaptively sampled bandit observations (see, e.g., Lemma 1 of Kaufmann, Cappé, and Garivier, 2016); see also Lattimore and Szepesvári (2020, Chapter 15).

Lemma A.1 (Binary reduction for KL)**.** *Let P and Q be two probability measures on the same measurable space, and let A be any measurable event. Then*

$$\mathrm{KL}(P,Q) \ge \mathrm{kl}(P(A),Q(A)),$$

where

$$\mathrm{kl}(p,q) := p\log\frac{p}{q} + (1-p)\log\frac{1-p}{1-q}$$

denotes the Bernoulli Kullback–Leibler divergence.

Proof. Let $\phi(x) = \mathbf{1}_A(x)$. Under P, the random variable $\phi(X)$ has Bernoulli distribution with parameter $P(A)$, while under Q it has Bernoulli distribution with parameter $Q(A)$. Denote these pushforward measures by $P\circ\phi^{-1}$ and $Q\circ\phi^{-1}$. Then

$$P\circ\phi^{-1} = \mathrm{Bernoulli}(P(A)), \qquad Q\circ\phi^{-1} = \mathrm{Bernoulli}(Q(A)).$$

By the data-processing inequality for relative entropy,

$$\mathrm{KL}(P, Q) \geq \mathrm{KL}\Big(P \circ \phi^{-1},\, Q \circ \phi^{-1}\Big).$$

The right-hand side is exactly the KL divergence between two Bernoulli distributions, namely

$$\mathrm{KL}\Big(P \circ \phi^{-1},\, Q \circ \phi^{-1}\Big) = P(A) \log \frac{P(A)}{Q(A)} + (1 - P(A)) \log \frac{1 - P(A)}{1 - Q(A)} = \mathrm{kl}(P(A), Q(A)).$$

This proves the claim. □

Lemma A.2 (A crude lower bound on Bernoulli KL)**.** *For any $p \in [0, 1]$ and $q \in (0, 1)$,*

$$\mathrm{kl}(p, q) \geq p \log \frac{1}{q} - \log 2.$$

Proof. By definition,

$$\mathrm{kl}(p, q) = p \log \frac{1}{q} + (1 - p) \log \frac{1}{1 - q} + p \log p + (1 - p) \log(1 - p).$$

The second term on the right-hand side is nonnegative, while

$$p \log p + (1 - p) \log(1 - p) \geq - \log 2.$$

Combining the two bounds gives the claim. □

Lemma A.3 (Bernoulli KL upper bound)**.** *For any $p \in [0, 1]$ and $q \in (0, 1)$,*

$$\mathrm{kl}(p, q) \leq \frac{(p - q)^2}{q(1 - q)}.$$

Consequently, if $p \in [0, 1]$ and $q \in [\gamma, 1 - \gamma]$ for some $\gamma \in (0, 1/2]$, then

$$\mathrm{kl}(p, q) \leq \frac{1}{\gamma(1 - \gamma)} (p - q)^2.$$

Proof. For $p \in (0, 1)$, the first inequality is standard. The boundary cases $p = 0$ and $p = 1$ follow by continuity. The second inequality is immediate from $q(1 - q) \geq \gamma(1 - \gamma)$. □

Lemma A.4 (Pinsker's inequality)**.** *For any probability measures P and Q,*

$$\mathrm{TV}(P, Q) \leq \sqrt{\frac{1}{2}\mathrm{KL}(P, Q)}.$$

Lemma A.5 (KL chain rule for adaptive sampling)**.** *Let ν and ν' be two bandit instances, and let $\mathbb{P}_\nu$ and $\mathbb{P}_{\nu'}$ denote the induced laws of the full history*

$$H_T := (I_1, X_1, \ldots, I_T, X_T).$$

Suppose that ν and ν' differ only in the reward law of arm j, whose distributions are P and Q on $[0, 1]$, respectively. Then

$$\mathrm{KL}(\mathbb{P}_\nu, \mathbb{P}_{\nu'}) = \mathbb{E}_\nu[N_j(T)]\, D(P\|Q),$$

where

$$N_j(T) := \sum_{t=1}^{T} \mathbf{1}\{I_t = j\}.$$

In particular, if $P = \mathrm{Bernoulli}(p)$ and $Q = \mathrm{Bernoulli}(q)$, then

$$\mathrm{KL}(\mathbb{P}_\nu, \mathbb{P}_{\nu'}) = \mathbb{E}_\nu[N_j(T)]\, \mathrm{kl}(p, q).$$

Proof. Let $K_t(\cdot \mid h_{t-1})$ denote the (possibly randomized) action kernel used by the policy at time t, given past history $H_{t-1} = h_{t-1}$. Since the policy is the same under ν and ν', the kernel K_t is identical under the two instances. By the chain rule for KL divergence,

$$\mathrm{KL}(\mathbb{P}_\nu, \mathbb{P}_{\nu'}) = \sum_{t=1}^{T} \mathbb{E}_\nu[\mathrm{KL}(\mathcal{L}((I_t, X_t) \mid H_{t-1})_\nu,\, \mathcal{L}((I_t, X_t) \mid H_{t-1})_{\nu'})]\,.$$

Conditional on $H_{t-1} = h_{t-1}$, the joint law of (I_t, X_t) under ν is given by

$$K_t(di \mid h_{t-1})\, M_i(dx),$$

where M_i is the reward law of arm i under ν; under ν' the same expression holds with M_i' in place of M_i. Since the two instances differ only on arm j, we have $M_i = M_i'$ for $i \neq j$, while $M_j = P$ and $M_j' = Q$. Therefore the conditional KL divergence equals

$$\mathbf{1}\{I_t = j\}\, D(P\|Q),$$

and hence

$$\mathrm{KL}(\mathbb{P}_\nu, \mathbb{P}_{\nu'}) = \sum_{t=1}^{T} \mathbb{E}_\nu[\mathbf{1}\{I_t = j\}\, D(P\|Q)] = \mathbb{E}_\nu[N_j(T)]\, D(P\|Q).$$

The Bernoulli specialization is immediate. □

A.2 Proof of Theorem 1

Proof of Theorem 1. We analyze the regret $\mathbb{E}(R^{(i)}(\pi))$ which is generated by each suboptimal arm $i \in A'(\lambda)$ by dividing the entire space into three cases. Define the event

$\tilde{E}_i = \{i \in A'(\lambda)$ has not been eliminated in or before $\overline{m}_i$-th elimination phase of round m_i $\}$,

$\tilde{F}_i = \{*(m_i) \in B_{m_i}^{\overline{m}_i}\}$.

Here, if $\overline{m}_i \geq m_i - 1$, then $\overline{m}_i$-th elimination phase is the formal elimination phase of round m_i; if $\overline{m}_i < m_i - 1$, then $\overline{m}_i$-th elimination phase is in the pre-elimination phase. $*(m_i) \in B_{m_i}^{\overline{m}_i}$ means the optimal arm at round m_i remains at the beginning of $\overline{m}_i$-th elimination phase.

Then, we divide the regret into three parts,

$$\mathbb{E}(R^{(i)}(\pi)) = \underbrace{\mathbb{E}\left(R^{(i)}(\pi) \cdot I_{\tilde{E}_i \cap \tilde{F}_i}\right)}_{\text{case (a)}} + \underbrace{\mathbb{E}\left(R^{(i)}(\pi) \cdot I_{\tilde{E}_i^c}\right)}_{\text{case (b1)}} + \underbrace{\mathbb{E}\left(R^{(i)}(\pi) \cdot I_{\tilde{E}_i \cap \tilde{F}_i^c}\right)}_{\text{case (b2)}}.$$

Now, we calculate the upper bounds for the regret of these three parts, respectively.

Case (a): *If suboptimal arm $i \in A'(\lambda)$ was not eliminated in or before the $\overline{m}_i$-th elimination phase of round m_i, and $*(m_i) \in B_{m_i}^{\overline{m}_i}$.*

If $i \in G_1(\lambda) \cup G_2(\lambda)$, then $\overline{m}_i = \tilde{m}_i$. We have $\sqrt{\frac{\log(N\tilde{\Delta}_{\tilde{m}_i}^2)}{2n_{\tilde{m}_i}}} \leq \frac{1}{2}\tilde{\Delta}_{\tilde{m}_i} < \frac{1}{4}\Delta_i(m_i)$ by the definition of $\tilde{m}_i$, if

$$\hat{r}_i(n_p) \leq r_i + \sqrt{\frac{\log(N\tilde{\Delta}_p^2)}{2n_p}} \tag{7}$$

and

$$\hat{r}_{*(m)}(n_p) \geq r_{*(m)} - \sqrt{\frac{\log(N\tilde{\Delta}_p^2)}{2n_p}} \tag{8}$$

hold for $m = m_i,\ p = \tilde{m}_i$, then we have:

$$\hat{r}_i(n_{\tilde{m}_i}) + \sqrt{\frac{\log(N\tilde{\Delta}^2_{\tilde{m}_i})}{2n_{\tilde{m}_i}}} \leq r_i + 2\sqrt{\frac{\log(N\tilde{\Delta}^2_{\tilde{m}_i})}{2n_{\tilde{m}_i}}} < r_i + \Delta_i(m_i) - 2\sqrt{\frac{\log(N\tilde{\Delta}^2_{\tilde{m}_i})}{2n_{\tilde{m}_i}}}$$
$$= r_{*(m_i)} - 2\sqrt{\frac{\log(N\tilde{\Delta}^2_{\tilde{m}_i})}{2n_{\tilde{m}_i}}} \leq \hat{r}_{*(m_i)}(n_{\tilde{m}_i}) - \sqrt{\frac{\log(N\tilde{\Delta}^2_{\tilde{m}_i})}{2n_{\tilde{m}_i}}}.$$

Further, by the assumption that arm i and $*(m_i)$ coexist and the elimination rule, i will be eliminated, and i is pulled no more than $n_{\tilde{m}_i}$ times. Besides, using the Chernoff-Hoeffding bound, we have for $p = 0, 1, \cdots$ and $m = 0, 1 \cdots$:

$$\mathbb{P}\left(\hat{r}_i(n_p) > r_i + \sqrt{\frac{\log(N\tilde{\Delta}^2_p)}{2n_p}}\right) \leq \frac{1}{N\tilde{\Delta}^2_p} \tag{9}$$

$$\mathbb{P}\left(\hat{r}_{*(m)}(n_p) < r_{*(m)} - \sqrt{\frac{\log(N\tilde{\Delta}^2_p)}{2n_p}}\right) \leq \frac{1}{N\tilde{\Delta}^2_p} \tag{10}$$

Take $p = \tilde{m}_i,\ m = m_i$, so the upper bound of the probability that arm i is not eliminated in or before the $\overline{m}_i$-th elimination phase of round m_i is $\frac{2}{N\tilde{\Delta}^2_{\tilde{m}_i}}$.

If $i \in G_3(\lambda)$, then $\overline{m}_i = m_i - 1$ and $m_i > t_i$, which means that the optimal arm $*(m_i)$ arrives at the beginning of round m_i, i.e. $t_{*(m_i)} = m_i$, while arm i is already present before round m_i. Note that $\Delta_i(m_i) = r_{*(m_i)} - r_i$. According to the workflow of UCB-AA, the suboptimal arm i is therefore expected to be eliminated during the $\overline{m}_i$-th elimination phase of round m_i. By the definition of $\tilde{m}_i$, if

$$\hat{r}_i(n_{m_i-1}) \leq r_i - \sqrt{\frac{\log(N\tilde{\Delta}^2_{m_i-1})}{2n_{m_i-1}}} + \tilde{\Delta}_{\tilde{m}_i} \tag{11}$$

and

$$\hat{r}_{*(m_i)}(n_{m_i-1}) \geq r_{*(m_i)} + \sqrt{\frac{\log(N\tilde{\Delta}^2_{m_i-1})}{2n_{m_i-1}}} - \tilde{\Delta}_{\tilde{m}_i} \tag{12}$$

hold, then we have:

$$\hat{r}_i(n_{m_i-1}) + \sqrt{\frac{\log(N\tilde{\Delta}^2_{m_i-1})}{2n_{m_i-1}}} \leq r_i + \tilde{\Delta}_{\tilde{m}_i}$$

$$\left(\tilde{\Delta}_{\tilde{m}_i} < \frac{1}{2}\Delta_i(m_i)\right) < r_{*(m_i)} - \tilde{\Delta}_{\tilde{m}_i} \leq \hat{r}_{*(m_i)}(n_{m_i-1}) - \sqrt{\frac{\log(N\tilde{\Delta}^2_{m_i-1})}{2n_{m_i-1}}}.$$

By the elimination rule, i will be eliminated by the optimal arm $*(m_i)$. Again, using the Chernoff-Hoeffding bound, we have:

$$\begin{aligned}&\mathbb{P}\left(\hat{r}_i(n_{m_i-1}) > r_i - \sqrt{\frac{\log(N\tilde{\Delta}^2_{m_i-1})}{2n_{m_i-1}}} + \tilde{\Delta}_{\tilde{m}_i}\right)\\ \leq &\exp\left(-2n_{m_i-1}\left(\tilde{\Delta}_{\tilde{m}_i} - \sqrt{\frac{\log(N\tilde{\Delta}^2_{m_i-1})}{2n_{m_i-1}}}\right)^2\right) \leq \frac{1}{N\tilde{\Delta}^2_{\tilde{m}_i}}.\end{aligned} \tag{13}$$

The last inequality of (13) comes from the fact that $n_{\tilde{m}_i} \leq n_{m_i-1}$ and

$$\sqrt{\frac{\log(N\tilde{\Delta}^2_{m_i-1})}{2n_{m_i-1}}} \leq \frac{1}{2}\tilde{\Delta}_{m_i-1} \leq \frac{1}{2}\tilde{\Delta}_{\tilde{m}_i}, \quad \sqrt{\frac{\log(N\tilde{\Delta}^2_{\tilde{m}_i})}{2n_{\tilde{m}_i}}} \leq \frac{1}{2}\tilde{\Delta}_{\tilde{m}_i}.$$

Similarly, we can conclude that

$$\mathbb{P}\left(\hat{r}_i(n_{m_i-1}) > r_i - \sqrt{\frac{\log(N\tilde{\Delta}^2_{m_i-1})}{2n_{m_i-1}}} + \tilde{\Delta}_{\tilde{m}_i}\right) \leq \frac{1}{N\tilde{\Delta}^2_{\tilde{m}_i}}. \tag{14}$$

Thus, the regret contributed by arm i, conditioned on case (a), is upper bounded by

$$R_i \coloneqq n_{t_i}\Delta_i(t_i) + \sum_{m=t_i+1}^{M}(n_m - n_{m-1})\Delta_i(m),$$

and the corresponding total expected regret contributed by each suboptimal arm $i \in A'(\lambda)$ in case (a) is upper-bounded by:

$$\sum_{i\in A'(\lambda)} \frac{2}{N\tilde{\Delta}^2_{\tilde{m}_i}} \cdot \left[n_{t_i}\Delta_i(t_i) + \sum_{m=t_i+1}^{M}(n_m - n_{m-1})\Delta_i(m)\right] = \sum_{i\in A'(\lambda)} \frac{2R_i}{N\tilde{\Delta}^2_{\tilde{m}_i}}.$$

Case (b): *For a suboptimal arm $i \in A'(\lambda)$, either it is eliminated in or before the $\overline{m}_i$-th elimination phase of round m_i, or $*(m_i) \notin B^{\overline{m}_i}_{m_i}$ (i.e., the optimal arm of round m_i is eliminated in the $\overline{m}_i$-th elimination phase of round m_i or earlier).*

We divide it into two smaller cases:

Case (b1): *i is eliminated in or before the $\overline{m}_i$-th elimination phase of round m_i.*

If $i \in G_1(\lambda)$, then $m_i = t_i$, and $\overline{m}_i = \tilde{m}_i$. In this case, arm i is eliminated in the $\tilde{m}_i$-th elimination phase of round m_i. Therefore, the random regret contributed by arm i is upper bounded by

$$\tilde{S}_i \coloneqq n_{\tilde{m}_i} \cdot \Delta_i(t_i), \quad i \in G_1(\lambda).$$

If $i \in G_2(\lambda)$, then $m_i > t_i$ and $\overline{m}_i = \tilde{m}_i$, and the total number of times arm i can be pulled does not exceed $n_{\tilde{m}_i}$. Hence, the random regret of arm i is upper bounded by

$$\tilde{S}_i \coloneqq n_{t_i}\Delta_i(t_i) + \sum_{m=t_i+1}^{\overline{m}_i} (n_m - n_{m-1})\Delta_i(m), \quad i \in G_2(\lambda).$$

If $i \in G_3(\lambda)$, then $m_i > t_i$ and $\overline{m}_i = m_i - 1$, arm i is eliminated in or before the $\overline{m}_i$-th elimination phase of round m_i. By the definition of the algorithm, arm i has already been pulled n_{m_i-1} times in total. Thus, the corresponding upper bound on the random regret of arm i is

$$\tilde{S}_i \coloneqq n_{t_i}\Delta_i(t_i) + \sum_{m=t_i+1}^{\overline{m}_i} (n_m - n_{m-1})\Delta_i(m), \quad i \in G_3(\lambda).$$

Therefore, the corresponding total expected regret in case (b1) is upper bounded by

$$\sum_{i \in A'(\lambda)} \tilde{S}_i.$$

Case (b2): *If suboptimal arm $i \in A'(\lambda)$ was not eliminated in or before the $\overline{m}_i$-th elimination phase of round m_i, and $*(m_i) \notin B_{m_i}^{\overline{m}_i}$.*

If $i \in G_2(\lambda)$, then $\overline{m}_i = \tilde{m}_i = m_i$. Under the assumption of case (b2), $*(m_i)$ must have been eliminated in round $t_{*(m_i)} \le m_* \le m_i$ by some suboptimal arm j. Note that if (7) and (8) hold at $p = m_{**}$ for j and $m = m_i$ respectively, then $*(m_i)$ will not be eliminated by j in the m_{**}-th elimination phase of round m_*, here $t_{*(m_i)} \le m_* \le m_i$. Therefore, the upper bound of the probability that $*(m_i)$ is eliminated by j in the m_{**}-th elimination phase of round m_* is $\frac{2}{N\tilde{\Delta}^2_{m_{**}}}$. We have $\Delta_j(M) > \Delta_{*(m_i)}(M) \ge 0$ and $t_j \le m_* \le m_i$.

On the other hand, we observe from the definitions that all possible elimination phases

in which the optimal arm $*(m_i)$ is eliminated in round $t_{*(m_i)} \le m_* \le m_i$ are as follows:

$$J_{m_*} := \{m_{**} : *(m_i) \text{ can be eliminated in the } m_{**}\text{-th elimination phase of round } m_*\}$$
$$= \begin{cases} \{0, 1, \cdots, m_*\}, & m_* = t_{*(m_i)}; \\ \{m_* - 1,\, m_*\}, & t_{*(m_i)} < m_* \le m_i. \end{cases}$$

Thus, we have:

$$\mathbb{P}\left(\tilde{E}_i \cap \tilde{F}_i^c\right)$$
$$\le \mathbb{P}\left(*(m_i) \text{ is eliminated by some suboptimal arm } j \text{ in some round } m_*,\ t_{*(m_i)} \le m_* \le m_i\right)$$
$$= \sum_{m_*=t_{*(m_i)}}^{m_i} \mathbb{P}\left(*(m_i) \text{ is eliminated by some suboptimal arm } j \text{ in round } m_*\right)$$
$$\le \sum_{m_*=t_{*(m_i)}}^{m_i} \sum_{\substack{j:\, t_j \le m_* \\ j \ne *(m_i)}} \sum_{m_{**} \in J_{m_*}} \mathbb{P}\left(*(m_i) \text{ is eliminated by } j \text{ in } m_{**}\text{-th elimination phase of round } m_*\right)$$
$$\le \sum_{m_*=t_{*(m_i)}}^{m_i} \sum_{\substack{j:\, t_j \le m_* \\ j \ne *(m_i)}} \sum_{m_{**}=0}^{m_*} \frac{2}{N\tilde{\Delta}_{m_{**}}^2} \le \sum_{m_*=t_{*(m_i)}}^{m_i} \sum_{\substack{j:\, t_j \le m_* \\ j \ne *(m_i)}} \frac{3}{N\tilde{\Delta}_{m_*}^2} \le \sum_{m_*=t_{*(m_i)}}^{m_i} \frac{3\left(K(m_i)-1\right)}{N\tilde{\Delta}_{m_*}^2}$$
$$< \frac{3\left(K(m_i)-1\right)}{N} \sum_{m_*=0}^{m_i} \frac{1}{\tilde{\Delta}_{m_*}^2} < \frac{3\left(K(m_i)-1\right)}{N} \cdot \frac{4}{3} \cdot 4^{m_i} = \frac{4\left(K(m_i)-1\right)}{N\tilde{\Delta}_{m_i}^2}$$
$$= \frac{4\left(K(m_i)-1\right)}{N\tilde{\Delta}_{\tilde{m}_i}^2}, \quad i \in G_2(\lambda). \tag{15}$$

If $i \in G_1(\lambda)$, then $t_i = m_i$ and $\overline{m}_i = \tilde{m}_i$. In this case, we further distinguish whether the optimal arm $*(m_i)$ arrives simultaneously with arm i; that is, whether $t_{*(m_i)} = t_i = m_i$.

- If $t_{*(m_i)} = t_i = m_i$, then we have

  $$\mathbb{P}\left(\tilde{E}_i \cap \tilde{F}_i^c\right)$$
  $$\le \mathbb{P}\left(*(m_i) \text{ is eliminated by some arm } j \text{ in or before } \tilde{m}_i\text{-th elimination phase in round } m_i\right)$$
  $$\le \sum_{\substack{j:\, t_j \le m_i \\ j \ne *(m_i)}} \sum_{m_*=0}^{\tilde{m}_i} \frac{2}{N\tilde{\Delta}_{m_*}^2} < \frac{4\left(K(m_i)-1\right)}{N\tilde{\Delta}_{\tilde{m}_i}^2}.$$

- If $t_{*(m_i)} < t_i = m_i$, by the definition of the UCB-AA algorithm, we know the optimal arm $*(m_i)$ must have been eliminated before the beginning of round m_i. We make

the following partition:

$$\begin{aligned}\mathbb{P}\left(\tilde{E}_i \cap \tilde{F}_i^c\right) =& \mathbb{P}\left(\tilde{E}_i \cap \tilde{F}_i^c \cap \{*(m_i) \text{ is eliminated in or before round } \tilde{m}_i\}\right) \\ &+\mathbb{P}\left(\tilde{E}_i \cap \tilde{F}_i^c \cap \{*(m_i) \text{ is eliminated after round } \tilde{m}_i\}\right). \end{aligned} \tag{16}$$

We analyze these two items separately. Without loss of generality, we assume $t_{*(m_i)} \leq \tilde{m}_i$; otherwise, the first term in (16) vanishes. Similar to the method used in (15), we obtain an upper bound of the first term:

$$\begin{aligned}&\mathbb{P}\left(\tilde{E}_i \cap \tilde{F}_i^c \cap \{*(m_i) \text{ is eliminated in or before round } \tilde{m}_i\}\right) \\ \leq& \sum_{m_*=t_{*(m_i)}}^{\tilde{m}_i} \mathbb{P}\left(*(m_i) \text{ is eliminated by some suboptimal arm } j \text{ in round } m_*\right) \\ \leq& \sum_{m_*=t_{*(m_i)}}^{\tilde{m}_i} \sum_{\substack{j:\, t_j \leq m_* \\ j \neq *(m_i)}} \sum_{m_{**} \in J_{m_*}} \frac{2}{N\tilde{\Delta}_{m_{**}}^2} < \frac{4\left(K(m_i)-1\right)}{N\tilde{\Delta}_{\tilde{m}_i}^2}. \end{aligned} \tag{17}$$

For the second term, we find that if $*(m_i)$ is eliminated after round $\tilde{m}_i$, then by the definition of the UCB-AA algorithm, the empirical average reward obtained from pulling $*(m_i)$ for $n_{\tilde{m}_i}$ times has already been recorded. Thus, we have:

$$\begin{aligned}&\mathbb{P}\left(\tilde{E}_i \cap \tilde{F}_i^c \cap \{*(m_i) \text{ is eliminated after round } \tilde{m}_i\}\right) \\ \leq& \mathbb{P}\left(\hat{r}_i(n_{\tilde{m}_i}) + \sqrt{\frac{\log(N\tilde{\Delta}_{\tilde{m}_i}^2)}{2n_{\tilde{m}_i}}} > \hat{r}_{*(m_i)}(n_{\tilde{m}_i}) - \sqrt{\frac{\log(N\tilde{\Delta}_{\tilde{m}_i}^2)}{2n_{\tilde{m}_i}}}\right) \leq \frac{2}{N\tilde{\Delta}_{\tilde{m}_i}^2}. \end{aligned} \tag{18}$$

Substituting (17) and (18) into (16), we can get:

$$\mathbb{P}\left(\tilde{E}_i \cap \tilde{F}_i^c\right) \leq \frac{4K(m_i)-2}{N\tilde{\Delta}_{\tilde{m}_i}^2}, \quad i \in G_1(\lambda). \tag{19}$$

If $i \in G_3(\lambda)$, then $t_i < m_i = t_{*(m_i)}$, $\tilde{m}_i < m_i$, and $\overline{m}_i = m_i - 1$. Thus, by the definition of the UCB-AA algorithm and $\tilde{F}_i^c$, $*(m_i)$ have been eliminated before the beginning of the (m_i-1)-th elimination phase of round m_i. By $\tilde{m}_i < m_i$, we have $*(m_i) \neq *(m_i-1)$. Recall from the definitions of m_i and $\tilde{m}_i$, we have:

$$r_{*(m_i)} - r_{*(m_i-1)} > 2\left(\tilde{\Delta}_{\tilde{m}_i} - \tilde{\Delta}_{m_i-1}\right) \geq 0.$$

We further distinguish whether $\tilde{m}_i \geq m_i - 2$.

- If $\tilde{m}_i \geq m_i - 2$, then

$$\begin{aligned}\mathbb{P}\left(\tilde{E}_i \cap \tilde{F}_i^c\right) &\leq \sum_{m_*=0}^{m_i-2} \mathbb{P}\left(*(m_i) \text{ is eliminated in } m_*\text{-th elimination phase of round } m_i\right) \\ &\leq \sum_{m_*=0}^{m_i-2} \sum_{\substack{j:\, t_j \leq m_* \\ j \neq *(m_i)}} \frac{2}{N\tilde{\Delta}_{m_*}^2} \leq \sum_{m_*=0}^{m_i-2} \frac{2\left(K(m_i)-1\right)}{N\tilde{\Delta}_{m_*}^2} < \frac{4\left(K(m_i)-1\right)}{N\tilde{\Delta}_{\tilde{m}_i}^2}. \end{aligned} \tag{20}$$

- If $\tilde{m}_i < m_i - 2$, we have the following partition:

$$\begin{aligned}&\mathbb{P}\left(\tilde{E}_i \cap \tilde{F}_i^c\right) \\ &= \sum_{m_*=0}^{m_i-2} \mathbb{P}\Big(\tilde{E}_i \cap \{*(m_i) \text{ is eliminated in } m_*\text{-th elimination phase of round } m_i\}\Big) \\ &\leq \sum_{m_*=0}^{m_i-2} \mathbb{P}(*(m_i) \text{ is eliminated in } m_*\text{-th elimination phase by some } j \in A(m_i-1)) \\ &+ \mathbb{P}\left(\tilde{E}_i \cap \{*(m_i) \text{ is eliminated after the } \tilde{m}_i\text{-th elimination phase by some } j \in A_{m_i}\}\right) \\ &+ \sum_{m_*=0}^{\tilde{m}_i} \mathbb{P}\left(*(m_i) \text{ is eliminated in the } m_*\text{-th elimination phase by some } j \in A_{m_i}\right). \end{aligned} \tag{21}$$

For the first summation term in (21), we know for each arm $j \in A(m_i - 1)$, the suboptimal gap of j satisfies:

$$\Delta_j(m_i) \geq r_{*(m_i)} - r_{*(m_i-1)} > \underbrace{2\left(\tilde{\Delta}_{\tilde{m}_i} - \tilde{\Delta}_{m_i-1}\right)}_{\text{defined as } \Delta} \geq \frac{3}{2}\tilde{\Delta}_{\tilde{m}_i}. \tag{22}$$

If

$$\hat{r}_j(n_{m_*}) \leq r_j + \frac{1}{2}\Delta + \sqrt{\frac{\log(N\tilde{\Delta}_{m_*}^2)}{2n_{m_*}}} \tag{23}$$

and

$$\hat{r}_{*(m_i)}(n_{m_*}) \geq r_{*(m_i)} - \frac{1}{2}\Delta - \sqrt{\frac{\log(N\tilde{\Delta}_{m_*}^2)}{2n_{m_*}}} \tag{24}$$

hold, then we have:

$$\hat{r}_j(n_{m_*}) - \sqrt{\frac{\log(N\tilde{\Delta}_{m_*}^2)}{2n_{m_*}}} \leq r_j + \frac{1}{2}\Delta < r_{*(m_i)} - \frac{1}{2}\Delta \leq \hat{r}_{*(m_i)}(n_{m_*}) + \sqrt{\frac{\log(N\tilde{\Delta}_{m_*}^2)}{2n_{m_*}}}.$$

By the elimination rule, $*(m_i)$ will be eliminated by the suboptimal arm j. Using the Chernoff-Hoeffding bound, we have

$$\mathbb{P}\left(\hat{r}_j(n_{m_*}) > r_j + \frac{1}{2}\Delta + \sqrt{\frac{\log(N\tilde{\Delta}_{m_*}^2)}{2n_{m_*}}}\right) \leq \exp\left(-2n_{m_*}\cdot\left(\frac{1}{2}\Delta + \sqrt{\frac{\log(N\tilde{\Delta}_{m_*}^2)}{2n_{m_*}}}\right)^2\right).$$

If $m_* \leq \tilde{m}_i$, it is not hard to find

$$\mathbb{P}\left(\hat{r}_j(n_{m_*}) > r_j + \frac{1}{2}\Delta + \sqrt{\frac{\log(N\tilde{\Delta}_{m_*}^2)}{2n_{m_*}}}\right) \leq \frac{1}{N\tilde{\Delta}_{m_*}^2}.$$

If $\tilde{m}_i < m_* \leq m_i - 2$, we first show

$$n_{m_*} \geq \frac{5}{2}n_{m_*-1} > 2n_{m_*-1}, \quad \forall\, m_* \leq M-2. \tag{25}$$

By the construction of the UCB-AA algorithm, $\{n_m\}_{m=0}^M$ is an increasing sequence, which means that

$$\begin{aligned} & 2\cdot 4^{m+1}\cdot\log\left(\frac{N}{4^{m+1}}\right) \geq 2\cdot 4^m\cdot\log\left(\frac{N}{4^m}\right), \quad 0\leq m\leq M-1 \\ \iff\ & 4\log N - 4(m+1)\log 4 \geq \log N - m\log 4 \ \Rightarrow\ N \geq 4^{M+\frac{1}{3}}. \end{aligned}$$

We know that when $n_* \leq M-2$,

$$n_{m_*} \geq \frac{5}{2}n_{m_*-1} \iff \log N \geq \left(m_* + \frac{5}{3}\right)\log 4.$$

Hence, (25) is correct. Therefore, when $\tilde{m}_i < m_* \leq m_i - 2$,

$$\begin{aligned} & \exp\left(-2n_{m_*}\cdot\left(\frac{1}{2}\Delta + \sqrt{\frac{\log(N\tilde{\Delta}_{m_*}^2)}{2n_{m_*}}}\right)^2\right) \leq \exp\left(-2^{m_*-\tilde{m}_i+1}n_{\tilde{m}_i}\cdot\frac{1}{4}\Delta^2\right) \\ & \text{(by (22))} \leq \exp\left(-2^{m_*-\tilde{m}_i}n_{\tilde{m}_i}\cdot\tilde{\Delta}_{\tilde{m}_i}^2\right) \leq \left(\frac{1}{N\tilde{\Delta}_{\tilde{m}_i}^2}\right)^{2^{m_*-\tilde{m}_i+1}}. \end{aligned} \tag{26}$$

The last inequality of (26) comes from the fact that $\sqrt{\frac{\log(N\tilde{\Delta}_{\tilde{m}_i}^2)}{2n_{\tilde{m}_i}}} \leq \frac{1}{2}\tilde{\Delta}_{\tilde{m}_i}$. Thus, we have

$$\mathbb{P}\left(\hat{r}_j(n_{m_*}) > r_j + \frac{1}{2}\Delta + \sqrt{\frac{\log(N\tilde{\Delta}_{m_*}^2)}{2n_{m_*}}}\right) \leq \begin{cases} \frac{1}{N\tilde{\Delta}_{m_*}^2}, & m_* \leq \tilde{m}_i; \\ \left(\frac{1}{N\tilde{\Delta}_{\tilde{m}_i}^2}\right)^{2^{m_*-\tilde{m}_i+1}}, & m_* > \tilde{m}_i. \end{cases} \tag{27}$$

With the same approach, we have

$$\mathbb{P}\left(\hat{r}_{*(m_i)}(n_{m_*}) < r_{*(m_i)} - \frac{1}{2}\Delta - \sqrt{\frac{\log(N\tilde{\Delta}^2_{m_*})}{2n_{m_*}}}\right) \leq \begin{cases} \frac{1}{N\tilde{\Delta}^2_{m_*}}, & m_* \leq \tilde{m}_i, \\ \left(\frac{1}{N\tilde{\Delta}^2_{\tilde{m}_i}}\right)^{2^{m_*-\tilde{m}_i+1}}, & m_* > \tilde{m}_i. \end{cases} \tag{28}$$

Substituting (27) and (28) into the first term of (21), we have

$$\begin{aligned} &\mathbb{P}\left(*(m_i) \text{ is eliminated in the } m_*\text{-th elimination phase by some } j \in A(m_i-1)\right) \\ \leq &\begin{cases} \frac{2K(m_i-1)}{N\tilde{\Delta}^2_{m_*}}, & m_* \leq \tilde{m}_i; \\ 2K(m_i-1)\cdot\left(\frac{1}{N\tilde{\Delta}^2_{\tilde{m}_i}}\right)^{2^{m_*-\tilde{m}_i+1}}, & m_* > \tilde{m}_i. \end{cases} \end{aligned} \tag{29}$$

For the second term in (21), define

$$\Delta_* := 2\left(\sqrt{\frac{\log(N\tilde{\Delta}^2_{\tilde{m}_i})}{2n_{m_i-1}}} + \sqrt{\frac{\log(N\tilde{\Delta}^2_{m_i-1})}{2n_{m_i-1}}}\right).$$

Next, we consider whether there exists an arm with mean reward greater than $r_i + \Delta_*$ at the start of the $m_i - 1$-th elimination phase of round m_i. Define the event of existence as $\mathcal{A}$. We further have a partition of the original term:

$$\begin{aligned} &\mathbb{P}\left(\tilde{E}_i \cap \{*(m_i) \text{ is eliminated after the } \tilde{m}_i\text{-th elimination phase by some } j \in A_{m_i}\}\right) \\ \leq &\mathbb{P}\left(\tilde{E}_i \cap \mathcal{A}^c \cap \{*(m_i) \text{ is eliminated after } \tilde{m}_i\text{-th elimination phase by some } j \in A_{m_i}\}\right) \\ &+\mathbb{P}\left(\tilde{E}_i \cap \mathcal{A}\right). \end{aligned}$$

If event $\tilde{E}_i \cap \mathcal{A}$ happens, take arm $*_0$ such that $r_{*_0} \geq r_i + \Delta_*$, then if

$$\hat{r}_i(n_{m_i-1}) \leq r_i + \frac{1}{2}\Delta_* - \sqrt{\frac{\log(N\tilde{\Delta}^2_{m_i-1})}{2n_{m_i-1}}}$$

and

$$\hat{r}_{*_0}(n_{m_i-1}) \geq r_{*_0} - \frac{1}{2}\Delta_* + \sqrt{\frac{\log(N\tilde{\Delta}^2_{m_i-1})}{2n_{m_i-1}}}$$

hold, then by the elimination rule, i will be eliminated by arm $*_0$. By Chernoff-Hoeffding bound, we have

$$\mathbb{P}\left(\hat{r}_i(n_{m_i-1}) > r_i + \frac{1}{2}\Delta_* - \sqrt{\frac{\log(N\tilde{\Delta}^2_{m_i-1})}{2n_{m_i-1}}}\right) \leq \frac{1}{N\tilde{\Delta}^2_{\tilde{m}_i}},$$

$$\mathbb{P}\left(\hat{r}_{*_0}(n_{m_i-1}) < r_{*_0} - \frac{1}{2}\Delta_* + \sqrt{\frac{\log(N\tilde{\Delta}^2_{m_i-1})}{2n_{m_i-1}}}\right) \leq \frac{1}{N\tilde{\Delta}^2_{\tilde{m}_i}}.$$

Thus, we have

$$\mathbb{P}\left(\tilde{E}_i \cap \mathcal{A}\right) \leq \frac{2}{N\tilde{\Delta}^2_{\tilde{m}_i}}.$$

If event $\tilde{E}_i \cap \mathcal{A}^c$ happens and $*(m_i)$ is eliminated after $\tilde{m}_i$-th elimination phase by some $j \in A_{m_i}$, we consider the arm with the highest mean reward at the end of the m_*th elimination phase, here $\tilde{m}_i \leq m_* \leq m_i - 2$. Assuming there have been $s+1$ distinct optimal arms with strictly decreasing mean reward during this period, from first to last, we denote them as $*_1 = *(m_i), *_2, \cdots, *_{s+1}$, where $r_{*_1} > r_{*_2} > \cdots > r_{*_{s+1}}$ and $r_{*_{s+1}} \leq r_i + \Delta_*$. Note that by (25),

$$\Delta_* \leq 2\left(\sqrt{\frac{n_{\tilde{m}_i}}{n_{m_i-1}}}\tilde{\Delta}_{\tilde{m}_i} + \tilde{\Delta}_{m_i-1}\right) \leq \frac{13}{10}\tilde{\Delta}_{\tilde{m}_i} < \Delta,$$

so it is not hard to find that $1 \leq s \leq m_i - \tilde{m}_i - 2$. We denote $*_k$ $(1 \leq k \leq s)$ is eliminated at the m^k-th elimination phase of round m_i, and Δ^k as the suboptimal gap between $*_k$ and the arm which eliminates it, denoted as $*^k$, we find that:

$$\Delta^k \geq r_{*_k} - r_{*_{k+1}} > 0, \quad \sum_{k=1}^{s-1}\Delta^k \geq r_{*(m_i)} - r_{*_s} \geq \Delta - \Delta_*.$$

Note if

$$\hat{r}_{*^k}(n_{m^k}) \leq r_i + \frac{1}{2}\Delta^k + \sqrt{\frac{\log(N\tilde{\Delta}^2_{m^k})}{2n_{m^k}}}$$

and

$$\hat{r}_{*_k}(n_{m^k}) \geq r_{*_k} - \frac{1}{2}\Delta^k - \sqrt{\frac{\log(N\tilde{\Delta}^2_{m^k})}{2n_{m^k}}}$$

hold, then by the elimination rule, $*_k$ will not be eliminated by arm $*^k$. By Chernoff-Hoeffding bound, we have

$$\mathbb{P}\left(*_k \text{ is eliminated by } *^k\right) \leq 2\exp\left(-2n_{m^k}\left(\frac{1}{2}\Delta^k + \sqrt{\frac{\log(N\tilde{\Delta}^2_{m^k})}{2n_{m^k}}}\right)^2\right).$$

Based on the above results, we find

$$\begin{aligned}
&\mathbb{P}\left(\tilde{E}_i \cap \mathcal{A}^c \cap \{*(m_i) \text{ is eliminated after } \tilde{m}_i\text{-th elimination phase by some } j \in A_{m_i}\}\right) \\
\leq &\mathbb{P}\left(\cap_{k=1}^{s}\left\{*_k \text{ is eliminated by } *^k\right\}\right) \\
\leq &\max_{\substack{1\leq s\leq m_i-\tilde{m}_i-2 \\ \tilde{m}_i+1\leq m^1<\cdots<m^s\leq m_i-2 \\ \sum_{k=1}^{s}\Delta^k\geq\Delta-\Delta_*,\,\Delta^k>0}} \min_{1\leq k\leq s} 2\exp\left(-2n_{m^k}\left(\frac{1}{2}\Delta^k+\sqrt{\frac{\log(N\tilde{\Delta}_{m^k}^2)}{2n_{m^k}}}\right)^2\right) \\
\leq &\max_{\substack{1\leq s\leq m_i-\tilde{m}_i-2 \\ \tilde{m}_i+1\leq m^1<\cdots<m^s\leq m_i-2 \\ \sum_{k=1}^{s}\Delta^k=\Delta-\Delta_*,\,\Delta^k>0}} \min_{1\leq k\leq s} \frac{2}{N\tilde{\Delta}_{m^k}^2}\exp\left(-2n_{m^k}\cdot\left(\frac{1}{2}\Delta^k\right)^2\right). \qquad (30)
\end{aligned}$$

By (25), we deduce that

$$\frac{2}{N\tilde{\Delta}_{m^k}^2}\exp\left(-2n_{m^k}\cdot\left(\frac{1}{2}\Delta^k\right)^2\right) \leq \frac{2}{N\tilde{\Delta}_{m^k}^2}\exp\left(-2n_{\tilde{m}_i}\cdot\left(\frac{5}{2}\right)^{m^k-\tilde{m}_i}\cdot\left(\frac{1}{2}\Delta^k\right)^2\right).$$

Denote $p_k \coloneqq m^k - \tilde{m}_i$, we have

$$\begin{aligned}
&\mathbb{P}\left(\tilde{E}_i \cap \mathcal{A}^c \cap \{*(m_i) \text{ is eliminated after } \tilde{m}_i\text{-th elimination phase by some } j \in A_{m_i}\}\right) \\
\leq &\max_{\substack{1\leq p_1<\cdots<p_s\leq m_i-\tilde{m}_i-2 \\ \sum_{k=1}^{s}\Delta^k=\Delta-\Delta_*,\,\Delta^k>0}} \min_{1\leq k\leq s} \frac{2\cdot 4^{p_k}}{N\tilde{\Delta}_{\tilde{m}_i}^2}\exp\left(-2n_{\tilde{m}_i}\cdot\left(\frac{5}{2}\right)^{p_k}\cdot\left(\frac{1}{2}\Delta^k\right)^2\right). \qquad (31)
\end{aligned}$$

By (25), we know

$$\frac{1}{2}\sum_{k=1}^{s}\Delta^k = \frac{1}{2}(\Delta-\Delta_*) \geq \left(1-\left(\frac{2}{5}\right)^{\frac{m_i-1-\tilde{m}_i}{2}}\right)\tilde{\Delta}_{\tilde{m}_i} - \tilde{\Delta}_{m_i-2}.$$

Below, we will discuss the values of $\tilde{m}_i$ case by case.

- If $\tilde{m}_i = m_i - 3$, then $p_1 = 1$, $\Delta^1 \geq \frac{1}{5}\tilde{\Delta}_{\tilde{m}_i}$. At this point, we know that the probability in (31) is upper bounded by

$$\frac{2\cdot 4}{N\tilde{\Delta}_{\tilde{m}_i}^2}\exp\left(-\frac{1}{20}n_{\tilde{m}_i}\cdot\tilde{\Delta}_{\tilde{m}_i}^2\right) \leq \frac{8}{N\tilde{\Delta}_{\tilde{m}_i}^2}.$$

- If $\tilde{m}_i = m_i - 4$, then if $p_1 = 1$, then we also deduce that the probability in (31) is upper bounded by $\frac{8}{N\tilde{\Delta}_{\tilde{m}_i}^2}$; if $p_1 = 2$, then $\Delta^1 \geq \sqrt{\frac{4}{5}}\tilde{\Delta}_{\tilde{m}_i}$, so the probability in (31) is upper bounded by

$$\frac{2\cdot 16}{N\tilde{\Delta}_{\tilde{m}_i}^2}\exp\left(-2n_{\tilde{m}_i}\cdot\frac{25}{4}\cdot\frac{1}{5}\tilde{\Delta}_{\tilde{m}_i}^2\right) \leq \frac{32}{N\tilde{\Delta}_{\tilde{m}_i}^2}\cdot\frac{1}{N\tilde{\Delta}_{\tilde{m}_i}^2} \leq \frac{2}{N\tilde{\Delta}_{\tilde{m}_i}^2},$$

the last inequality comes from $N\tilde{\Delta}_{\tilde{m}_i}^2 \geq 4^{M+\frac{1}{3}-(M-2)} > 16$.

– If $\tilde{m}_i \le m_i - 5$, then we have

$$(\Delta - \Delta_*) = \frac{1}{2}\sum_{k=1}^{s} \Delta^k \ge \left(1 - \left(\frac{2}{5}\right)^2 - \frac{1}{8}\right)\tilde{\Delta}_{\tilde{m}_i} \ge \frac{1}{\sqrt{2}}\tilde{\Delta}_{\tilde{m}_i}.$$

If we let $\frac{1}{2}\Delta^k = \lambda_k(\Delta - \Delta_*),\ \lambda_k > 0$, then we know that (31) can be transformed into the following form:

$$\begin{aligned}
&\mathbb{P}\left(\tilde{E}_i \cap \mathcal{A}^c \cap \{*(m_i) \text{ is eliminated after } \tilde{m}_i\text{-th elimination phase by some } j \in A_{m_i}\}\right) \\
&\le \max_{\substack{1\le p_1<\cdots<p_s\le m_i-\tilde{m}_i-2 \\ \sum_{k=1}^s \lambda_k=1,\ \lambda_k>0}} \min_{1\le k\le s} \frac{2\cdot 4^{p_k}}{N\tilde{\Delta}_{\tilde{m}_i}^2} \exp\left(-2n_{\tilde{m}_i}\cdot\left(\frac{5}{2}\right)^{p_k}\cdot\lambda_k^2(\Delta-\Delta_*)^2\right) \\
&\le \frac{2}{N\tilde{\Delta}_{\tilde{m}_i}^2} \max_{\substack{1\le p_1<\cdots<p_s\le m_i-\tilde{m}_i-2 \\ \sum_{k=1}^s \lambda_k=1,\ \lambda_k>0}} \min_{1\le k\le s} \exp\left(\left(p_k - \frac{8\lambda_k^2}{3}\cdot\left(\frac{5}{2}\right)^{p_k}\right)\log 4\right). \qquad (32)
\end{aligned}$$

Next, we establish a lemma to estimate the above term:

Lemma A.6. *Let $n \ge 3$ be an arbitrary integer. For any integers $1 \le p_1 < \cdots < p_s \le n$ (where $1 \le s \le n$) and any positive numbers $\lambda_1, \ldots, \lambda_s > 0$ satisfying $\sum_{k=1}^s \lambda_k = 1$, define*

$$\Psi(p_{1:s}, \lambda_{1:s}) \coloneqq \min_{1\le k\le s} \exp\left(\left(p_k - \frac{8}{3}\left(\frac{5}{2}\right)^{p_k}\lambda_k^2\right)\log 4\right).$$

Then for all $n \ge 3$,

$$\max_{\substack{1\le p_1<\cdots<p_s\le n \\ \sum_{k=1}^s \lambda_k=1,\ \lambda_k>0}} \Psi(p_{1:s}, \lambda_{1:s}) \le 10.$$

Proof of Lemma A.6. We focus on the logarithmic scale (equivalently, the exponent in the exponential expression), which constitutes the main object of interest. For notational simplicity, we absorb all constants into a single coefficient and define

$$S_p(\lambda) = p - \frac{8}{3}\left(\frac{5}{2}\right)^p \lambda^2.$$

Given an index set $p_{1:s}$ and parameters $\lambda_{1:s}$, our objective is to characterize

$$\min_{1\le k\le s} S_{p_k}(\lambda_k).$$

Since the global objective is to take the supremum over all possible index sets and admissible λ, it suffices to show that for any such choice, the above minimum is uniformly bounded from above by some constant t that is independent of n. To this end, we reformulate the problem as a **feasibility problem** for a given real number t. Specifically, we ask whether there exists a collection of positive weights $\{\lambda_k\}_{k=1}^{s}$, satisfying $\sum_{k=1}^{s} \lambda_k = 1$, such that

$$S_{p_k}(\lambda_k) \geq t, \quad \forall\, 1 \leq k \leq s,$$

or equivalently,

$$p_k - \frac{8}{3}\left(\frac{5}{2}\right)^{p_k} \lambda_k^2 \geq t.$$

Rearranging yields the necessary condition

$$\lambda_k^2 \leq \frac{3}{8} \cdot \frac{p_k - t}{\left(\frac{5}{2}\right)^{p_k}}, \quad \forall\, 1 \leq k \leq s.$$

In particular, if $p_k \leq t$, the above inequality forces $\lambda_k = 0$. For indices satisfying $p_k > t$, we may define the upper bound

$$U_p(t) \coloneqq \sqrt{\frac{3}{8}}\sqrt{p - t}\left(\frac{5}{2}\right)^{-p/2}.$$

Hence, for a given index set $p_{1:s}$ (restricted to those $p_k > t$), a necessary and sufficient condition for the existence of a feasible λ with $\sum_k \lambda_k = 1$ is

$$\sum_{k=1}^{s} \lambda_k \leq \sum_{k=1}^{s} U_{p_k}(t).$$

In particular, feasibility requires

$$\sum_{k=1}^{s} U_{p_k}(t) \geq 1.$$

We now take the supremum over all possible index sets. For a fixed t, regardless of the choice of the finite index set $\{p_k : p_k > t\}$, the total "capacity" is upper bounded by

$$F_\infty(t) \coloneqq \sum_{p=\lfloor t \rfloor + 1}^{\infty} U_p(t),$$

since the sum over any finite subset cannot exceed the sum over the entire tail. Consequently, if $F_\infty(t) < 1$, then no feasible λ satisfying $\sum_k \lambda_k = 1$ exists for any finite n or any index set, and thus t is infeasible. It follows that the maximal feasible value t^* satisfies

$$t^* \le t_\infty,$$

where t_∞ is the unique solution to $F_\infty(t_\infty) = 1$. In what follows, we construct an explicit upper bound t_0 such that $t_\infty \le t_0$. Let

$$r \coloneqq \left(\frac{5}{2}\right)^{-1/2} = \sqrt{\frac{2}{5}} \in (0,1).$$

Then

$$F_\infty(t) = \sqrt{\frac{3}{8}} \sum_{m=1}^{\infty} \sqrt{m} \left(\frac{5}{2}\right)^{-(t+m)/2} = \sqrt{\frac{3}{8}} r^t \sum_{m=1}^{\infty} \sqrt{m} r^m.$$

Thus, it remains to bound the tail sum $\sum_{m\ge1} \sqrt{m} r^m$.

Define $f(x) \coloneqq \sqrt{x} r^x$ for $x \ge 1$. Since f is positive, integrable, and monotonically decreasing on $[1, \infty)$, a standard comparison argument yields

$$\sum_{m=1}^{\infty} \sqrt{m} r^m \le r + \int_1^{\infty} \sqrt{x} r^x dx.$$

Let $\alpha \coloneqq -\ln r > 0$. Then

$$\int_1^{\infty} \sqrt{x} r^x dx = \int_1^{\infty} x^{1/2} e^{-\alpha x} dx \le \int_0^{\infty} x^{1/2} e^{-\alpha x} dx = \frac{\Gamma(3/2)}{\alpha^{3/2}} = \frac{\sqrt{\pi}/2}{\alpha^{3/2}}.$$

Define the constants (independent of n and t)

$$S \coloneqq r + \frac{\sqrt{\pi}/2}{(-\ln r)^{3/2}}, \quad C_1 \coloneqq \sqrt{\frac{3}{8}} S.$$

Then for all t,

$$F_\infty(t) \le C_1 r^t.$$

Now let t_0 be the solution to $C_1 r^{t_0} = 1$ (which exists and is unique, since $r \in (0,1)$). Then for all $t \ge t_0$, $F_\infty(t) \le 1$, so $t_\infty \le t_0$. Finally, for all n and all

index choices,

$$\min_{1\le k\le s} S_{p_k}(\lambda_k) \le t_0,$$

and thus the maximum possible value of the original exponential expression is bounded above by 4^{t_0}, by calculations, we have

$$\max_{\substack{1\le p_1<\cdots<p_s\le n\\ \lambda_k>0,\ \sum\lambda_k=1}} \Psi(p_{1:s},\lambda_{1:s}) \le 4^{t_0} < 10.$$

□

With Lemma A.6, we find that for all $\tilde{m}_i \le m_i - 3$,

$$\mathbb{P}\left(\tilde{E}_i \cap \mathcal{A}^c \cap \{*(m_i) \text{ is eliminated after } \tilde{m}_i\text{-th elimination phase by some } j \in A_{m_i}\}\right) \le \frac{20}{N\tilde{\Delta}^2_{\tilde{m}_i}}. \tag{33}$$

For the third summation term in (21):

$$\begin{aligned}&\sum_{m_*=0}^{\tilde{m}_i} \mathbb{P}(*(m_i) \text{ is eliminated in } m_*\text{-th elimination phase by some } j \in A_{m_i})\\ &\le \sum_{m_*=0}^{\tilde{m}_i} \sum_{\substack{j\in A_{m_i}\\ j\ne *(m_i)}} \frac{2}{N\tilde{\Delta}^2_{m_*}} \le \frac{4(K(m_i)-K(m_i-1)-1)}{N\tilde{\Delta}^2_{\tilde{m}_i}}.\end{aligned} \tag{34}$$

Combining (29), (33) and (34), we have:

$$\begin{aligned}\mathbb{P}\left(\tilde{E}_i \cap \tilde{F}_i^c\right) &\le \sum_{m_*=0}^{\tilde{m}_i} \frac{2K(m_i-1)}{N\tilde{\Delta}^2_{m_*}} + \frac{4(K(m_i)-K(m_i-1)-1)}{N\tilde{\Delta}^2_{\tilde{m}_i}}\\ &\quad + \sum_{m_*=\tilde{m}_i+1}^{m_i-2} \frac{2K(m_i-1)}{\left(N\tilde{\Delta}^2_{\tilde{m}_i}\right)^{2^{m_*-\tilde{m}_i+1}}} + \frac{20}{N\tilde{\Delta}^2_{\tilde{m}_i}} \le \frac{4\left(K(m_i)+4\right)}{N\tilde{\Delta}^2_{\tilde{m}_i}}.\end{aligned} \tag{35}$$

The second-to-last inequality of (35) comes from the fact below:

$$N\tilde{\Delta}^2_{\tilde{m}_i} > 16, \quad \frac{2K(m_i-1)}{\left(N\tilde{\Delta}^2_{\tilde{m}_i}\right)^{2^{m_*-\tilde{m}_i+1}}} \le 2K(m_i-1)\cdot\left(\frac{1}{N\tilde{\Delta}^2_{\tilde{m}_i}}\right)^{m_*-\tilde{m}_i+1}.$$

In case (b2), the random regret generated by arm i does not exceed R_i. Summing over all $i \in A'(\lambda)$, we find the corresponding total expected regret in case (b2) is upper bounded by

$$\sum_{i\in A'(\lambda)} \frac{4\left(K(m_i)+5\right)\cdot R_i}{N\tilde{\Delta}^2_{\tilde{m}_i}}.$$

For all arms not belonging to $A'(\lambda)$, the total regret provided by this part can be easily upper-bounded as

$$T \max_{i:\Delta_i(M)\leq\lambda} \Delta_i(M).$$

Therefore, the upper bound of the regret is:

$$\sum_{i:\Delta_i(M)\geq\lambda} \left[\tilde{S}_i + \frac{4\left(K(m_i)+5\right)\cdot R_i}{N\tilde{\Delta}^2_{\tilde{m}_i}}\right] + T \max_{i:\Delta_i(M)\leq\lambda} \Delta_i(M).$$

The result follows. □

A.3 Proof of Corollary 1

Proof of Corollary 1. By definition of $\tilde{m}_i$, we have:

$$\tilde{\Delta}_{\tilde{m}_i} \leq \frac{1}{2}\Delta_i(m_i) \leq \tilde{\Delta}_{\tilde{m}_i-1} = 2\tilde{\Delta}_{\tilde{m}_i}.$$

For suboptimal arm $i \in G_1(\lambda) \cup G_2(\lambda)$, we have:

$$\begin{aligned}\tilde{S}_i &\leq n_{\tilde{m}_i}\Delta_i(m_i) \leq 4\left\lceil\frac{2\log(N\tilde{\Delta}^2_{\tilde{m}_i})}{\tilde{\Delta}^2_{\tilde{m}_i}}\right\rceil\tilde{\Delta}_{\tilde{m}_i} \leq \frac{8\log(N\tilde{\Delta}^2_{\tilde{m}_i})}{\tilde{\Delta}_{\tilde{m}_i}} + 4\tilde{\Delta}_{\tilde{m}_i}\\ &\leq \frac{32\log(N\Delta_i(m_i)^2)}{\Delta_i(m_i)} + 4\Delta_i(m_i) \leq \frac{32\tilde{C}\log(N\Delta_i(M)^2)}{\Delta_i(M)} + 4\Delta_i(M).\end{aligned}$$

Since function $\frac{\log(Ns^2)}{s}$ is decreasing in s for $\frac{1}{2^M} \leq s \leq 1$, and $\frac{\log(Ns^2)}{s} \geq \log N > 1$ when $N > e^2 \cdot 4^M$, we can bound S_i:

$$\tilde{S}_i \leq \frac{36\tilde{C}\log(N\lambda^2)}{\lambda}, \qquad i \in G_1(\lambda) \cup G_2(\lambda)$$

For suboptimal arm $i \in G_3(\lambda)$, we have:

$$\begin{aligned}\tilde{S}_i &\leq n_{m_i-1}\Delta_i(m_i-1) \leq 2\tilde{\Delta}_{m_i-1}\left\lceil\frac{2\log(N\tilde{\Delta}^2_{m_i-1})}{\tilde{\Delta}^2_{m_i-1}}\right\rceil \leq \frac{4\log(N\tilde{\Delta}^2_{m_i-1})}{\tilde{\Delta}_{m_i-1}} + 2\tilde{\Delta}_{m_i-1}\\ &\leq \frac{4\log(N\tilde{\Delta}^2_{m_i-1})}{\tilde{\Delta}_{m_i-1}} + 2\tilde{\Delta}_{m_i-1} = \frac{2\log(4N\tilde{\Delta}^2_{m_i})}{\tilde{\Delta}_{m_i}} + 4\tilde{\Delta}_{m_i} \leq \frac{8\log(N\tilde{\Delta}^2_{m_i})}{\tilde{\Delta}_{m_i}} + 4\tilde{\Delta}_{m_i}\\ &\leq \frac{32\log(N\Delta_i(m_i)^2)}{\Delta_i(m_i)} + 4\Delta_i(m_i) \leq \frac{32\tilde{C}\log(N\Delta_i(M)^2)}{\Delta_i(M)} + 4\Delta_i(M).\end{aligned}$$

Similar to before, we can obtain:

$$\tilde{S}_i \leq \frac{36\tilde{C}\log(N\lambda^2)}{\lambda}, \qquad i \in G_3(\lambda)$$

Thus, for all arm $i \in A'(\lambda) = \{i \in A(M) : \Delta_i(M) > \lambda\}$,

$$\tilde{S}_i \leq \frac{36\tilde{C}\log(N\lambda^2)}{\lambda}.$$

Besides,

$$\frac{K(m_i)\cdot R_i}{N\tilde{\Delta}^2_{\tilde{m}_i}} \leq \frac{16K(m_i)\cdot n_M\Delta_i(M)}{N\Delta_i(m_i)^2} \leq \frac{16\tilde{C}^2K(m_i)\cdot n_M}{N\Delta_i(M)} \leq \frac{16\tilde{C}^2K(M)\cdot n_M}{N\lambda}.$$

Based on the inequalities given earlier, we have a simplified upper bound of Theorem 1:

$$\begin{aligned}\mathbb{E}R_M(\pi) &\leq \sum_{i:\Delta_i(M)\geq\lambda}\left[\tilde{S}_i + \frac{4\left(K(m_i)+5\right)\cdot R_i}{N\tilde{\Delta}^2_{\tilde{m}_i}}\right] + T\max_{i:\Delta_i(M)\leq\lambda}\Delta_i(M)\\ &\leq \frac{684\tilde{C}^2K^2(M)n_M}{N\lambda} + \frac{36\tilde{C}K(M)\log(N\lambda^2)}{\lambda} + \lambda T.\end{aligned}$$

We take parameter $N = 10K(M)\cdot n_M$ and an approximation of λ:

$$\lambda \approx 10\sqrt{\frac{K\log(K)+K}{T}} \geq 10\sqrt{\frac{\log(K)+1}{n_M}} > \frac{1}{2^M},$$

Then, three parts of the upper bound coincide with $\Theta\left(\sqrt{K\log K\cdot T}\right)$.

□

A.4 Proof of Proposition 1

Proof of Proposition 1. By definition,

$$D_M = \sum_{k=0}^{M} H_k, \qquad H_k := L_k\rho_k\delta_k. \tag{36}$$

Fix $k \leq M-1$ such that $\rho_k\delta_k > \lambda$. By the definition of $m_{*(k)}$, for every $\ell = k+1,\ldots,m_{*(k)}-1$,

$$\Delta_{*(k)}(\ell) \leq 2\tilde{\Delta}_\ell.$$

Hence, the good-event accounting in Section 3.3 yields

$$\tilde{S}_{*(k)} \geq \sum_{\ell=k+1}^{m_{*(k)}-1} (n_\ell - n_{\ell-1})\Delta_{*(k)}(\ell).$$

Using

$$\Delta_{*(k)}(\ell) = \sum_{i=k}^{\ell-1} \delta_i$$

and exchanging the order of summation, we obtain

$$\begin{aligned}
\tilde{S}_{*(k)} &\geq \sum_{\ell=k+1}^{m_{*(k)}-1} (n_\ell - n_{\ell-1}) \sum_{i=k}^{\ell-1} \delta_i \\
&= \sum_{i=k}^{m_{*(k)}-2} \delta_i \sum_{\ell=i+1}^{m_{*(k)}-1} (n_\ell - n_{\ell-1}) \\
&= \sum_{i=k}^{m_{*(k)}-2} \delta_i \, (n_{m_{*(k)}-1} - n_i) \\
&\geq \frac{1}{2} \, n_{m_{*(k)}-1} \sum_{i=k}^{m_{*(k)}-2} \delta_i = \frac{1}{2} \, \Delta_{*(k)}(m_{*(k)} - 1) n_{m_{*(k)}-1},
\end{aligned} \tag{37}$$

where the last inequality uses the doubling property of $\{n_m\}$ established in the proof of Theorem 1. Moreover, the monotonicity of $\{n_m\}_{m=0}^{M}$ implies $N \geq 4^{M+\frac{1}{3}}$. Therefore,

$$\frac{\log(N\tilde{\Delta}_k^2)}{\tilde{\Delta}_k^2} \geq \log N \geq \left(M + \tfrac{1}{3}\right) \log 4 > 1,$$

and hence

$$n_k = \left\lceil \frac{2\log(N\tilde{\Delta}_k^2)}{\tilde{\Delta}_k^2} \right\rceil \leq \frac{3\log(N\tilde{\Delta}_k^2)}{\tilde{\Delta}_k^2}.$$

Also, by the definition of $m_{*(k)}$,

$$\tilde{\Delta}_{m_{*(k)}-1} = 2\tilde{\Delta}_{m_{*(k)}} < \Delta_{*(k)}\big(m_{*(k)}\big),$$

so

$$n_{m_{*(k)}-1} = \left\lceil \frac{2\log(N\tilde{\Delta}_{m_{*(k)}-1}^2)}{\tilde{\Delta}_{m_{*(k)}-1}^2} \right\rceil \geq \frac{2\log(N\tilde{\Delta}_{m_{*(k)}-1}^2)}{\Delta_{*(k)}\big(m_{*(k)}\big)^2}.$$

If (2) holds, then by the doubling property again,

$$n_{m_{*(k)}-1} \geq 2^{\,m_{*(k)}-k-1} n_k,$$

and therefore (37) gives

$$\tilde{S}_{*(k)} \geq 2^{m_{*(k)}-k-2}\Delta_{*(k)}(m_{*(k)}-1)n_k \geq \rho_k K(k) n_k \delta_k.$$

Since $L_k \leq K(k)n_k$, it follows that

$$\tilde{S}_{*(k)} \geq \rho_k K(k) n_k \delta_k \geq H_k.$$

If instead (1) holds, then

$$\frac{\log(N\tilde{\Delta}^2_{m_{*(k)}-1})}{\log(N\tilde{\Delta}^2_k)} = \frac{\log N - (m_{*(k)}-1)\log 4}{\log N - k\log 4} \geq \frac{M - m_{*(k)} + 1}{M-k+1}.$$

Combining this with (1) and the bound

$$\delta_k \leq \Delta_{*(k)}(m_{*(k)}-1),$$

we obtain

$$\Delta_{*(k)}(m_{*(k)}-1)\frac{\log(N\tilde{\Delta}^2_{m_{*(k)}-1})}{\Delta_{*(k)}(m_{*(k)})^2} \geq 3\rho_k K(k)\,\frac{\log(N\tilde{\Delta}^2_k)}{\tilde{\Delta}^2_k}\,\delta_k.$$

Substituting this bound into (37) yields

$$\tilde{S}_{*(k)} \geq \rho_k K(k) n_k \delta_k \geq H_k.$$

Therefore,

$$H_k \leq \tilde{S}_{*(k)}, \qquad \forall\, k \leq M-1 :\ \rho_k\delta_k > \lambda.$$

For rounds satisfying $\rho_k\delta_k \leq \lambda$, we have

$$H_k = L_k\rho_k\delta_k \leq \lambda L_k,$$

and thus, by (36),

$$\sum_{k:\rho_k\delta_k\leq\lambda} H_k \leq \lambda \sum_{k=0}^{M} L_k = \lambda T_0(\pi).$$

Taking expectations gives

$$\mathbb{E}\left[\sum_{k:\rho_k\delta_k\leq\lambda} H_k\right] \leq \lambda T.$$

For the terminal round,

$$H_M = L_M \rho_M \delta_M \leq \lambda L_M$$

by the assumption $\lambda \geq \rho_M \delta_M$.

Finally, if $k < \ell$ and both satisfy $\rho_k \delta_k > \lambda$ and $\rho_\ell \delta_\ell > \lambda$, then $*(k) \neq *(\ell)$, since once the benchmark strictly increases after round k, arm $*(k)$ can no longer remain optimal thereafter. Moreover,

$$\Delta_{*(k)}(M) \geq \delta_k > \lambda,$$

because $\rho_k \leq 1$ and $\rho_k \delta_k > \lambda$. Hence the arms

$$\{\, *(k) : \rho_k \delta_k > \lambda \,\}$$

are distinct and all belong to the set $\{i : \Delta_i(M) > \lambda\}$. Therefore,

$$\sum_{k:\rho_k \delta_k > \lambda} \tilde{S}_{*(k)} \leq \sum_{i:\Delta_i(M) > \lambda} \tilde{S}_i.$$

Combining the preceding bounds completes the proof. □

A.5 Proof of Corollary 2

In this section, we extend UCB-AA to an online regime in which M is not known in advance. Our approach relies on geometrically increasing guesses of the horizon and is closely related in spirit to the classical *doubling trick* used to convert fixed-horizon bandit algorithms into anytime (horizon-free) procedures: we run the base algorithm over successive stages with geometrically increasing horizons and restart it at each stage. Accordingly, the resulting procedure operates in stages indexed by $l = 1, 2, 3, \ldots$.

We summarize the online procedure in Algorithm 2. The online version of UCB-AA removes the fixed-horizon requirement: the learner does *not* know the total number of rounds M in advance, and therefore cannot tune the algorithm with M a priori. To handle this, we

adopt a standard doubling-type schedule and run UCB-AA over stages with geometrically increasing stage horizons. Specifically, at stage $l \geq 1$ we set

$$M_l \coloneqq 2^l, \qquad \tilde{M}_l \coloneqq \sum_{j=1}^{l} M_j,$$

and choose the stage parameter N_l according to the implicit relation $N_l = 10\, K(\tilde{M}_l)\, n_{M_l}(N_l)$ (cf. Algorithm 2). For a suboptimal arm i, define the first stage at which its gap becomes resolvable by

$$l_i \coloneqq \min\Big\{ l : \; 2^{M_l}\, \Delta_i(\tilde{M}_l) \geq 2 \Big\},$$

and let $L \coloneqq \min\{l : \tilde{M}_l \geq M\}$ denote the (random) index of the stage that contains the terminal round M.

We use the following restart convention. Stage l starts after global round $\tilde{M}_{l-1}$, with $\tilde{M}_0 = 0$. At this boundary, the active set, buffer set, and empirical means of the base UCB-AA instance are reset. Arms already available by round $\tilde{M}_{l-1}$ form the initial available set of the restarted instance, and arms arriving during the stage are admitted according to the usual UCB-AA rule. Thus eliminations do not persist across restarts, whereas regret is accumulated globally over all stages. Hence each stage may be analyzed as a fixed-horizon UCB-AA run with horizon M_l, and the online bound is obtained by summing stagewise bounds.

If the terminal round M lies inside stage L, we complete that stage only for analysis. Define an auxiliary arrival process by $A^\dagger_m = A_m$ for $m \leq M$ and $A^\dagger_m = \varnothing$ for $M < m \leq \tilde{M}_L$, leaving all reward distributions unchanged. The online algorithm and this auxiliary run coincide up to round M; after M, the auxiliary run continues the same restarted UCB-AA instance on the frozen available set until stage L ends. Since the additional virtual pulls have nonnegative roundwise gaps, the regret up to round M is bounded by the regret of the auxiliary completed run. Below, T denotes the expected number of pulls in this auxiliary completed run.

Theorem A.1 (Roundwise regret bound for Online UCB-AA)**.** *Suppose that Assumption 1 holds. Then, for any $\lambda \geq 2^{-M_L+1}$, the regret of Online UCB-AA up to the (unknown) terminal round M satisfies*

$$\mathbb{E}R_M(\pi) \leq \sum_{i:\,\Delta_i(M)\geq\lambda} \left(U_i \; + \frac{684\,\tilde{C}^2\,\Gamma_M}{\Delta_i(\tilde{M}_{l_i})} \right) \; + \; \lambda T,$$

where

$$\Gamma_M \coloneqq 2(M+2)\log 4 + \log_2(M+2)\log\{420K(M)\log K(M)\},$$

and

$$U_i \coloneqq \begin{cases} \dfrac{336\,\log K(M)}{\Delta_i(\tilde{M}_{l_i-1})}, & l_i > 1 \text{ and } \tilde{M}_{l_i-1} \geq t_i, \\ 0, & l_i = 1 \text{ or } \tilde{M}_{l_i-1} < t_i. \end{cases}$$

Here U_i is the pre-identification contribution of arm i.

The first two terms in Theorem A.1 split the regret contribution of arm i into two stages: up to stage $l_i - 1$ and from stage l_i onward. By definition of l_i, before stage l_i arm i cannot be reliably identified as suboptimal under the stage-based sampling budget, whereas from stage l_i onward such identification is feasible. The term U_i accounts for the arrival time and is defined to be zero when $l_i = 1$. The factor Γ_M collects the overhead incurred when converting the fixed-horizon UCB-AA analysis into an anytime procedure via geometric restarts. In particular, it reflects that the final guarantee is obtained by aggregating bounds across stages up to L, which introduces an additional logarithmic dependence relative to the fixed-M setting.

Lemma A.7. *If Assumption 1 holds and set $N = 10K(M) \cdot n_M$, the regret of UCB-AA satisfies:*

$$\mathbb{E}R_M(\pi) \leq \sum_{i:\Delta_i(M)\geq\lambda} 36\tilde{C}\left(\frac{12\tilde{C}}{\Delta_i(M)} + \frac{\log(N\Delta_i(M)^2)}{\Delta_i(M)}\right) + \lambda T,$$

for $\lambda \geq 2^{-M+1}$.

Proof of Lemma A.7. By definition of $\tilde{m}_i$, we have:

$$\tilde{\Delta}_{\tilde{m}_i} \le \frac{1}{2}\Delta_i(m_i) \le \tilde{\Delta}_{\tilde{m}_i-1} = 2\tilde{\Delta}_{\tilde{m}_i}.$$

For suboptimal arm $i \in G_1(\lambda) \cup G_2(\lambda)$, we have:

$$\begin{aligned}\tilde{S}_i &\le n_{\tilde{m}_i}\Delta_i(m_i) \le 4\left\lceil\frac{2\log(N\tilde{\Delta}^2_{\tilde{m}_i})}{\tilde{\Delta}^2_{\tilde{m}_i}}\right\rceil\tilde{\Delta}_{\tilde{m}_i} \le \frac{8\log(N\tilde{\Delta}^2_{\tilde{m}_i})}{\tilde{\Delta}_{\tilde{m}_i}} + 4\tilde{\Delta}_{\tilde{m}_i}\\ &\le \frac{32\log(N\Delta_i(m_i)^2)}{\Delta_i(m_i)} + 4\Delta_i(m_i) \le \frac{32\tilde{C}\log(N\Delta_i(M)^2)}{\Delta_i(M)} + 4\Delta_i(M).\end{aligned}$$

For suboptimal arm $i \in G_3(\lambda)$, we have:

$$\begin{aligned}\tilde{S}_i &\le n_{m_i-1}\Delta_i(m_i-1) \le 2\tilde{\Delta}_{m_i-1}\left\lceil\frac{2\log(N\tilde{\Delta}^2_{m_i-1})}{\tilde{\Delta}^2_{m_i-1}}\right\rceil \le \frac{4\log(N\tilde{\Delta}^2_{m_i-1})}{\tilde{\Delta}_{m_i-1}} + 2\tilde{\Delta}_{m_i-1}\\ &\le \frac{4\log(N\tilde{\Delta}^2_{m_i-1})}{\tilde{\Delta}_{m_i-1}} + 2\tilde{\Delta}_{m_i-1} = \frac{2\log(4N\tilde{\Delta}^2_{m_i})}{\tilde{\Delta}_{m_i}} + 4\tilde{\Delta}_{m_i} \le \frac{8\log(N\tilde{\Delta}^2_{m_i})}{\tilde{\Delta}_{m_i}} + 4\tilde{\Delta}_{m_i}\\ &\le \frac{32\log(N\Delta_i(m_i)^2)}{\Delta_i(m_i)} + 4\Delta_i(m_i) \le \frac{32\tilde{C}\log(N\Delta_i(M)^2)}{\Delta_i(M)} + 4\Delta_i(M).\end{aligned}$$

Besides,

$$\frac{K(m_i)\cdot R_i}{N\tilde{\Delta}^2_{\tilde{m}_i}} \le \frac{16K(m_i)\cdot n_M\Delta_i(M)}{N\Delta_i(m_i)^2} \le \frac{16\tilde{C}^2K(M)\cdot n_M}{N\Delta_i(M)} \le \frac{16\tilde{C}^2}{\Delta_i(M)}.$$

We have a simplified upper bound:

$$\begin{aligned}\mathbb{E}R_M(\pi) &\le \sum_{i:\Delta_i(M)\ge\lambda}\left[\tilde{S}_i + \frac{4\left(K(m_i)+5\right)\cdot R_i}{N\tilde{\Delta}^2_{\tilde{m}_i}}\right] + T\max_{i:\Delta_i(M)\le\lambda}\Delta_i(M)\\ &\le \sum_{i:\Delta_i(M)\ge\lambda}\left[\frac{384\tilde{C}^2}{\Delta_i(M)} + \frac{36\tilde{C}\log(N\Delta_i(M)^2)}{\Delta_i(M)}\right] + \lambda T\\ &\le \sum_{i:\Delta_i(M)\ge\lambda} 36\tilde{C}\left(\frac{12\tilde{C}}{\Delta_i(M)} + \frac{\log(N\Delta_i(M)^2)}{\Delta_i(M)}\right) + \lambda T,\end{aligned}$$

where constants are enlarged in the last line, and the stability constant is taken to satisfy $\tilde{C} \ge 1$ without loss of generality. □

We then prove a lemma to characterize the strict upper bounds of N and n_M.

Lemma A.8. *Let $K \ge 2$ and $y > 1$ be positive real numbers. If the inequality $y \le 30K\ln y$ holds, then y is bounded by:*

$$y \le 420K\ln K.$$

Remark A.1. *If* $N = 10Kn_M = 10K\left\lceil 2 \cdot 4^M \log\left(\frac{N}{4^M}\right)\right\rceil$ *, then* $N \le 30K \cdot 4^M \log\left(\frac{N}{4^M}\right)$*. Let* $y = \frac{N}{4^M}$ *in Lemma A.8, we have:*

$$N \le 420K\log(K) \cdot 4^M \quad \textit{and} \quad n_M \le 42\log(K) \cdot 4^M.$$

Proof of Lemma A.8. Let $f(t) = t - 30K\ln t$. The premise of the lemma is $f(y) \le 0$.The derivative is $f'(t) = 1 - \frac{30K}{t}$. $f(t)$ has a global minimum at $t = 30K$. Since $K \ge 2$, the minimum value $f(30K) = 30K(1 - \ln(30K)) < 0$. Therefore, $f(t) = 0$ has two roots, $t_1 < 30K < t_2$. The inequality $f(t) \le 0$ holds on the interval $[t_1, t_2]$.

We show that $t_0 = 60K\ln(60K)$ is an upper bound for t_2 by proving $f(t_0) > 0$.

$$f(t_0) = 30K[\ln(60K) - \ln(\ln(60K))]$$

Let $x = 60K$. Since $K \ge 2$, $x \ge 120 > e$, which implies $\ln x > \ln(\ln x)$. Thus, $f(t_0) > 0$. Since $t_0 > 30K$ and $f(t_0) > 0$, t_0 must be greater than the larger root t_2. It follows that $y \le t_2 < t_0 = 60K\ln(60K)$. Then,

$$y < 60K\ln(60K) < 60K(7\ln K) = 420K\ln K$$

This completes the proof. □

Proof of Theorem A.1. We adopt a geometrically increasing stage schedule $M_l = 2^l$ for $l \ge 1$, and run UCB-AA at stage l with parameter $N_l = 10K(\tilde{M}_l)n_{M_l}$, where $\tilde{M}_l = \sum_{i=1}^{l} M_i$. Define $l_i = \min\{l : 2^{M_l}\Delta_i(\tilde{M}_l) \ge 2\}$ and $L = \min\{l : \tilde{M}_l \ge M\}$.

Then, we focus on the regret caused by some arm $i \in A'(\lambda)$. If $l_i = 1$, the pre-identification contribution is zero. Otherwise, for stage $l = 1, 2, \ldots, l_i - 1$, we have $\Delta_i(\tilde{M}_l) < 2^{-M_l+1}$, which means that the algorithm's rounds may be insufficient to effectively identify the suboptimal interval for arm i in round M_l, potentially preventing its elimination. The regret caused by arm i in these stages has an upper bound:

$$\sum_{l=1}^{l_i-1} \Delta_i(\tilde{M}_l)n_{M_l} \le \Delta_i(\tilde{M}_{l_i-1}) \sum_{l=1}^{l_i-1} n_{M_l} \le 42\Delta_i(\tilde{M}_{l_i-1}) \sum_{l=1}^{l_i-1} \log(K(\tilde{M}_{l_i-1})) \cdot 4^{2^l}$$

$$
\begin{aligned}
&\leq 42\Delta_i(\tilde{M}_{l_i-1})\log(K(\tilde{M}_{l_i-1}))\sum_{l=1}^{l_i-1}\cdot 4^{2^l} \\
&\leq 84\Delta_i(\tilde{M}_{l_i-1})\log(K(\tilde{M}_{l_i-1}))\cdot 4^{2^{l_i-1}} \\
&\leq \begin{cases} \dfrac{336\log(K(\tilde{M}_{l_i-1}))}{\Delta_i(\tilde{M}_{l_i-1})}, & \tilde{M}_{l_i-1}\geq t_i, \\ 0, & \tilde{M}_{l_i-1}<t_i, \end{cases} \\
&\leq U_i.
\end{aligned}
$$

For stage $l = l_i, l_i+1, \ldots, L$, we have $\Delta_i(\tilde{M}_l) \geq 2^{-M_l+1}$, then

$$
N_l\Delta_i(\tilde{M}_l)^2 \geq 4^{M_l}K(\tilde{M}_l)\Delta_i(\tilde{M}_l)^2 \geq K(\tilde{M}_l)(2^{M_l}\Delta_i(\tilde{M}_l))^2 \geq 4K(\tilde{M}_l) \geq 2
$$

Under the auxiliary completed-stage convention, the terminal stage contains no new arrivals after the actual terminal round M. Hence the number of arms available throughout the completed terminal stage is bounded by $K(M)$. For all earlier stages $l < L$, $\tilde{M}_l < M$, so $K(\tilde{M}_l) \leq K(M)$ as well. Thus, in the stagewise bounds below we may uniformly replace the stage-specific arm-count factors by $K(M)$. If Assumption 1 holds, we have $\Delta_i(\tilde{M}_l) \leq \Delta_i(M) \leq \tilde{C}\Delta_i(m_i)$. Therefore, using Lemma A.7, the regret caused by arm i in these stages has an upper bound:

$$
\begin{aligned}
&\sum_{l=l_i}^{L} 36\tilde{C}\left(\frac{12\tilde{C}}{\Delta_i(\tilde{M}_l)}+\frac{\log(N_l\Delta_i(\tilde{M}_l)^2)}{\Delta_i(\tilde{M}_l)}\right) \leq \sum_{l=l_i}^{L} 684\tilde{C}^2\frac{\log(N_l\Delta_i(\tilde{M}_l)^2)}{\Delta_i(\tilde{M}_l)} \\
&\leq \frac{684\tilde{C}^2}{\Delta_i(\tilde{M}_{l_i})}\sum_{l=l_i}^{L}\log\Big(N_l\Delta_i(M)^2\Big) \\
&\leq \frac{684\tilde{C}^2}{\Delta_i(\tilde{M}_{l_i})}\sum_{l=l_i}^{L}\log(N_l) \\
&\leq \frac{684\tilde{C}^2}{\Delta_i(\tilde{M}_{l_i})}\sum_{l=l_i}^{L}\log\Big(420K(M)\log(K(M))\cdot 4^{M_l}\Big) \\
&\leq \frac{684\tilde{C}^2}{\Delta_i(\tilde{M}_{l_i})}\left[\log_2(M+2)\log\{420K(M)\log K(M)\}+2(M+2)\log 4\right] \\
&= \frac{684\tilde{C}^2\,\Gamma_M}{\Delta_i(\tilde{M}_{l_i})}.
\end{aligned}
$$

The penultimate inequality uses $L \leq \log_2(M+2)$ and $\sum_{l=1}^{L} M_l = \tilde{M}_L < 2(M+2)$, which follow from the definition of L.

For arms not belonging to $A'(\lambda)$, there exists an upper bound λT. Therefore, we obtain an upper bound for online UCB-AA:

$$\sum_{i:\Delta_i(M)\geq\lambda} \left[U_i + \frac{684\tilde{C}^2\Gamma_M}{\Delta_i(\tilde{M}_{l_i})} \right] + \lambda T. \tag{38}$$

We now relate the gap at the identification stage, $\Delta_i(\tilde{M}_{l_i})$, to the final gap $\Delta_i(M)$. Assumption 1 imposes a stability condition on the environment: $\Delta_i(M) \leq \tilde{C}\Delta_i(m_i)$. The online algorithm performs checks at the end of each geometrically increasing stage $M_l = 2^l$. $\tilde{M}_{l_i}$ is the cumulative number of rounds at stage l_i where the arm is actually identified, i.e.$\Delta(\tilde{M}_{l_i}) \geq \frac{2}{2^{\tilde{M}_{l_i}}} > \frac{2}{2^{\tilde{M}_{l_i}}}$. The theoretical identification round m_i represents the *earliest possible* round for elimination. Due to the discrete and sparse nature of the online checks, the actual elimination in the online algorithm cannot happen before m_i. Thus, we must have $\tilde{M}_{l_i} \geq m_i$. Since the suboptimality gap $\Delta_i(m)$ is non-decreasing with respect to round m, it follows that:

$$\Delta_i(\tilde{M}_{l_i}) \geq \Delta_i(m_i).$$

Then, we derive the lower bound for the identifiable gap:

$$\frac{1}{\Delta_i(\tilde{M}_{l_i})} \leq \frac{\tilde{C}}{\Delta_i(M)}. \tag{39}$$

Substituting (39) into (38) yields:

$$\sum_{i:\Delta_i(M)\geq\lambda} \left(U_i + \frac{684\tilde{C}^3\Gamma_M}{\Delta_i(M)} \right) + \lambda T.$$

□

Proof of Corollary 2. **The Analysis of R_M.** We derive the gap-independent bound by separating the pre-identification and post-identification contributions. For the post-identification

contribution, we use the second term in Theorem A.1. For the pre-identification contribution, however, we do not apply the crude bound U_i on the small-gap part; instead, we re-bound the actual regret accumulated before arm i becomes identifiable. Let this actual pre-identification regret be denoted by $R_U^{(i)}$. The terms with $l_i = 1$ are zero by definition, so it is enough to consider arms with $l_i > 1$. We distinguish two cases:

$$\begin{aligned}\sum_{i:\Delta_i(M)\geq\lambda} R_U^{(i)} &= \sum_{\substack{i:\Delta_i(M)\geq\lambda,\, l_i>1,\\ \Delta_i(\tilde{M}_{l_i-1})\leq\lambda}} R_U^{(i)} + \sum_{\substack{i:\Delta_i(M)\geq\lambda,\, l_i>1,\\ \Delta_i(\tilde{M}_{l_i-1})\geq\lambda}} R_U^{(i)} \\ &\leq \lambda T + \sum_{\substack{i:\Delta_i(M)\geq\lambda,\, l_i>1,\\ \Delta_i(\tilde{M}_{l_i-1})\geq\lambda}} \frac{336\log(K(M))I_{\{\tilde{M}_{l_i-1}>t_i\}}}{\Delta_i(\tilde{M}_{l_i-1})} \\ &\leq \lambda T + \frac{336K(M)\log(K(M))}{\lambda}\end{aligned}$$

In the first inequality, the term λT comes from a direct bound on the actual regret in the small-gap case $\Delta_i(\tilde{M}_{l_i-1}) \leq \lambda$: by monotonicity of the gaps, all earlier pre-identification pulls of such arms have instantaneous gap at most λ, and their total number is at most the sampling budget T. For the second term, by (39),

$$\sum_{i:\Delta_i(M)\geq\lambda} \frac{684\tilde{C}^3\Gamma_M}{\Delta_i(M)} \leq \frac{684\tilde{C}^3K(M)\Gamma_M}{\lambda}.$$

Combining the two bounds gives

$$\mathbb{E}R_M(\pi) \leq \frac{CK(M)\{\log K(M) + \Gamma_M\}}{\lambda} + 2\lambda T,$$

for a constant $C > 0$. In the present setting, the expected sampling budget T dominates the round and arm-count scales, so the logarithmic restart factor satisfies $\log K(M)+\Gamma_M \lesssim \log T$. Taking

$$\lambda \asymp \sqrt{\frac{K(M)\log T}{T}},$$

provided this choice satisfies the admissibility condition $\lambda \geq 2^{-M_L+1}$, yields

$$\mathbb{E}R_M(\pi) \lesssim \sqrt{K(M)T\log T}.$$

The Analysis of R_{T_0}. Decompose the online regret stagewise: let $T_0^{(l)}$ be the number of pulls made during stage l, and let $R^{(l)}_{T_0^{(l)}}(\pi)$, $R^{(l)}_{M_l}(\pi)$, and $D^{(l)}_{M_l}(\pi)$ denote, respectively, the dynamic regret, roundwise regret, and delay regret of the restarted UCB-AA run in that stage. For the terminal stage, these quantities are interpreted through the completed-stage upper-bound convention described above.

$$R_{T_0}(\pi) \le \sum_{l=1}^{L} R^{(l)}_{T_0^{(l)}}(\pi) = \sum_{l=1}^{L} \Big(R^{(l)}_{M_l}(\pi) + D^{(l)}_{M_l}(\pi) \Big).$$

Applying Proposition 1 on each stage and summing over l yields

$$\mathbb{E} \sum_{l=1}^{L} D^{(l)}_{M_l}(\pi) \lesssim \sum_{l=1}^{L} \mathbb{E} R^{(l)}_{M_l}(\pi) = \mathbb{E} R_M(\pi).$$

Hence

$$\mathbb{E} R_{T_0}(\pi) \lesssim \mathbb{E} R_M(\pi) \lesssim \sqrt{K(M)\, T \, \log T}.$$

□

A.6 Proof of Theorem 3

Proof of Theorem 3. Fix $i \in \{1, \dots, K\}$, and define

$$m_i(T) := \left\lfloor \frac{\log T}{8 C_i \Delta_i^2} \right\rfloor, \qquad A_i := \{ N_i(T) \le m_i(T) \}.$$

We first bound the probability of A_i under the alternative instance $\nu^{(i)}$. Under $\nu^{(i)}$, arm i has mean

$$\mathbb{E}_{Q_i}[X] = r^\star + \eta_i > r^\star,$$

whereas every other arm has mean at most $r^\star$. Therefore arm i is uniquely optimal under $\nu^{(i)}$, and each round in which the learner does not pull arm i incurs regret at least η_i. On the event A_i, the learner pulls arm i at most $m_i(T)$ times over the whole horizon, and hence fails to pull arm i at least $T - m_i(T)$ times. Thus

$$R_T(\pi; \nu^{(i)}) \ge \eta_i \Big(T - m_i(T) \Big) \mathbf{1}_{A_i}.$$

Taking expectations gives

$$\mathbb{E}_{\nu^{(i)}}[R_T(\pi;\nu^{(i)})] \geq \eta_i\Big(T - m_i(T)\Big)\,\mathbb{P}_{\nu^{(i)}}(A_i),$$

and therefore

$$\mathbb{P}_{\nu^{(i)}}(A_i) \leq \frac{\mathbb{E}_{\nu^{(i)}}[R_T(\pi;\nu^{(i)})]}{\eta_i\Big(T - m_i(T)\Big)}.$$

Since $\eta_i > 0$ is fixed, $m_i(T) = O(\log T) = o(T)$, and

$$\mathbb{E}_{\nu^{(i)}}[R_T(\pi;\nu^{(i)})] = o(T^{1/2})$$

by uniform goodness (taking $a = 1/2$), it follows that

$$\mathbb{P}_{\nu^{(i)}}(A_i) = o(T^{-1/2}).$$

In particular, for all sufficiently large T,

$$\mathbb{P}_{\nu^{(i)}}(A_i) \leq T^{-1/3}. \tag{40}$$

Next, let $\mathbb{P}^{(0)}$ and $\mathbb{P}^{(i)}$ denote the full-history laws under $\nu^{(0)}$ and $\nu^{(i)}$, respectively. Since the two instances differ only in the reward law of arm i, Lemma A.5 yields

$$\mathrm{KL}(\mathbb{P}^{(0)}, \mathbb{P}^{(i)}) = \mathbb{E}_{\nu^{(0)}}[N_i(T)]\, D(P_i \| Q_i) \leq C_i \Delta_i^2\, \mathbb{E}_{\nu^{(0)}}[N_i(T)].$$

Hence

$$\mathrm{KL}(\mathbb{P}^{(0)}, \mathbb{P}^{(i)}) \leq C_i \Delta_i^2\, \mathbb{E}_{\nu^{(0)}}[N_i(T)]. \tag{41}$$

We now lower bound the same KL divergence by considering the event A_i. There are two cases.

Case 1:

$$\mathbb{E}_{\nu^{(0)}}[N_i(T)] \geq \frac{1}{2} m_i(T).$$

Since

$$m_i(T) = \left\lfloor \frac{\log T}{8 C_i \Delta_i^2} \right\rfloor,$$

we have, for all sufficiently large T,

$$m_i(T) \geq \frac{\log T}{16 C_i \Delta_i^2}.$$

Therefore

$$\mathbb{E}_{\nu^{(0)}}[N_i(T)] \geq \frac{1}{32 C_i} \frac{\log T}{\Delta_i^2}.$$

Case 2:

$$\mathbb{E}_{\nu^{(0)}}[N_i(T)] < \frac{1}{2} m_i(T).$$

By Markov's inequality,

$$\mathbb{P}_{\nu^{(0)}}(A_i^c) = \mathbb{P}_{\nu^{(0)}}(N_i(T) > m_i(T)) \leq \frac{\mathbb{E}_{\nu^{(0)}}[N_i(T)]}{m_i(T)} < \frac{1}{2}.$$

Hence

$$\mathbb{P}_{\nu^{(0)}}(A_i) \geq \frac{1}{2}. \tag{42}$$

Applying Lemma A.1 and Lemma A.2, we obtain

$$\mathrm{KL}(\mathbb{P}^{(0)}, \mathbb{P}^{(i)}) \geq \mathrm{kl}\left(\mathbb{P}_{\nu^{(0)}}(A_i), \mathbb{P}_{\nu^{(i)}}(A_i)\right) \geq \mathbb{P}_{\nu^{(0)}}(A_i) \log \frac{1}{\mathbb{P}_{\nu^{(i)}}(A_i)} - \log 2.$$

Using (40) and (42), we get

$$\mathrm{KL}(\mathbb{P}^{(0)}, \mathbb{P}^{(i)}) \geq \frac{1}{2} \cdot \frac{1}{3} \log T - \log 2 = \frac{1}{6} \log T - \log 2.$$

Combining this with (41), we obtain

$$C_i \Delta_i^2 \, \mathbb{E}_{\nu^{(0)}}[N_i(T)] \geq \frac{1}{6} \log T - \log 2.$$

Therefore, for all sufficiently large T,

$$\mathbb{E}_{\nu^{(0)}}[N_i(T)] \geq c_1 \frac{\log T}{C_i \Delta_i^2}$$

for some absolute constant $c_1 > 0$.

Combining the two cases, we conclude that there exists an absolute constant $c_0 > 0$ such that

$$\mathbb{E}_{\nu^{(0)}}[N_i(T)] \geq c_0 \frac{\log T}{C_i \Delta_i^2} \qquad \text{for every } i = 1, \ldots, K.$$

Under the baseline instance $\nu^{(0)}$, arm 0 is optimal and arm i has gap Δ_i. Hence each pull of arm i incurs regret exactly Δ_i, and therefore

$$R_T(\pi; \nu^{(0)}) = \sum_{i=1}^{K} \Delta_i N_i(T).$$

Taking expectations yields

$$\mathbb{E}_{\nu^{(0)}}[R_T(\pi; \nu^{(0)})] = \sum_{i=1}^{K} \Delta_i \, \mathbb{E}_{\nu^{(0)}}[N_i(T)] \geq c_0 \sum_{i=1}^{K} \frac{\log T}{C_i \Delta_i}.$$

□

A.7 Proof of Theorem 4

We first prove a lemma that establishes a lower bound on the regret resulting from perturbing a single distractor.

Lemma A.9 (Armwise distractor lower bound under a local alternative condition)**.** *Consider a MAB-AA family with rewards supported on* $[0, 1]$*. For each horizon* T*, let* $\nu^{(0)}$ *be a baseline instance in which arm* 0 *is the unique optimal arm, available from time* 1*, with mean* $r^\star \in (0, 1)$*. Let* $\mathcal{I}_T$ *be a collection of arriving distractor arms. Fix a sequence of indices* $i = i(T) \in \mathcal{I}_T$*, and write* i *in place of* $i(T)$ *below. Let arm* i *arrive at time* $\tau_i \leq T$*, have reward law* P_i*, and mean*

$$r_i = r^\star - \Delta_i, \qquad \Delta_i > 0.$$

Define the post-arrival horizon $H_i(T) := T - \tau_i + 1$*. Assume that* $H_i(T) \to \infty$ *as* $T \to \infty$*. Assume moreover that there exists a law* $Q_i \in \mathcal{P}([0, 1])$ *such that*

$$\eta_i := \mathbb{E}_{Q_i}[X] - r^\star > 0, \qquad d_i := D(P_i \| Q_i) < \infty.$$

Let $\nu^{(i)}$ denote the instance obtained from $\nu^{(0)}$ by replacing only the reward law of arm i from P_i to Q_i, while keeping all other arms and the arrival process unchanged. Set

$$\mathcal{V}_i := \{\nu^{(0)}, \nu^{(i)}\}.$$

If there exist constants $\underline{\Delta}, \underline{\eta} > 0$ such that

$$\Delta_i \geq \underline{\Delta}, \qquad \eta_i \geq \underline{\eta}$$

for all sufficiently large T, and if a policy sequence $\{\pi^{(T)}\}_{T\geq 1}$ is uniformly good on the local family $\mathcal{V}_i$ at the scale $H_i(T)$, in the sense that

$$\sup_{\nu\in\mathcal{V}_i} \mathbb{E}_\nu[R_T(\pi^{(T)};\nu)] = o\Big(H_i(T)^a\Big) \qquad \text{for every } a > 0,$$

then

$$\mathbb{E}_{\nu^{(0)}}[N_i(T)] \geq c\,\frac{\log H_i(T)}{d_i}$$

for all sufficiently large T, where $c > 0$ is an absolute constant. Consequently,

$$\mathbb{E}_{\nu^{(0)}}[R_T(\pi^{(T)};\nu^{(0)})] \geq c\,\frac{\Delta_i}{d_i}\log H_i(T)$$

for all sufficiently large T.

Proof of Lemma A.9. Fix T and define

$$m_i(T) := \left\lfloor \frac{\log H_i(T)}{8d_i} \right\rfloor, \qquad A_i := \{N_i(T) \leq m_i(T)\}.$$

We first bound the probability of A_i under the alternative instance $\nu^{(i)}$. Under $\nu^{(i)}$, arm i has mean

$$\mathbb{E}_{Q_i}[X] = r^\star + \eta_i > r^\star,$$

whereas every other arm has mean at most $r^\star$. Therefore arm i is uniquely optimal after time τ_i, and each post-arrival round in which the learner does not pull arm i incurs regret at least η_i. On the event A_i, the learner pulls arm i at most $m_i(T)$ times over the whole

post-arrival horizon, and hence fails to pull arm i at least $H_i(T) - m_i(T)$ times after its arrival. Thus

$$R_T(\pi^{(T)}; \nu^{(i)}) \geq \eta_i \Big(H_i(T) - m_i(T) \Big) \mathbf{1}_{A_i}.$$

Taking expectations gives

$$\mathbb{E}_{\nu^{(i)}}[R_T(\pi^{(T)}; \nu^{(i)})] \geq \eta_i \Big(H_i(T) - m_i(T) \Big) \, \mathbb{P}_{\nu^{(i)}}(A_i),$$

and therefore

$$\mathbb{P}_{\nu^{(i)}}(A_i) \leq \frac{\mathbb{E}_{\nu^{(i)}}[R_T(\pi^{(T)}; \nu^{(i)})]}{\eta_i \Big(H_i(T) - m_i(T) \Big)}.$$

By Lemma A.4,

$$d_i = D(P_i \| Q_i) \geq 2 \, \mathrm{TV}(P_i, Q_i)^2 \geq 2 \left(\mathbb{E}_{Q_i}[X] - \mathbb{E}_{P_i}[X] \right)^2 = 2(\Delta_i + \eta_i)^2 \geq 2(\underline{\Delta} + \underline{\eta})^2.$$

Hence $m_i(T) = O(\log H_i(T)) = o(H_i(T))$. Since

$$\mathbb{E}_{\nu^{(i)}}[R_T(\pi^{(T)}; \nu^{(i)})] = o(H_i(T)^{1/2})$$

by uniform goodness (taking $a = 1/2$), it follows that

$$\mathbb{P}_{\nu^{(i)}}(A_i) = o(H_i(T)^{-1/2}).$$

In particular, for all sufficiently large T,

$$\mathbb{P}_{\nu^{(i)}}(A_i) \leq H_i(T)^{-1/3}. \tag{43}$$

Next, let $\mathbb{P}^{(0)}$ and $\mathbb{P}^{(i)}$ denote the full-history laws under $\nu^{(0)}$ and $\nu^{(i)}$, respectively. Since the two instances differ only in the reward law of arm i, Lemma A.5 yields

$$\mathrm{KL}(\mathbb{P}^{(0)}, \mathbb{P}^{(i)}) = \mathbb{E}_{\nu^{(0)}}[N_i(T)] \, D(P_i \| Q_i) = d_i \, \mathbb{E}_{\nu^{(0)}}[N_i(T)]. \tag{44}$$

We now lower bound the same KL divergence by considering the event A_i.

Case 1:

$$\mathbb{E}_{\nu^{(0)}}[N_i(T)] \geq \frac{1}{2} m_i(T).$$

Then, for all sufficiently large T,

$$\mathbb{E}_{\nu^{(0)}}[N_i(T)] \geq \frac{1}{2}\left(\frac{\log H_i(T)}{8d_i} - 1\right) \geq \frac{1}{32}\frac{\log H_i(T)}{d_i}.$$

Case 2:

$$\mathbb{E}_{\nu^{(0)}}[N_i(T)] < \frac{1}{2}m_i(T).$$

By Markov's inequality,

$$\mathbb{P}_{\nu^{(0)}}(A_i^c) = \mathbb{P}_{\nu^{(0)}}(N_i(T) > m_i(T)) \leq \frac{\mathbb{E}_{\nu^{(0)}}[N_i(T)]}{m_i(T)} < \frac{1}{2}.$$

Hence

$$\mathbb{P}_{\nu^{(0)}}(A_i) \geq \frac{1}{2}. \tag{45}$$

Applying Lemma A.1 and Lemma A.2, we obtain

$$\mathrm{KL}(\mathbb{P}^{(0)}, \mathbb{P}^{(i)}) \geq \mathrm{kl}\left(\mathbb{P}_{\nu^{(0)}}(A_i), \mathbb{P}_{\nu^{(i)}}(A_i)\right) \geq \mathbb{P}_{\nu^{(0)}}(A_i)\log\frac{1}{\mathbb{P}_{\nu^{(i)}}(A_i)} - \log 2.$$

Using (43) and (45), we get

$$\mathrm{KL}(\mathbb{P}^{(0)}, \mathbb{P}^{(i)}) \geq \frac{1}{2}\cdot\frac{1}{3}\log H_i(T) - \log 2 = \frac{1}{6}\log H_i(T) - \log 2.$$

Combining this with (44), we obtain

$$d_i\,\mathbb{E}_{\nu^{(0)}}[N_i(T)] \geq \frac{1}{6}\log H_i(T) - \log 2.$$

Therefore, for all sufficiently large T,

$$\mathbb{E}_{\nu^{(0)}}[N_i(T)] \geq c_1\frac{\log H_i(T)}{d_i}$$

for some absolute constant $c_1 > 0$.

Combining the two cases, we conclude that there exists an absolute constant $c > 0$ such that

$$\mathbb{E}_{\nu^{(0)}}[N_i(T)] \geq c\,\frac{\log H_i(T)}{d_i}$$

for all sufficiently large T. Since arm 0 is available from time 1 and is uniquely optimal under the baseline instance $\nu^{(0)}$, each pull of arm i incurs regret exactly

$$r^\star - r_i = \Delta_i.$$

Therefore

$$\mathbb{E}_{\nu^{(0)}}[R_T(\pi^{(T)}; \nu^{(0)})] \geq \Delta_i \, \mathbb{E}_{\nu^{(0)}}[N_i(T)] \geq c \, \frac{\Delta_i}{d_i} \log H_i(T),$$

which completes the proof. □

Now, we prove the theorem.

Proof of Theorem 4. For each $i \in \mathcal{I}_T$, Lemma A.9 applies uniformly, since $H_i(T) \geq H_{\min}(T)$,

$$\sup_{\nu \in \mathcal{V}_i} \mathbb{E}_\nu[R_T(\pi^{(T)}; \nu)] \leq \sup_{\nu \in \mathcal{V}_T} \mathbb{E}_\nu[R_T(\pi^{(T)}; \nu)] = o\Big(H_{\min}(T)^a\Big) = o\Big(H_i(T)^a\Big)$$

for every $a > 0$. Hence, for all sufficiently large T,

$$\mathbb{E}_{\nu^{(0)}}[N_i(T)] \geq c \, \frac{\log H_i(T)}{d_i} \qquad \text{for every } i \in \mathcal{I}_T,$$

where $c > 0$ is an absolute constant independent of i and T.

Under the baseline instance $\nu^{(0)}$, each pull of arm i incurs regret exactly Δ_i. Therefore

$$R_T(\pi^{(T)}; \nu^{(0)}) \geq \sum_{i \in \mathcal{I}_T} \Delta_i N_i(T),$$

and taking expectations yields

$$\mathbb{E}_{\nu^{(0)}}[R_T(\pi^{(T)}; \nu^{(0)})] \geq \sum_{i \in \mathcal{I}_T} \Delta_i \, \mathbb{E}_{\nu^{(0)}}[N_i(T)] \geq c \sum_{i \in \mathcal{I}_T} \frac{\Delta_i}{d_i} \log H_i(T).$$

The corresponding supremum lower bound follows immediately because $\nu^{(0)} \in \mathcal{V}_T$.

For the final simplification, note that

$$\log H_i(T) \geq \log H_{\min}(T) \qquad \text{for every } i \in \mathcal{I}_T.$$

Hence, if $\sup_{i \in \mathcal{I}_T} d_i \leq \overline{d}$, then

$$\mathbb{E}_{\nu^{(0)}}[R_T(\pi^{(T)}; \nu^{(0)})] \geq c \, \underline{\Delta} \, \overline{d}^{-1} \, |\mathcal{I}_T| \, \log H_{\min}(T),$$

which is $\Omega(K_T \log H_{\min}(T))$ when $|\mathcal{I}_T| \asymp K_T$. □

A.8 Proof of Theorem 5

Proof of Theorem 5. Fix any non-anticipating policy π. For each $m \in \{1, \dots, K_T\}$ and each configuration $\tau \in \{-1, +1\}^{K_T - 1}$, let $\nu_\tau^{m,+}$ and $\nu_\tau^{m,-}$ denote the two instances that agree on all blocks except block m, where the challenger arm d_m has mean $\theta_m + \varepsilon_m$ or $\theta_m - \varepsilon_m$, respectively.

Let $\mathcal{R}_m$ denote the random regret accumulated during block m, so that

$$R_T(\pi; \nu) = \mathbb{E}_\nu \left[\sum_{m=1}^{K_T} \mathcal{R}_m \right].$$

Let

$$N_m := \sum_{t \in \mathcal{B}_m} \mathbf{1}\{I_t = d_m\}$$

be the number of pulls of the challenger arm d_m during block m. We first identify the benchmark level in block m. By construction, $\mu(c_m) = \theta_m$. Moreover, for each $j < m$,

$$\mu(c_j) = \theta_j \le \theta_m,$$

because (4) implies that $\{\theta_m\}$ is nondecreasing, and

$$\mu(d_j) \le \theta_j + \varepsilon_j \le \theta_{j+1} \le \theta_m, \qquad j < m.$$

Hence every arm available before the arrival of d_m has mean at most θ_m.

Therefore, under $\nu_\tau^{m,-}$, the challenger arm d_m is suboptimal by exactly ε_m throughout block m, so

$$\mathcal{R}_m \ge \varepsilon_m N_m.$$

Under $\nu_\tau^{m,+}$, the challenger arm d_m is optimal in block m with advantage exactly ε_m, so every pull of an arm other than d_m incurs regret at least ε_m:

$$\mathcal{R}_m \ge \varepsilon_m (H_m - N_m).$$

Taking expectations yields

$$\mathbb{E}_\tau^{m,-}[\mathcal{R}_m] + \mathbb{E}_\tau^{m,+}[\mathcal{R}_m] \ge \varepsilon_m \Big(\mathbb{E}_\tau^{m,-}[N_m] + \mathbb{E}_\tau^{m,+}[H_m - N_m] \Big). \tag{46}$$

Set $\alpha_m := N_m/H_m \in [0,1]$. To compare the two expectations, consider the prefix transcript up to the end of block m,

$$H_{s_m} := (I_1, X_1, \ldots, I_{s_m}, X_{s_m}),$$

and let $P^{m,+}_{\tau,[m]}$ and $P^{m,-}_{\tau,[m]}$ denote the laws of H_{s_m} under $\nu^{m,+}_\tau$ and $\nu^{m,-}_\tau$, respectively. Since α_m is measurable with respect to H_{s_m} and takes values in $[0,1]$, the variational characterization of total variation gives

$$\mathbb{E}^{m,-}_\tau[\alpha_m] + \mathbb{E}^{m,+}_\tau[1-\alpha_m] \geq 1 - \mathrm{TV}(P^{m,-}_{\tau,[m]}, P^{m,+}_{\tau,[m]}).$$

Substituting into (46), we obtain

$$\mathbb{E}^{m,-}_\tau[\mathcal{R}_m] + \mathbb{E}^{m,+}_\tau[\mathcal{R}_m] \geq H_m\varepsilon_m\Big(1 - \mathrm{TV}(P^{m,-}_{\tau,[m]}, P^{m,+}_{\tau,[m]})\Big). \tag{47}$$

The two instances $\nu^{m,-}_\tau$ and $\nu^{m,+}_\tau$ differ only in the reward law of arm d_m, and only pulls of d_m up to time s_m contribute to the KL divergence between the corresponding prefix laws. Therefore, by Lemma A.5,

$$\mathrm{kl}(P^{m,-}_{\tau,[m]} \,\|\, P^{m,+}_{\tau,[m]}) = \mathbb{E}^{m,-}_\tau[N_m]\,\mathrm{kl}(\theta_m - \varepsilon_m, \theta_m + \varepsilon_m) \leq H_m\,\mathrm{kl}(\theta_m - \varepsilon_m, \theta_m + \varepsilon_m),$$

since $N_m \leq H_m$. Pinsker's inequality then yields

$$\mathrm{TV}(P^{m,-}_{\tau,[m]}, P^{m,+}_{\tau,[m]}) \leq \sqrt{\frac{1}{2}H_m\,\mathrm{kl}(\theta_m - \varepsilon_m, \theta_m + \varepsilon_m)}.$$

Combining this with (47), we get

$$\mathbb{E}^{m,-}_\tau[\mathcal{R}_m] + \mathbb{E}^{m,+}_\tau[\mathcal{R}_m] \geq H_m\varepsilon_m\left(1 - \sqrt{\frac{1}{2}H_m\,\mathrm{kl}(\theta_m - \varepsilon_m, \theta_m + \varepsilon_m)}\right)_+.$$

Averaging over the two signs in block m gives

$$2^{-K_T}\sum_{\sigma\in\{-1,+1\}^{K_T}} \mathbb{E}_{\nu_\sigma}[\mathcal{R}_m] \geq \frac{1}{2}H_m\varepsilon_m\left(1 - \sqrt{\frac{1}{2}H_m\,\mathrm{kl}(\theta_m - \varepsilon_m, \theta_m + \varepsilon_m)}\right)_+.$$

Summing over $m = 1, \ldots, K_T$, we have

$$\sup_{\nu\in\mathcal{V}^{\mathrm{prog}}_T} \mathbb{E}_\nu[R_T(\pi;\nu)] \geq 2^{-K_T}\sum_{\sigma\in\{-1,+1\}^{K_T}} \mathbb{E}_{\nu_\sigma}[R_T(\pi;\nu_\sigma)]$$

$$\geq \frac{1}{2}\sum_{m=1}^{K_T} H_m\varepsilon_m\left(1-\sqrt{\frac{1}{2}H_m\,\mathrm{kl}(\theta_m-\varepsilon_m,\theta_m+\varepsilon_m)}\right)_+. \tag{48}$$

Next, by (3) and Lemma A.3,

$$\mathrm{kl}(\theta_m-\varepsilon_m,\theta_m+\varepsilon_m)\leq\frac{(2\varepsilon_m)^2}{\gamma(1-\gamma)}=\frac{4\varepsilon_m^2}{\gamma(1-\gamma)}.$$

Substituting this bound into (48) yields (5). Finally, if $H_m\varepsilon_m^2\leq\kappa\,\gamma(1-\gamma)$ for every m, then

$$\sqrt{\frac{2H_m\varepsilon_m^2}{\gamma(1-\gamma)}}\leq\sqrt{2\kappa},$$

and each summand in (5) is bounded below by

$$\frac{1}{2}H_m\varepsilon_m\left(1-\sqrt{2\kappa}\right)=c_\kappa H_m\varepsilon_m.$$

This proves (6). □

B Analysis of Examples

Before analyzing the examples, we first establish an upper bound for UCB-AA without pre-elimination mechanism. At this point, the method can be viewed as a direct generalization of the improved UCB algorithm in Auer and Ortner (2010).

Lemma B.1 (Regret upper bound of UCB-AA without pre-elimination)**.** *Up to round M,*

$$\mathbb{E}R_M(\pi)\leq\sum_{i:\Delta_i(M)\geq\lambda}\left[S_i+\frac{4K(m_i)\cdot R_i}{N\tilde{\Delta}_{m_i}^2}\right]+T\max_{i:\Delta_i(M)\leq\lambda}\Delta_i(M).$$

for $\lambda\geq\frac{1}{2^M}$, where T is the total number of pulls, and

$$R_i\coloneqq n_{t_i}\Delta_i(t_i)+\sum_{m=t_i+1}^{M}(n_m-n_{m-1})\Delta_i(m),$$
$$S_i\coloneqq n_{t_i}\Delta_i(t_i)+\sum_{m=t_i+1}^{m_i}(n_m-n_{m-1})\Delta_i(m).$$

It's obvious that $S_i\leq R_i$ when $\Delta_i(M)\geq\lambda$.

Algorithm 3 UCB-AA without pre-elimination mechanism

Input: Initial arm set A_0, number of rounds M, parameter $N > 0$.

Initialization: Set $\tilde{\Delta}_0 \coloneqq 1$, and $B_0 \coloneqq \varnothing$, $C_{-1} \coloneqq \varnothing$, $T_0 \coloneqq 0$.

for rounds $m = 0, 1, 2, \cdots, M$ **do**

Arm update and selection:

Receive the new arm set A_m, $B_m = C_{m-1} \bigcup A_m$.

choose each arm in B_m until total number of times it has been chosen is

$$n_m \coloneqq \left\lceil \frac{2\log(N\tilde{\Delta}_m^2)}{\tilde{\Delta}_m^2} \right\rceil.$$

After each arm pull, let $T_0 \leftarrow T_0 + 1$.

Arm elimination:

delete all arms i from B_m for which

$$\left\{ \hat{r}_i + \sqrt{\frac{\log(N\tilde{\Delta}_m^2)}{2n_m}} \right\} < \max_{j \in B_m} \left\{ \hat{r}_j - \sqrt{\frac{\log(N\tilde{\Delta}_m^2)}{2n_m}} \right\}.$$

in order to obtain C_m. Here $\hat{r}_j$ is the average reward obtained from arm j.

Reset $\tilde{\Delta}_m$:

Set $\tilde{\Delta}_{m+1} \coloneqq \tilde{\Delta}_m / 2$.

end for

Output T_0.

Proof. By definition of m_i, we know $\tilde{\Delta}_{m_i} < \frac{1}{2}\Delta_i(m_i)$, and

$$\frac{4}{\Delta_i(m_i)} < \frac{2}{\tilde{\Delta}_{m_i}} = \frac{1}{\tilde{\Delta}_{m_i+1}} = 2^{m_i+1}.$$

Now we consider suboptimal arms in $A'(\lambda) \coloneqq \{i \in A(M) : \Delta_i(M) > \lambda\}$ for some fixed $\lambda \geq \frac{1}{2^M}$. Then, we first decompose the total regret into the regret for each arm:

$$\mathbb{E}R_M(\pi) = \sum_{i \in A'(\lambda)} \mathbb{E}(R^{(i)}(\pi)) + \sum_{j \notin A'(\lambda)} \mathbb{E}(R^{(j)}(\pi)),$$

where $R^{(i)}(\pi)$ is the regret caused by arm i.

We analyze the regret $\mathbb{E}(R^{(i)}(\pi))$ generated by each suboptimal arm $i \in A'(\lambda)$ by dividing the entire space into three cases. Define the event

$$E_i = \{i \in A'(\lambda) \text{ has not been eliminated in or before round } m_i\},$$
$$F_i = \{*(m_i) \in C_{m_i}\}.$$

Then, we divide the regret into three parts,

$$\mathbb{E}(R^{(i)}(\pi)) = \underbrace{\mathbb{E}\left(R^{(i)}(\pi) \cdot I_{E_i \cap F_i}\right)}_{\text{case (a)}} + \underbrace{\mathbb{E}\left(R^{(i)}(\pi) \cdot I_{E_i^c}\right)}_{\text{case (b1)}} + \underbrace{\mathbb{E}\left(R^{(i)}(\pi) \cdot I_{E_i \cap F_i^c}\right)}_{\text{case (b2)}}.$$

Below, we calculate the upper bounds for the regret of these three parts, respectively.

Case (a): *If $i \in A'(\lambda)$ has not been eliminated in or before round m_i, and $*(m_i) \in C_{m_i}$.* We have $\sqrt{\frac{\log(N\tilde{\Delta}_{m_i}^2)}{2n_{m_i}}} \leq \frac{1}{2}\tilde{\Delta}_{m_i} < \frac{1}{4}\Delta_i(m_i)$ by the definition of m_i and n_m. Notice that if

$$\hat{r}_i \leq r_i + \sqrt{\frac{\log(N\tilde{\Delta}_m^2)}{2n_m}} \tag{49}$$

and

$$\hat{r}_{*(m)} \geq r_{*(m)} - \sqrt{\frac{\log(N\tilde{\Delta}_m^2)}{2n_m}} \tag{50}$$

hold for $m = m_i$, then we have:

$$\hat{r}_i + \sqrt{\frac{\log(N\tilde{\Delta}_{m_i}^2)}{2n_{m_i}}} \leq r_i + 2\sqrt{\frac{\log(N\tilde{\Delta}_{m_i}^2)}{2n_{m_i}}} < r_i + \Delta_i(m_i) - 2\sqrt{\frac{\log(N\tilde{\Delta}_{m_i}^2)}{2n_{m_i}}}$$

$$= r_{*(m_i)} - 2\sqrt{\frac{\log(N\tilde{\Delta}_{m_i}^2)}{2n_{m_i}}} \leq \hat{r}_{*(m_i)} - \sqrt{\frac{\log(N\tilde{\Delta}_{m_i}^2)}{2n_{m_i}}}.$$

Further, by the assumption that $*(m_i)$, $i \in C_{m_i}$ and the elimination rule, arm i will be eliminated. Besides, using the Chernoff-Hoeffding bound, we have for $m = 0, 1, \cdots$:

$$\mathbb{P}\left(\hat{r}_i > r_i + \sqrt{\frac{\log(N\tilde{\Delta}_m^2)}{2n_m}}\right) \leq \frac{1}{N\tilde{\Delta}_m^2}, \tag{51}$$

$$\mathbb{P}\left(\hat{r}_{*(m)} < r_{*(m)} - \sqrt{\frac{\log(N\tilde{\Delta}_m^2)}{2n_m}}\right) \leq \frac{1}{N\tilde{\Delta}_m^2}. \tag{52}$$

Further, the upper bound of the probability that arm i is not eliminated in or before round m_i is $\frac{2}{N\tilde{\Delta}_{m_i}^2}$. Thus, the regret generated by arm i conditioned on case (a) does not exceed

$$R_i \coloneqq n_{t_i}\Delta_i(t_i) + \sum_{m=t_i+1}^{M}(n_m - n_{m-1})\Delta_i(m).$$

Summing over all $i \in A'(\lambda)$, we find the corresponding total expected regret for each suboptimal $i \in A'(\lambda)$ in case (a) is upper bounded by:

$$\sum_{i\in A'(\lambda)} \frac{2}{N\tilde{\Delta}_{m_i}^2} \cdot \left[n_{t_i}\Delta_i(t_i) + \sum_{m=t_i+1}^{M}(n_m - n_{m-1})\Delta_i(m)\right] = \sum_{i\in A'(\lambda)} \frac{2R_i}{N\tilde{\Delta}_{m_i}^2}.$$

Case (b): *For suboptimal arm $i \in A'(\lambda)$, either it is eliminated in or before round m_i, or $*(m_i) \notin C_{m_i}$ with i not eliminated in or before round m_i.*

We divide it into two smaller cases:

Case (b1): *If $i \in A'(\lambda)$ is eliminated in or before round m_i.*

At this point, we know each suboptimal arm $i \in A'(\lambda)$ is eliminated in or before round m_i, so the number of times arm i is pulled does not exceed

$$n_{m_i} = \left\lceil \frac{2\log(N\tilde{\Delta}_{m_i}^2)}{\tilde{\Delta}_{m_i}^2} \right\rceil \leq \left\lceil \frac{2\log\left(N \cdot \frac{\Delta_i^2(m_i)}{4}\right)}{\tilde{\Delta}_{m_i}^2} \right\rceil.$$

Thus, the upper bound of the random regret of arm i in this case is

$$S_i \coloneqq n_{t_i}\Delta_i(t_i) + \sum_{m=t_i+1}^{m_i}(n_m - n_{m-1})\Delta_i(m) \leq R_i$$

Summing over all $i \in A'(\lambda)$, we find that the corresponding total expected regret in case (b1) is upper bounded by

$$\sum_{i\in A'(\lambda)} \left[n_{t_i}\Delta_i(t_i) + \sum_{m=t_i+1}^{m_i} (n_m - n_{m-1})\Delta_i(m) \right] = \sum_{i\in A'(\lambda)} S_i.$$

Case (b2): *If $*(m_i) \notin C_{m_i}$ with i is not eliminated in or before round m_i.*

Note that if (49) and (50) hold at round m_* for j and $*(m_i)$ respectively, then $*(m_i)$ will not be eliminated by j in round m_*. Therefore, the upper bound of the probability that $*(m_i)$ is eliminated by j at round m_* is $\frac{2}{N\tilde{\Delta}^2_{m_*}}$.

Now, under the assumption of case (b2), $*(m_i)$ must have been eliminated by some suboptimal arm j in some round m_* satisfying $t_{*(m_i)} \le m_* \le m_i$, and j should appear before round m_*. We have $\Delta_j(M) > \Delta_{*(m_i)}(M) \ge 0$ and $t_j \le m_* \le m_i$. Thus, we have:

$$\begin{aligned}
&\mathbb{P}\left(*(m_i) \text{ is eliminated by some suboptimal arm } j \text{ in some round } m_*,\ t_{*(m_i)} \le m_* \le m_i\right)\\
=&\sum_{m_*=t_{*(m_i)}}^{m_i} \mathbb{P}\left(*(m_i) \text{ is eliminated by some suboptimal arm } j \text{ in round } m_*\right)\\
\le&\sum_{m_*=t_{*(m_i)}}^{m_i} \sum_{\substack{j:\, t_j\le m_*\\ j\neq *(m_i)}} \mathbb{P}\left(*(m_i) \text{ is eliminated by suboptimal arm } j \text{ in round } m_*\right)\\
\le&\sum_{m_*=t_{*(m_i)}}^{m_i} \sum_{\substack{j:\, t_j\le m_*\\ j\neq *(m_i)}} \frac{2}{N\tilde{\Delta}^2_{m_*}} \le \sum_{m_*=t_{*(m_i)}}^{m_i} \frac{2\left(K(m_i)-1\right)}{N\tilde{\Delta}^2_{m_*}} < \frac{2\left(K(m_i)-1\right)}{N} \sum_{m_*=0}^{m_i} \frac{1}{\tilde{\Delta}^2_{m_*}}\\
=&\frac{2\left(K(m_i)-1\right)}{N} \sum_{m_*=0}^{m_i} 4^{m_*} < \frac{2\left(K(m_i)-1\right)}{N}\cdot 2\cdot 4^{m_i} = \frac{4\left(K(m_i)-1\right)}{N\tilde{\Delta}^2_{m_i}}.
\end{aligned}$$

In case (b2), the random regret generated by arm i does not exceed R_i. Summing over all $i \in A'(\lambda)$, we find the corresponding total expected regret in case (b2) is upper bounded by

$$\sum_{i\in A'(\lambda)} \frac{4\left(K(m_i)-1\right)\cdot R_i}{N\tilde{\Delta}^2_{m_i}}.$$

For all suboptimal arms not belonging to $A'(\lambda)$, the total regret provided by this part can be easily upper-bounded as

$$\mathbb{E}(T_0 \max_{i:\Delta_i(M)\le\lambda} \Delta_i(M)) = T \max_{i:\Delta_i(M)\le\lambda} \Delta_i(M).$$

Therefore, the upper bound of the regret is:

$$\sum_{i:\Delta_i(M)\geq\lambda}\left[S_i+\frac{4K(m_i)\cdot R_i}{N\tilde{\Delta}^2_{m_i}}\right]+T\max_{i:\Delta_i(M)\leq\lambda}\Delta_i(M).$$

The result follows. □

B.1 Analysis of Example 1

If no new arms arrive, i.e. $A_m = \varnothing$ for all $m \geq 1$, the gaps are time-invariant, Assumption 1 holds trivially with stability constant $\widetilde{C} = 1$, and there is no delay regret. When the horizon T is known, we choose the stage parameter $M = \lfloor \frac{1}{2}\log_2 T \rfloor$ and set N via $N = 10K\, n_M(N)$. In this case, for any $\lambda \geq \sqrt{1/T}$, the bound in Theorem 1 simplifies to

$$\mathbb{E}R_T(\pi_{\text{UCB-AA}}) = \mathbb{E}R_M(\pi_{\text{UCB-AA}}) \leq \sum_{i:\,\Delta_i\geq\lambda} 36\left(\frac{12+\log\big(420K\log K\big)}{\Delta_i}+\frac{\log\big(T\Delta_i^2\big)}{\Delta_i}\right) + \lambda T$$

with leading order $\sum_i \frac{\log(T\Delta_i^2)}{\Delta_i}$. Using Lemma A.7, we can easily obtain the conclusion.

B.2 Analysis of Example 2

The suboptimality gap for any arm $i \geq 0$ is given by

$$\Delta_i(m) = \begin{cases} \mu_* - \mu_i = \frac{1}{4} + 2^{-i-2}, & m \geq i \\ 0, & m < i \end{cases}, \quad m \geq 0.$$

A crucial property of this setting is that all suboptimality gaps are lower-bounded by a constant: $\Delta_i(m) > 1/4$ for all i and $m \geq i$.

Analysis of UCB-AA without Pre-elimination.

The definition of the elimination round for arm i is

$$m_i = \min\left\{m \leq M : \tilde{\Delta}_m < \frac{1}{2}\Delta_i(m)\right\}.$$

Substituting $\tilde{\Delta}_m = 2^{-m}$ and the gap expression, we require

$$2^{-m} < \frac{1}{2}\left(\frac{1}{4} + 2^{-(i+2)}\right) = \frac{1}{8} + 2^{-(i+3)}.$$

For any arm i arriving at a round $i \geq 3$, the elimination round is determined by the arrival time, i.e., $m_i = i$. According to Lemma B.1, the dominant term in the regret bound for arm i is

$$S_i = n_{m_i}\Delta_i(m_i) = \left\lceil \frac{2\log(N2^{-2m_i})}{2^{-2m_i}} \right\rceil \Delta_i(m_i).$$

Substituting $m_i = i$ and $\Delta_i(m_i) > 1/4$ yields

$$S_i \geq \frac{2(\log N - 2i\ln 2)}{4^{-i}} \cdot \frac{1}{4} = \frac{1}{2}4^i(\log N - 2i\ln 2).$$

We donate the regret upper bound in Lemma B.1 as B_{A2}. Then,

$$B_{A2} \geq \sum_{i=3}^{M} S_i \geq \frac{1}{2}\sum_{i=3}^{M} 4^i(\log N - 2i\ln 2).$$

We always choose $N = 10K(M)n_M = 10K(M)\left\lceil 2\cdot 4^M \log\left(\frac{N}{4^M}\right)\right\rceil$ in the algorithm, so $N \geq 4^M$. Therefore,

$$\begin{aligned} B_{A2} &\geq \frac{1}{2}\sum_{i=3}^{M} 4^i \cdot 2\ln 2 \cdot (M-i) \\ &= \ln 2 \sum_{i=3}^{M} 4^i(M-i) = 4^M \sum_{j=0}^{M-3} \frac{j}{4^j} = \Theta(4^M). \end{aligned}$$

We know

$$4^M \leq n_M \leq T \leq K(M)\cdot n_M = \Theta(M\log(M)4^M).$$

Therefore, in the sense of ignoring logarithmic terms, we have $B_{A2} = \tilde{\Omega}(T)$.

Analysis of UCB-AA.

The identification phase $\tilde{m}_i$ is determined solely by the gap size: $\tilde{\Delta}_{\tilde{m}_i} < \Delta_i(m)/2$. Given $\Delta_i(m) > 1/4$, the condition $2^{-\tilde{m}_i} < 1/8 + \epsilon$ is satisfied at a constant phase index $\tilde{m}_i = 3$, independent of i. Then, for all $i \geq 0$, we have

$$\tilde{S}_i = n_{\tilde{m}_i}\Delta_i(M) = \left\lceil \frac{2\log(N2^{-6})}{2^{-6}} \right\rceil \left(\frac{1}{4} + 2^{-i-2}\right) \leq 64\log(N),$$

$$\frac{K(m_i)\cdot R_i}{N\tilde{\Delta}^2_{\tilde{m}_i}} = \left(\frac{1}{4}+2^{-i-2}\right)\left\lceil\frac{2\log(N2^{-2M})}{2^{-2M}}\right\rceil\frac{i+2}{2^{-6}N} \le \frac{16\cdot 4^M M}{N}\log(4^{-M}N).$$

Choose $N = 10K(M)n_M = 10K(M)\left\lceil 2\cdot 4^M\log\left(\frac{N}{4^M}\right)\right\rceil$ in the algorithm as before; then $N=\tilde{\Theta}(4^M K(M))=\tilde{\Theta}(4^M M)$. We donate the regret upper bound in Theorem 1 as B_{AA}.

$$\begin{aligned} B_{AA} &\le \sum_{i=0}^{M}\left(\tilde{S}_i + 24\frac{K(m_i)\cdot R_i}{N\tilde{\Delta}^2_{\tilde{m}_i}}\right) \\ &\le (M+2)\left(64\log(N)+24\frac{16\cdot 4^M M}{N}\log(4^{-M}N)\right) \\ &= \tilde{\Theta}(M^2)=\tilde{\Theta}(K(M)\log T). \end{aligned}$$

B.3 Analysis of Example 3

Let $K := K(M) = 2M+1$. Recall that

$$\tilde{\Delta}_m = 2^{-m}, \qquad n_m = \left\lceil\frac{2\log(N\tilde{\Delta}^2_m)}{\tilde{\Delta}^2_m}\right\rceil.$$

Set

$$\alpha_M := \log(N\tilde{\Delta}^2_M) = \log(N4^{-M}).$$

Since

$$n_M = \Theta(4^M\alpha_M)$$

and $N = 10Kn_M$, we have

$$N = \Theta(K\,4^M\alpha_M).$$

Substituting this into the definition of α_M gives

$$\alpha_M = \log\Big(\Theta(K\alpha_M)\Big) = \Theta(\log K) = \Theta(\log M).$$

Consequently,

$$n_M = \Theta(4^M\log M), \qquad N = \Theta(M\,4^M\log M).$$

More generally, for each $m \le M$,

$$n_m = \Theta\Big(4^m(\log M + M - m)\Big). \tag{53}$$

We next compute the suboptimality gaps. For the arm c_t,

$$\Delta_{c_t}(m) = \theta_m - \theta_{t-1} = 2^{-t-1} - 2^{-m-2}, \qquad m \geq t.$$

In particular,

$$\Delta_{c_t}(M) \in [2^{-t-2},\, 2^{-t-1}].$$

For the arm d_t, which is optimal at round t and becomes suboptimal afterward,

$$\Delta_{d_t}(m) = \theta_m - \theta_t = 2^{-t-2} - 2^{-m-2}, \qquad m \geq t+1,$$

so

$$\Delta_{d_t}(M) \in [2^{-t-3},\, 2^{-t-2}] \qquad (t \leq M-1).$$

Now consider the elimination rounds from Section 3.3. For c_t, one checks directly that

$$m_{c_t} = t+3, \qquad \tilde{m}_{c_t} = t+3.$$

Similarly, for d_t,

$$m_{d_t} = t+4, \qquad \tilde{m}_{d_t} = t+4.$$

Thus both families belong to the class G_2.

Take $\lambda = 2^{-M+1}$, which is the smallest admissible threshold in Theorem 1. Then all but the last $O(1)$ suboptimal arms satisfy $\Delta_i(M) \geq \lambda$.

We first bound the leading term $\sum_i \tilde{S}_i$ in Theorem 1. Since $c_t \in G_2$,

$$\tilde{S}_{c_t} = n_t \Delta_{c_t}(t) + \sum_{m=t+1}^{t+3} (n_m - n_{m-1}) \Delta_{c_t}(m).$$

For $m \in \{t, t+1, t+2, t+3\}$ we have $\Delta_{c_t}(m) = \Theta(2^{-t})$, hence

$$\tilde{S}_{c_t} = O\Big(2^{-t} n_{t+3}\Big) = O\bigg((\log M + M - t) 2^t\bigg),$$

where we used (53).

Similarly,

$$\tilde{S}_{d_t} = \sum_{m=t+1}^{t+4} (n_m - n_{m-1}) \Delta_{d_t}(m),$$

and on these rounds $\Delta_{d_t}(m) = \Theta(2^{-t})$, so

$$\tilde{S}_{d_t} = O\Big(2^{-t} n_{t+4}\Big) = O\Big((\log M + M - t)2^t\Big).$$

Summing over t yields

$$\sum_{i:\Delta_i(M)\geq\lambda} \tilde{S}_i = O\left(\sum_{t=1}^{M} (\log M + M - t)2^t\right) = O(2^M \log M). \tag{54}$$

We now turn to the bad-event term in Theorem 1:

$$\sum_{i:\Delta_i(M)\geq\lambda} \frac{4(K(m_i)+5)R_i}{N\tilde{\Delta}^2_{\tilde{m}_i}}.$$

For both c_t and d_t, the gap is uniformly bounded by $C2^{-t}$ for all $m \geq t$, and hence

$$R_i = O(2^{-t} n_M) = O(2^{-t} 4^M \log M).$$

Moreover, for both families $\tilde{m}_i = t + O(1)$, so

$$N\tilde{\Delta}^2_{\tilde{m}_i} = \Theta(M\, 4^{M-t} \log M).$$

Since $K(m_i) = O(t)$, each summand is

$$\frac{(K(m_i)+5)R_i}{N\tilde{\Delta}^2_{\tilde{m}_i}} = O\left(\frac{t}{M} 2^t\right).$$

Summing over all suboptimal arms gives

$$\sum_{i:\Delta_i(M)\geq\lambda} \frac{4(K(m_i)+5)R_i}{N\tilde{\Delta}^2_{\tilde{m}_i}} = O(2^M). \tag{55}$$

Finally, we bound the truncation term. Since, in the present instance, each suboptimal arm is eliminated by round $t + O(1)$, the total expected number of pulls is dominated by the last few elimination scales. Therefore

$$T = \Theta(n_M) = \Theta(4^M \log M).$$

Hence

$$\lambda T = 2^{-M+1} \cdot \Theta(4^M \log M) = \Theta(2^M \log M). \tag{56}$$

Combining (54), (55), and (56) in Theorem 1 yields

$$\mathbb{E}R_M(\pi_{\text{UCB-AA}}) = O(2^M \log M).$$

Since $T = \Theta(4^M \log M)$, we also obtain

$$\mathbb{E}R_M(\pi_{\text{UCB-AA}}) = O\Big(\sqrt{T \log M}\Big) = O\Big(\sqrt{T \log K(M)}\Big),$$

as claimed.

B.4 The Case that Gaps Vary Dramatically

Example B.1 (Gaps Vary Dramatically but Assumption 1 hold)**.** *Let the initial set of arms be $A_0 = \{*_1\}$, where arm $*_1$ is the optimal arm between round 1 and round $M-1$ whose mean reward is $\mu_{*_1} = 1/2$, and arm 0 whose mean reward is $\mu_0 = 1/4$. In each subsequent round $m = 1, \ldots, M-1$, a new arm $i = m$ arrives with mean reward*

$$\mu_i = \frac{1}{2} - \frac{1}{2^M}.$$

In round M, arm $*_2$ arrives with mean reward $\mu_{*_2} = 1$. In addition, we simply set the upper bound of the number of arms in round m as $K(m) = m + 1$.

By construction, the suboptimality gap for any arm $i \geq 0$ is given by

$$\Delta_i(m) = \begin{cases} 0 & m \leq i-1 \\ \mu_{*_1} - \mu_i = \frac{1}{2^M} & i \leq m \leq M-1 \\ \mu_{*_2} - \mu_i = \frac{1}{2} + \frac{1}{2^M} & m = M \end{cases}$$

Analysis of UCB-AA: By the definition of the elimination round for arm $i \geq 1$:

$$m_i = \min\left\{m \leq M : \tilde{\Delta}_m < \frac{1}{2}\Delta_i(m)\right\} = M.$$

By the definition of $\tilde{m}_i$:

$$\tilde{m}_i = \min\left\{m : \tilde{\Delta}_m < \frac{1}{2}\Delta_i(m_i)\right\} = \min\left\{m : \tilde{\Delta}_m < \frac{1}{2}\Delta_i(M)\right\} = 2.$$

Then, the main part of the upper bound:

$$\tilde{S}_i = n_{M-1}\Delta_i(M-1) = 2^{-M}\left\lceil \frac{2\log(N2^{-2M+2})}{2^{-2M+2}} \right\rceil \leq 2^M \log(N2^{-2M+2}),$$

$$R_i = S_i \leq n_M \Delta_i(M) \leq \frac{3\log(N\cdot 2^{-2M})}{2^{-2M}} = 3\log(N\cdot 2^{-2M})\cdot 2^{2M},$$

$$\frac{(K(m_i)+5)\cdot R_i}{N\tilde{\Delta}^2_{\tilde{m}_i}} = \frac{(K(M)+5)\cdot R_i}{N\tilde{\Delta}_2^2} \leq \frac{48\cdot 2^{2M}(M+6)\log(N\cdot 2^{-2M})}{N}.$$

We know $N = \tilde{\Theta}(4^M K(M)) = \tilde{\Theta}(4^M M)$ and $T = (M+1)n_{M-1} = \tilde{\Theta}(4^M M)$, then the upper bound of UCB-AA:

$$\begin{aligned}
B_{AA} &\leq \sum_{i=1}^{M}\left(\tilde{S}_i + \frac{4\,(K(m_i)+5)\cdot R_i}{N\tilde{\Delta}^2_{\tilde{m}_i}}\right) \\
&\leq M\left(2^M\log(N2^{-2M+2}) + \frac{192\cdot 2^{2M}(M+6)\log(N\cdot 2^{-2M})}{N}\right) \\
&\leq 192\cdot M\cdot 2^M\log(N2^{-2M+2})\left(1+\frac{2^M(6+M)}{N}\right) \\
&= \tilde{\Theta}\left(M\cdot 2^M\right) = \tilde{\Theta}\left(\sqrt{K(M)T}\right).
\end{aligned}$$

Example B.2 (Gaps Vary Dramatically and Assumption 1 is Violated)**.** *Let the initial set of arms be $A_0 = \{*_1\}$, where arm $*_1$ is the optimal arm between round 1 and round $M-1$ whose mean reward is $\mu_{*_1} = 1/2$, and arm 0 whose mean reward is $\mu_0 = 1/4$. In each subsequent round $m = 1,\ldots,M-1$, a new arm $i = m$ arrives with mean reward*

$$\mu_i = \frac{1}{2} - \frac{3}{2^{M-1}}.$$

In round M, arm $*_2$ arrives with mean reward $\mu_{*_2} = 1$. In addition, we simply set the upper bound of the number of arms in round m as $K(m) = m+1$.

By construction, the suboptimality gap for any arm $i \geq 0$ is given by

$$\Delta_i(m) = \begin{cases} 0 & m \leq i-1 \\ \mu_{*_1} - \mu_i = \frac{3}{2^{M-1}} & i \leq m \leq M-1 \\ \mu_{*_2} - \mu_i = \frac{1}{2} + \frac{3}{2^{M-1}} & m = M \end{cases}$$

Analysis of UCB-AA: By the definition of the elimination round for arm $i \geq 1$:

$$m_i = \min\left\{m \leq M : \tilde{\Delta}_m < \frac{1}{2}\Delta_i(m)\right\} = M - 1.$$

By the definition of $\tilde{m}_i$:

$$\tilde{m}_i = \min\left\{m : \tilde{\Delta}_m < \frac{1}{2}\Delta_i(m_i)\right\} = \min\left\{m : \tilde{\Delta}_m < \frac{1}{2}\Delta_i(M-1)\right\} = M - 1.$$

Then, the main part of the upper bound:

$$\begin{aligned} R_i &= n_{M-1}\Delta_i(M-1) + (n_M - n_{M-1})\Delta_i(M) \geq (n_M - n_{M-1})\Delta_i(M) \\ &\geq \frac{1}{2}\left(\left\lceil\frac{2\log(N2^{-2M})}{2^{-2M}}\right\rceil - \left\lceil\frac{2\log(N2^{-2M+2})}{2^{-2M+2}}\right\rceil\right) \geq \frac{3\log(N2^{-2M})}{2^{-2M+2}}. \end{aligned}$$

$$\frac{(K(m_i)+5)\cdot R_i}{N\tilde{\Delta}^2_{\tilde{m}_i}} = \frac{(K(M-1)+5)\cdot R_i}{N\tilde{\Delta}^2_{M-1}} \geq \frac{3\cdot 2^{4M} M\log(N\cdot 2^{-2M})}{64N}.$$

Since $N = 10K(M)\left\lceil 2\cdot 4^M \log\left(\frac{N}{4^M}\right)\right\rceil$, we have $N \geq K(M)4^M = (M+1)4^M$ and $N \leq 420(M+1)\log(M+1)\cdot 4^M$ by Lemma A.8. Then the upper bound of UCB-AA:

$$B_{AA} \geq \sum_{i=1}^{M} \frac{4\,(K(m_i)+5)\cdot R_i}{N\tilde{\Delta}^2_{\tilde{m}_i}} \geq \frac{3\cdot 2^{4M} M^2\log(N\cdot 2^{-2M})}{64N} = \tilde{\Theta}\left(4^M\cdot M\right)$$

we know:

$$4^M \leq n_M \leq T \leq K(M)\cdot n_M = \Theta(M\log(M)4^M),$$

Therefore, in the sense of ignoring logarithmic terms, we have:

$$B_{AA} = \tilde{\Omega}(T).$$

Discussion

Here we compare the asymptotic orders of the UCB-AA regret upper bound given by Theorem 1 for Example B.1 and Example B.2. The two settings are highly similar: the only difference is a very small change in the means of arms arriving in rounds 1 to $M-1$. However, this slight change shifts the elimination stage of these arms: in Example B.1 they

are removed in the pre-elimination step of round M, whereas in Example B.2 they are removed in the formal elimination phase of round $M-1$. In both cases, these arms are expected to be pulled a comparable number of times, because the pre-elimination step in round M does not pull the incumbent arms that survived from rounds 1 to $M-1$. Hence, the actual regrets in the two examples should be of similar magnitude. Nevertheless, our bound analysis yields a sublinear simplified order in Example B.1 but a linear simplified order in Example B.2. This discrepancy stems from the fact that when gaps vary sharply, the bound constructed via R_i is not tight and may fail to accurately reflect the actual regret. Therefore, when gap variation is large enough to violate Assumption 1, a linear simplified upper bound does not imply that the algorithm itself incurs linear regret. We acknowledge that our current regret analysis has limitations in this regime.

C Details of Simulation

In this study, two challenging MAB-AA scenarios are employed to evaluate the performance of algorithms under arm arrivals and optimal arm shifts. All experiments are conducted over discrete time steps $t = 1, \ldots, T$.

C.1 Scenario 1: Deterministic Step Growth

It is an environment with background stochastic noise where significantly superior arms appear at specific, pre-defined intervals. It aims to test the algorithm's responsiveness to sudden shifts in the optimal arm.

- **Background Arrival Mechanism:**
  - **Arrival Process:** New arms arrive according to a Poisson process with rate ρ. The number of new arms at each time step t is independently sampled from Poisson(ρ).

  - **Mean Generation:** The expected reward μ of background arms follows a bi-modal mixture distribution:
    * With probability π_{good} (default 0.1), μ is sampled from Uniform$(\mu_h - \delta, \mu_h)$.
    * With probability $1 - \pi_{good}$, μ is sampled from Uniform(μ_l, μ_m).
- **Forced Step Arms:**
  - **Trigger Points:** At pre-defined proportions $t/T \in \{0.3, 0.6, 0.9\}$, a high-performance arm is forced into the system.
  - **Step Mean Calculation:** The mean of the k-th forced arm is determined by $\mu_k = \text{clip}(r_0 + k \cdot \Delta_{step}, [0, 1])$.

C.2 Scenario 2: Near-Optimal Arrivals

It is a highly competitive environment where newly arriving arms frequently possess rewards extremely close to the current optimum. It requires algorithms to maintain high sensitivity to detect new superior arms while exploiting existing ones.

- **Initial Configuration ($t = 1$):**
  - The system starts with K_0 arms. The first m_0 arms are "near-optimal", with means sampled using the near-optimal distribution described below. The remaining $K_0 - m_0$ arms are sampled using the mixture distribution from Scenario 1.
- **Dynamic Arrival and Mean Mechanism:**
  - **Arrival Process:** Arms arrive via a Bernoulli process with probability λ at each step.
  - **Mean Generation:**
    * **Near-Optimal Arms:** Each new arm has a probability p_{join_near} of being "near-optimal", with $\mu \sim$ Uniform$(\mu_{top} - \Delta_{min}, \mu_{top})$.
    * **Base Arms:** With probability $1 - p_{join_near}$, the mean is generated via the

standard mixture distribution.

The following table summarizes the key parameters used for the experimental environments in Scenarios 1 and 2.

Table 1: Hyper-parameter Configurations for Simulation Scenarios

Category	**Symbol**	**Description**	**Default Value**
General	T	Total time steps	Variable
	K_0	Initial number of arms	Variable
Scenario 1	ρ	Poisson arrival rate	1/200
	r_0	Base mean for step arms	0.7
	Δ_{step}	Increment for step arms	0.05
	γ	Trigger points for step arms	$\{0.3, 0.6, 0.9\}$
Scenario 2	λ	Bernoulli arrival probability	1/200
	m_0	Number of initial near-optimal arms	5
	μ_{top}	Upper bound for near-optimal means	0.9
	Δ_{min}	Maximum deviation for near-optimal arms	0.01
	p_{join_near}	Probability of a new arm being near-optimal	0.3
Mixture Dist.	π_{good}	Probability of generating a "good" arm	0.1
	μ_h, δ	High-quality arm mean and range	$0.8, 0.1$
	μ_l, μ_m	Base arm mean lower/upper bounds	$0.1, 0.4$

C.3 Details of Algorithms

Here we present the pseudocode of other algorithms used in the simulations.

Algorithm 4 Growing UCB1 for Arriving Arms

Input: Initial arm set A_0, horizon T.

Initialize $K \leftarrow A_0$; for $i \in K$: $N_i \leftarrow 0,\ S_i \leftarrow 0$.
while the environment is not done **do**
  if $\exists\, i \in K$ with $N_i = 0$ **then**
    Select smallest-index arm in $\{i \in K : N_i = 0\}$.
  else
    $t \leftarrow \max\{2, \text{current time} + 1\}$.
    for each arm $i \in K$ **do**
      $U_i \leftarrow S_i/N_i + \sqrt{2 \log t / N_i}$.
    end for
    Select $I_t \in \arg\max_{i \in K} U_i$ (smallest-index tie break).
  end if
  Pull I_t, observe $X_t \in \{0, 1\}$, and update N_{I_t}, S_{I_t}.
  Receive V_t and set $K \leftarrow K \cup V_t$.
  for each newly visible arm $j \in V_t$ **do**
    Initialize $N_j \leftarrow 0,\ S_j \leftarrow 0$.
  end for
end while
return $\{N_i, S_i\}_{i \in K}$.

Algorithm 5 Growing MOSS for Arriving Arms

Input: Initial arm set A_0, horizon T.

Initialize $K \leftarrow A_0$; for $i \in K$: $N_i \leftarrow 0,\ S_i \leftarrow 0$.
while the environment is not done **do**
  if $\exists\, i \in K$ with $N_i = 0$ **then**
    Select smallest-index arm in $\{i \in K : N_i = 0\}$.
  else
    $t \leftarrow \max\{2, \text{current time} + 1\},\ K_t \leftarrow |K|$.
    for each arm $i \in K$ **do**
      $M_i \leftarrow S_i/N_i + \sqrt{\log(\max\{1, t/(K_t N_i)\})/N_i}$.
    end for
    Select $I_t \in \arg\max_{i \in K} M_i$ (smallest-index tie break).
  end if
  Pull I_t, observe $X_t \in \{0, 1\}$, and update N_{I_t}, S_{I_t}.
  Receive V_t and set $K \leftarrow K \cup V_t$.
  for each newly visible arm $j \in V_t$ **do**
    Initialize $N_j \leftarrow 0,\ S_j \leftarrow 0$.
  end for
end while
return $\{N_i, S_i\}_{i \in K}$.

Algorithm 6 Dynamic SE

Input: Initial arms A_0, horizon T, parameter $\gamma > 0$.

Initialize $C \leftarrow A_0$, $A \leftarrow \varnothing$; for $i \in C$: $n_i \leftarrow 0, R_i \leftarrow 0$.
while the environment is not done **do**
  $B \leftarrow (C \cup A) \cap \mathcal{A}_{\text{arrived}}$; if $B = \varnothing$, set $B \leftarrow \mathcal{A}_{\text{arrived}}$.
  if $B = \varnothing$ **then**
    break.
  end if
  Initialize unseen $i \in B$: $n_i \leftarrow 0, R_i \leftarrow 0$; $\mathcal{B}uf \leftarrow \varnothing$.
  $O \leftarrow \{i \in B : n_i > 0\}$, $N \leftarrow B \setminus O$.
  if $O \neq \varnothing$ and $N \neq \varnothing$ **then**
    $\tau \leftarrow \min_{i \in O} n_i$; pull each $j \in N$ until $n_j \geq \tau$ (new arrivals $\rightarrow \mathcal{B}uf$).
    $\tau_B \leftarrow \min_{i \in B} n_i$; $\beta \leftarrow 2\gamma\sqrt{2\log(\max\{1, T/\tau_B\})/\tau_B}$; run SE-elimination on B.
  end if
  for each $i \in B$ **do**
    Pull i, update (n_i, R_i), and buffer new arrivals in $\mathcal{B}uf$.
  end for
  $C \leftarrow$ SE-survivors of B; $A \leftarrow \mathcal{B}uf$.
end while
$E \leftarrow C$ if nonempty, else $\mathcal{A}_{\text{arrived}}$.
while the environment is not done **do**
  Pull $\arg\max_{i \in E}(R_i/n_i)$ (smallest-index tie break); update.
end while
return C.